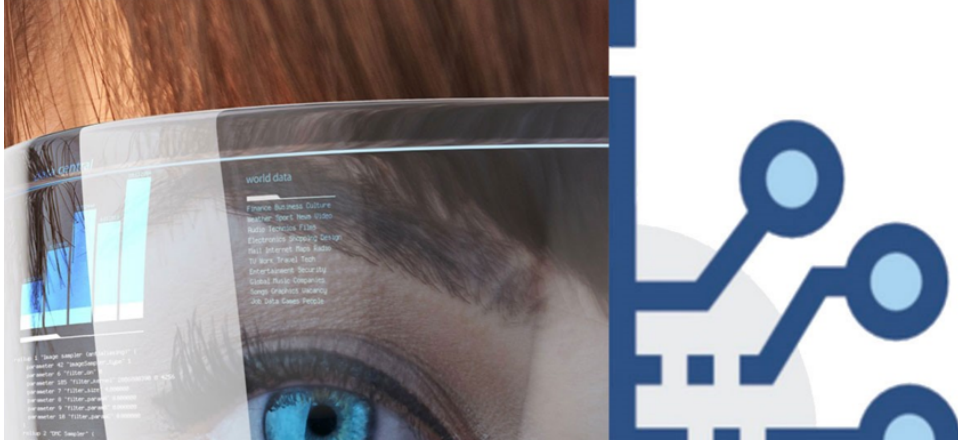




Multi-Asset Signal Generation

Principal Financial Group
Carnegie Mellon University

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1 Asset Classes

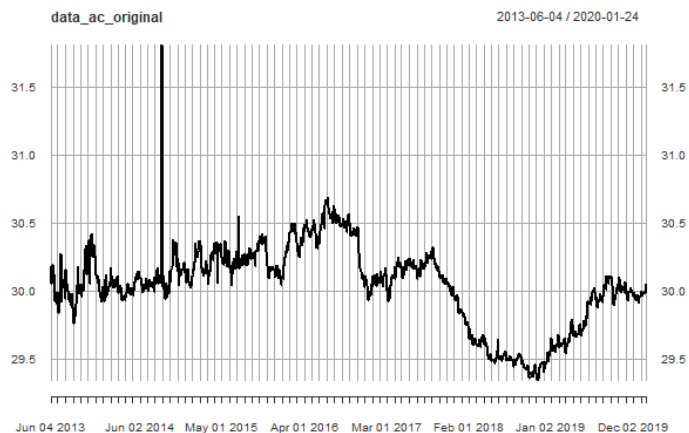
1.1 Summary of Asset Classes

Table 1: FixedIncome

	AssetClass	AC_ID	CSI_ID	Description	SOURCE	MOD_ID
1	FixedIncome	FixedIncome	FI_treas_short	Fixed Income Treasury short term	Principle	FI_treas_short
2	FixedIncome	FixedIncome	FI_treas_long	Fixed Income Treasury long term	Principle	FI_treas_long
3	FixedIncome	FixedIncome	FI_IG_short	Fixed Income IG short term	Principle	FI_IG_short
4	FixedIncome	FixedIncome	FI_IG_long	Fixed Income IG long term	Principle	FI_IG_long
5	FixedIncome	FixedIncome	FI_IG_mid	Fixed Income IG mid term	Principle	FI_IG_mid
6	FixedIncome	FixedIncome	FI_HY	Fixed Income High Yield	Principle	FI_HY
7	FixedIncome	FixedIncome	FI_deved_intl	Fixed Income Developed International	Principle	FI_deved_intl
8	FixedIncome	FixedIncome	FI_deving_intl	Fixed Income Developing International	Principle	FI_deving_intl

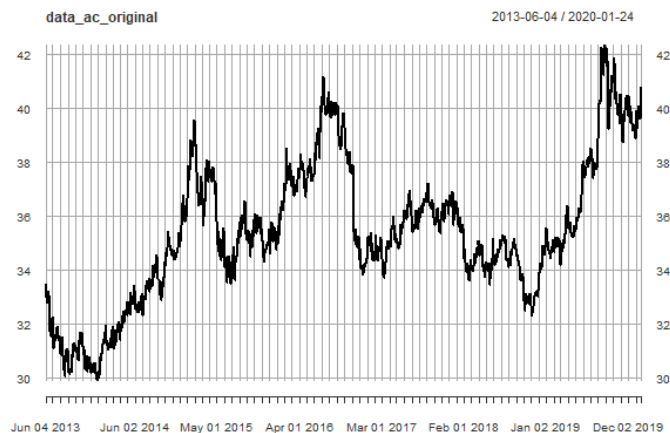
1.2 History: FI_{treas_hort}

Current Version is $FI_{development.PriorVersionis}$.



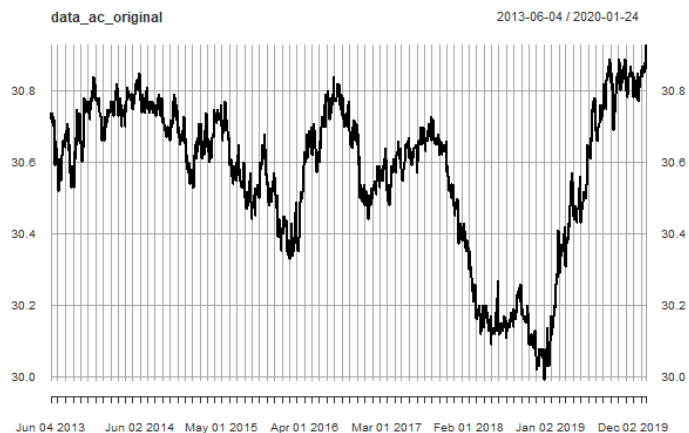
1.3 History: $FI_{t_{reasong}}$

Current Version is $FI_{Development.PriorVersionis}$.



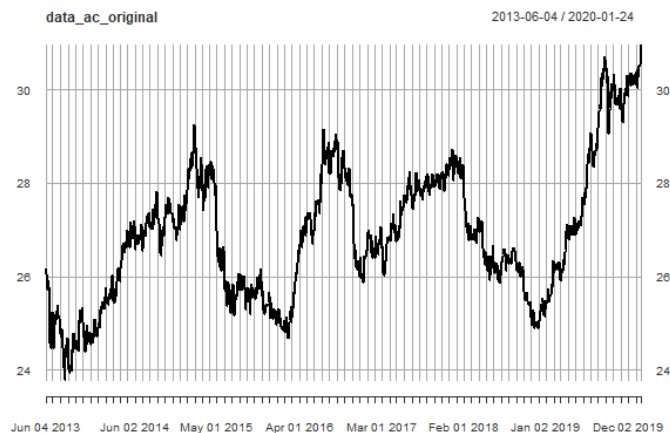
1.4 History: $FI_{IGshort}$

Current Version is $FI_{development.PriorVersionis}$.



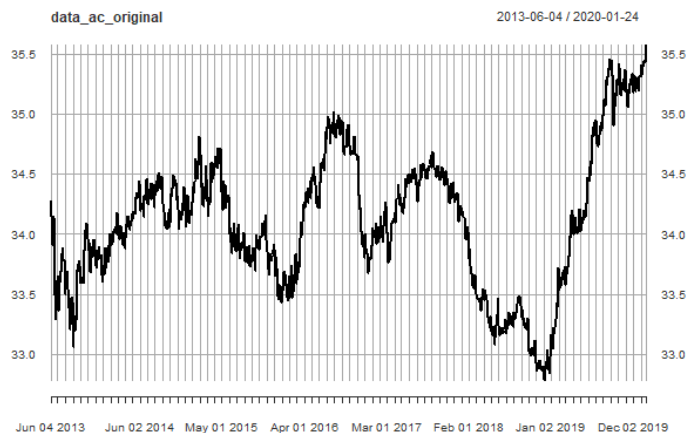
1.5 History: FI_{long}

Current Version is $FI_{development.PriorVersionis}$.



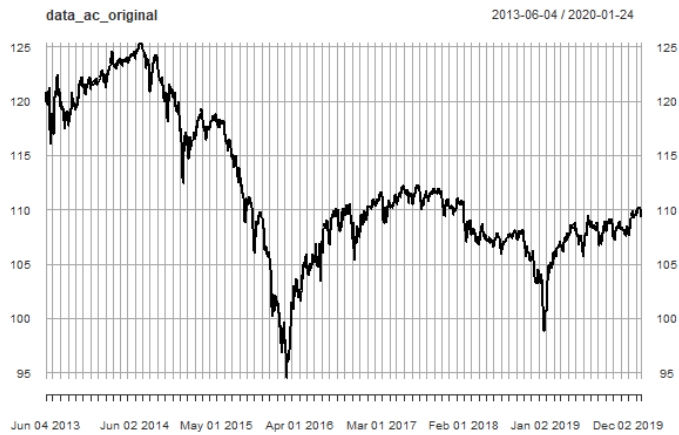
1.6 History: FI_{IGmid}

Current Version is $FI_{Development.PriorVersionis}$.



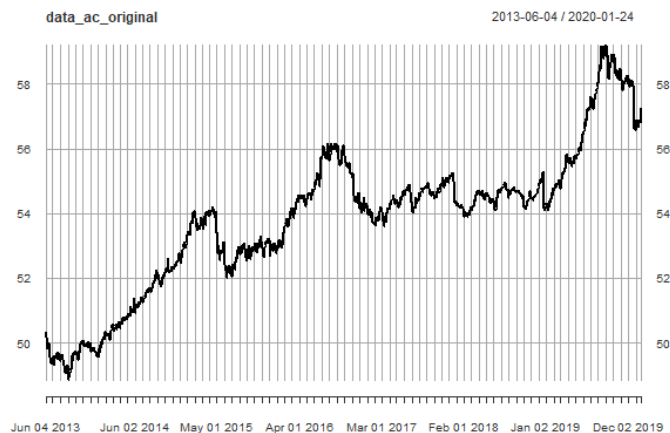
1.7 History: FI_{HY}

Current Version is $FI_{Development.PriorVersionis}$.



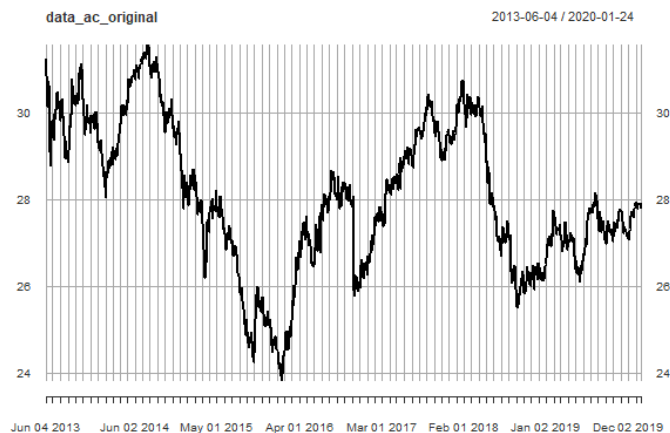
1.8 History: FI_{dev}^{intl}

Current Version is $FI_{development.PriorVersionis}$.



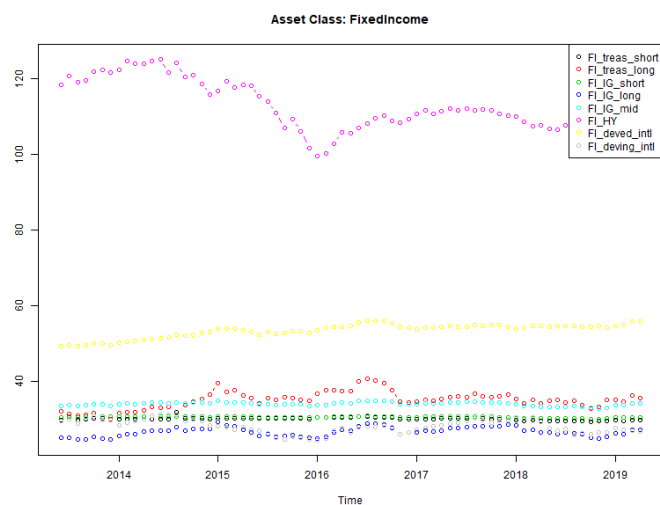
1.9 History: $FI_{deving;ntl}$

Current Version is $FI_{development.PriorVersionis}$.



2 Clusters Analysis of Asset Classes

2.1 Asset Class: *FixedIncome*



FixedIncome

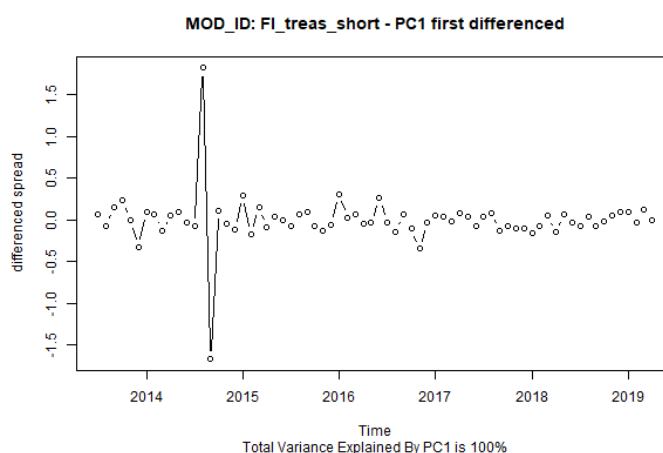
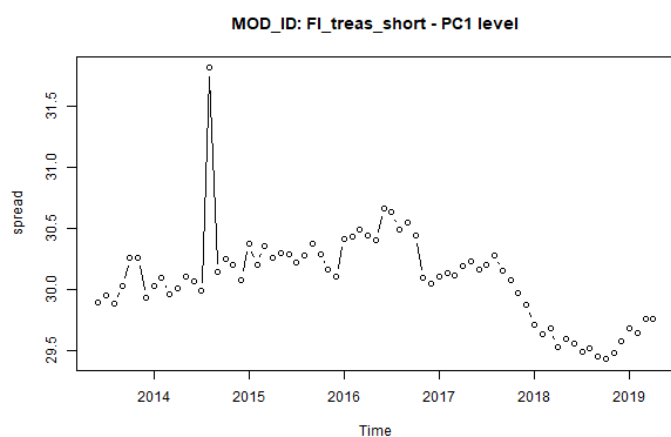
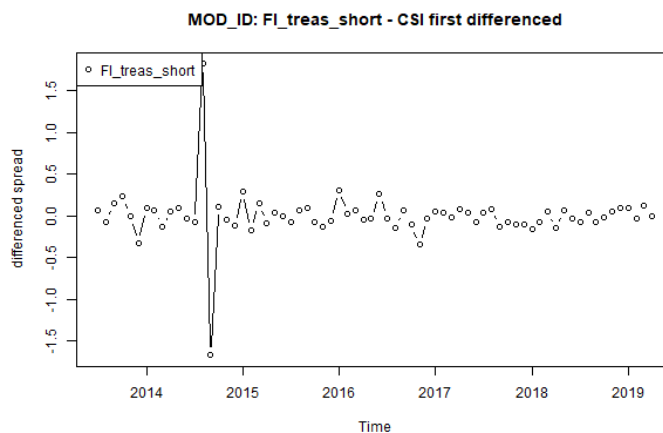
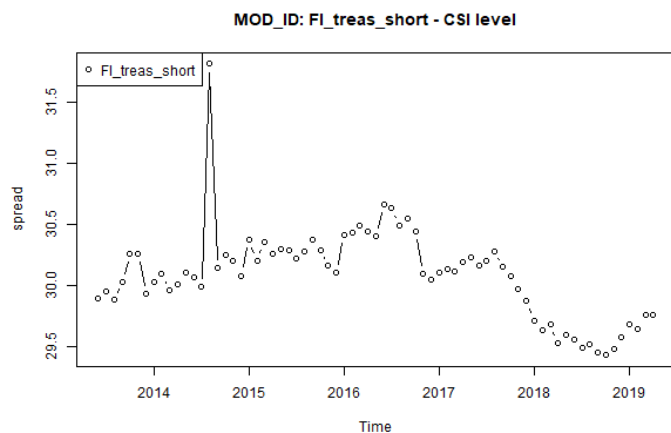
ariance%	FI_deved_intl	FI_deving_intl	FI_HY	FI_IG_long	FI_IG_mid	FI_IG_short	FI_treas_long	FI_treas_short	Spearman	Kendall
64.9%	1	1	1	1	1	1	1	1	11%	7%
86.9%	2	1	1	2	2	2	2	1	27%	18%
86.9%	2	3	3	2	2	2	2	1	55%	38%
88%	2	4	4	2	3	3	2	1	59%	40%
88%	2	5	4	2	3	3	2	1	78%	63%
91.7%	5	6	4	2	3	3	2	1	83%	64%
97.6%	6	7	5	4	3	3	2	1	86%	69%
100%	7	8	6	4	5	3	2	1	100%	100%

Table 2: Cluster Analysis - FixedIncome

	1	2	3	4	5	6	7	8
FI_treas_short	1	1	1	1	1	1	1	1
FI_HY	1	1	3	4	4	4	5	6
FI_deving_intl	1	1	3	4	5	6	7	8
FI_treas_long	1	2	2	2	2	2	2	2
FI_IG_long	1	2	2	2	2	2	4	4
FI_deved_intl	1	2	2	2	2	5	6	7
FI_IG_short	1	2	2	3	3	3	3	3
FI_IG_mid	1	2	2	3	3	3	3	5
Variance% of PC1	64.9%	86.9%	86.9%	88%	88%	91.7%	97.6%	100%
Retained Correlation	11%	27%	55%	59%	78%	83%	86%	100%

3 PCA for Clusters

3.1 Cluster: FI_{treas_short}



	EIGENVECTOR	MULTIPLIER
FI_treas_short	1.00	1.00

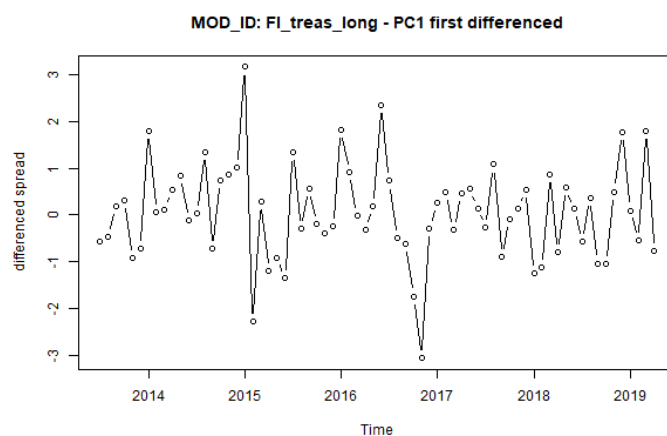
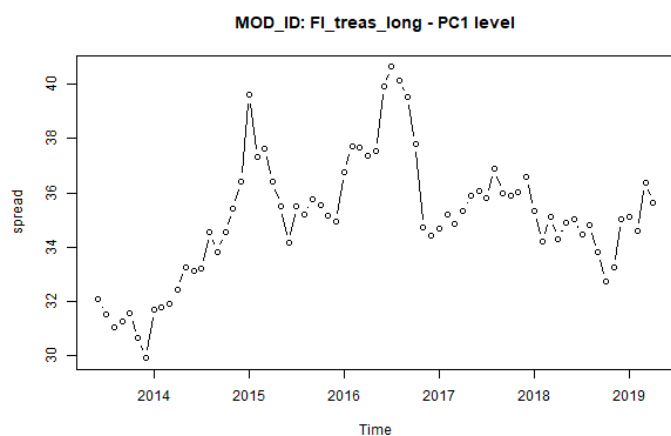
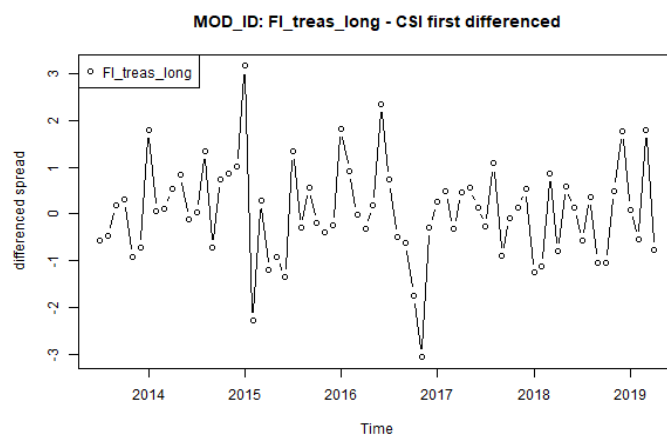
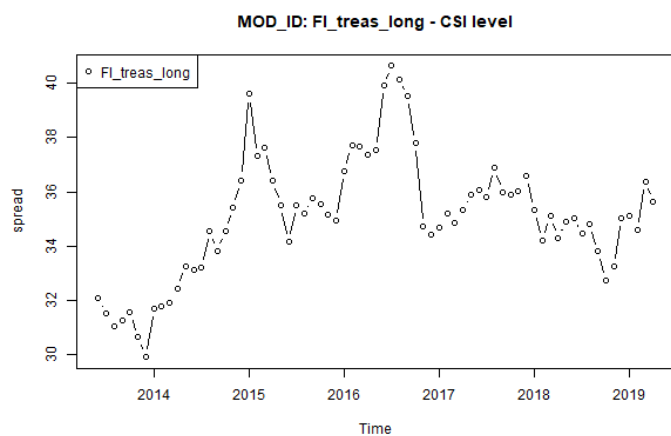
Kendall Correlation

	FI_treas_short	pc1
FI_treas_short	1	1
pc1	1	1

Spearman Correlation

	FI_treas_short	pc1
FI_treas_short	1	1
pc1	1	1

3.2 Cluster: FI_{treas_long}



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
FI_treas_long	1.00	1.00

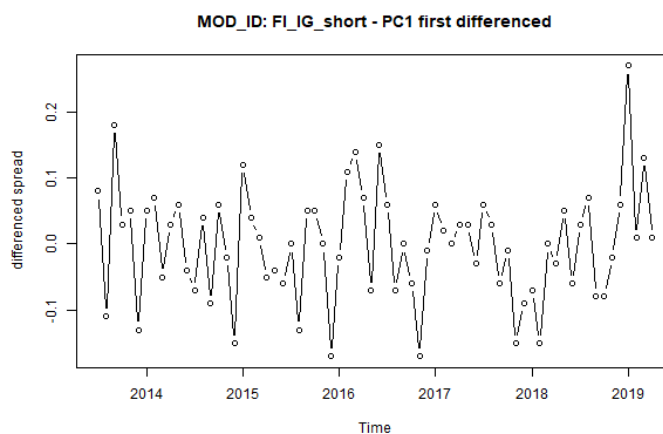
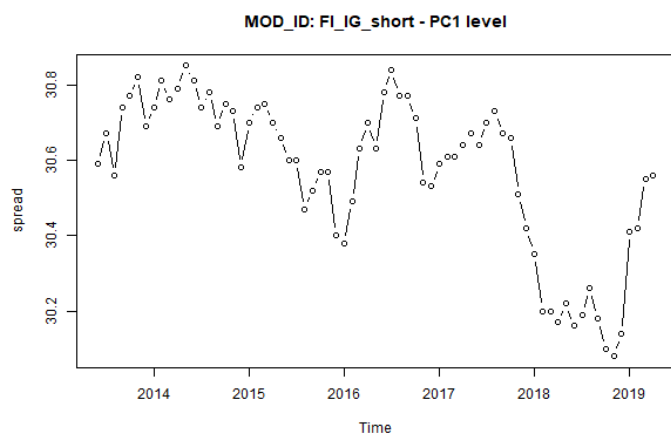
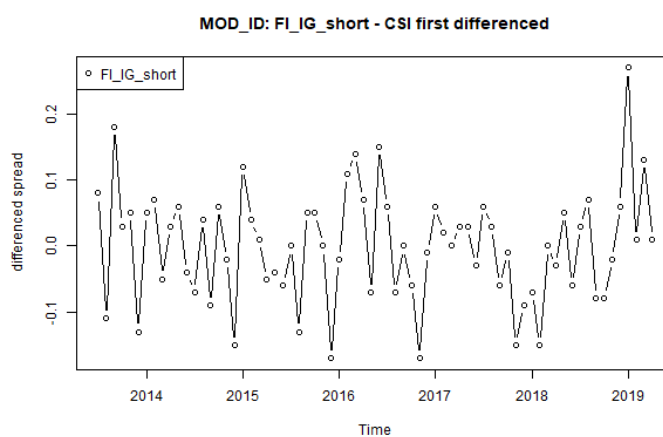
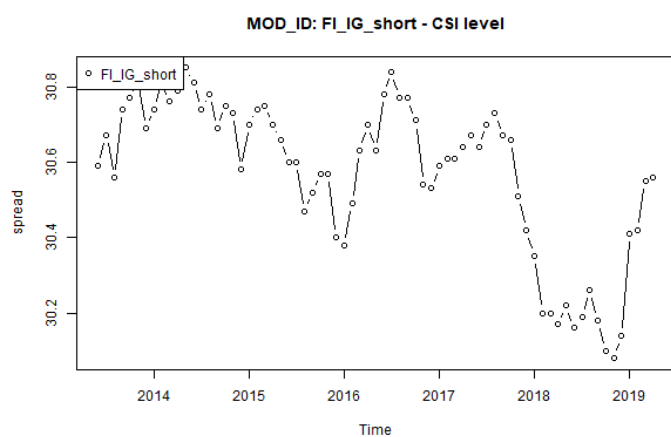
Kendall Correlation

	FI_treas_long	pc1
FI_treas_long	1	1
pc1	1	1

Spearman Correlation

	FI_treas_long	pc1
FI_treas_long	1	1
pc1	1	1

3.3 Cluster: FI_{IG_short}



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
FI_IG_short	1.00	1.00

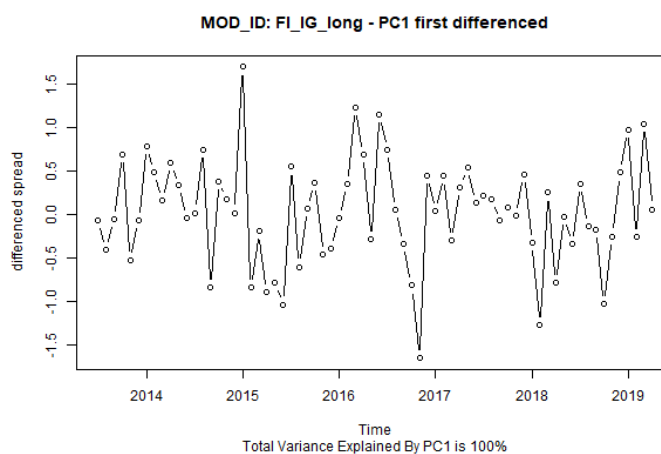
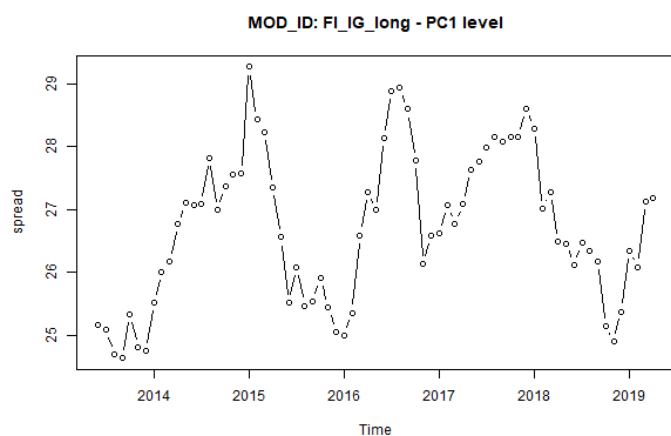
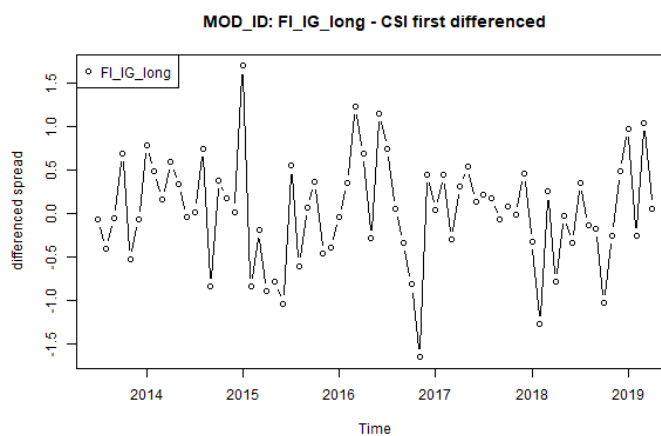
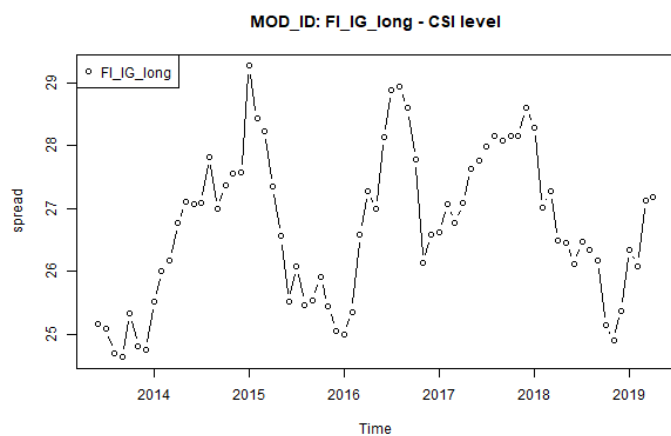
Kendall Correlation

	FI_IG_short	pc1
FI_IG_short	1	1
pc1	1	1

Spearman Correlation

	FI_IG_short	pc1
FI_IG_short	1	1
pc1	1	1

3.4 Cluster: FI_{IG_long}



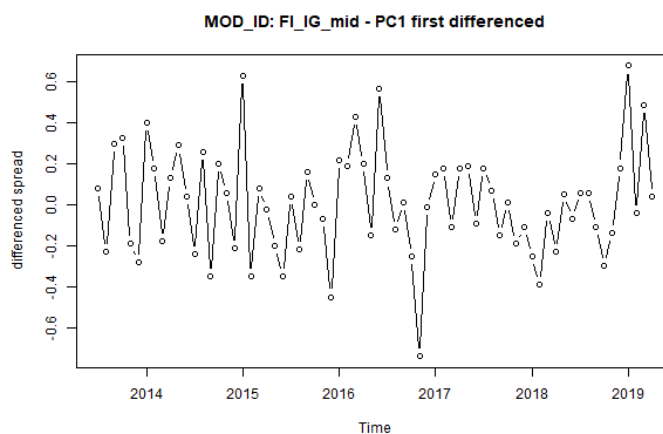
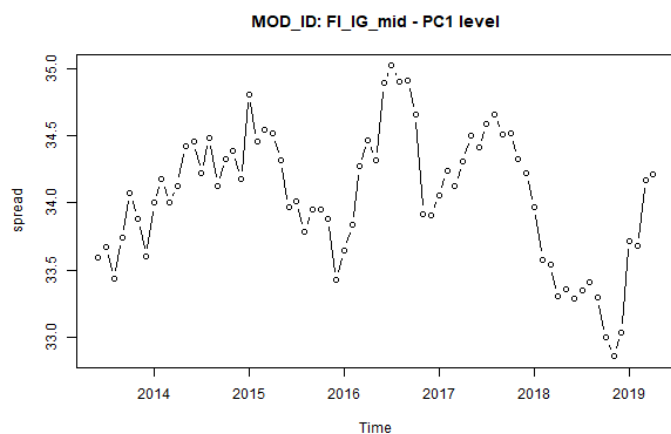
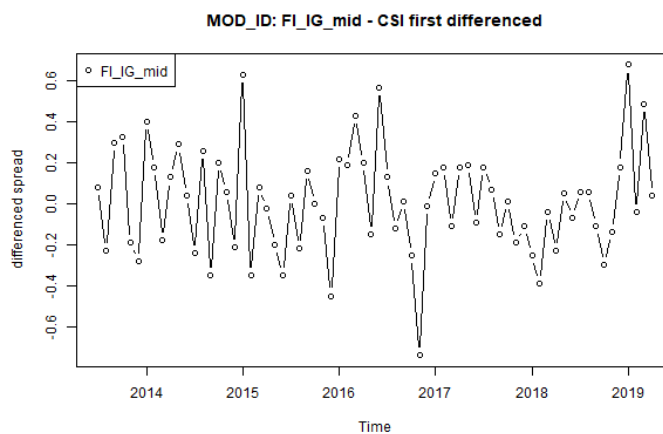
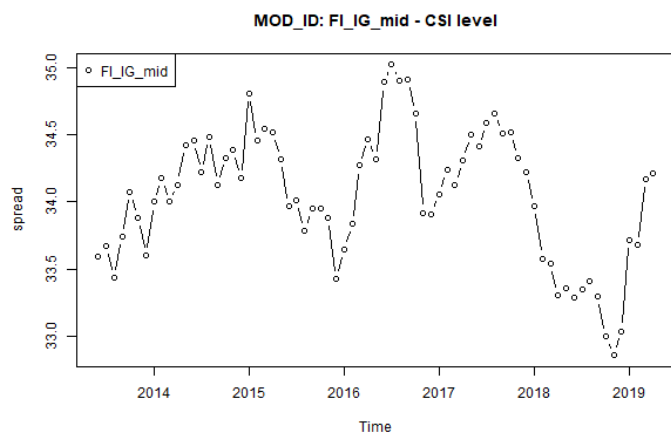
	EIGENVECTOR	MULTIPLIER
FI_IG_long	1.00	1.00

Kendall Correlation

	FI_IG_long	pc1
FI_IG_long	1	1
pc1	1	1

Spearman Correlation

	FI_IG_long	pc1
FI_IG_long	1	1
pc1	1	1

3.5 Cluster: $FI_{IG_{mid}}$ 

Total Variance Explained By PC1 is 100%

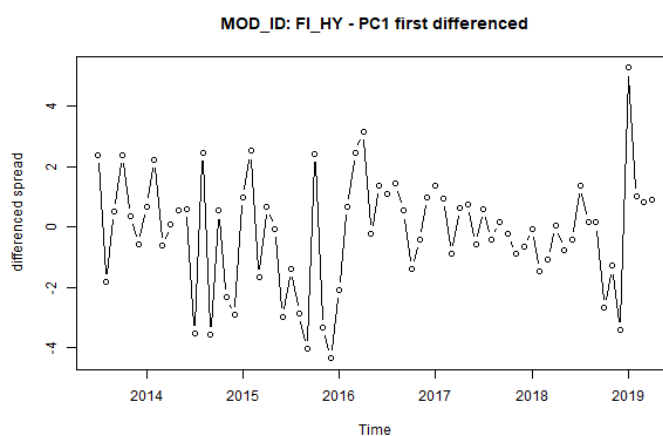
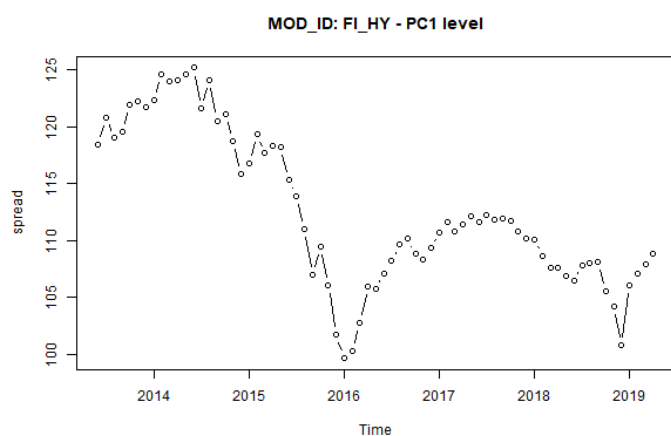
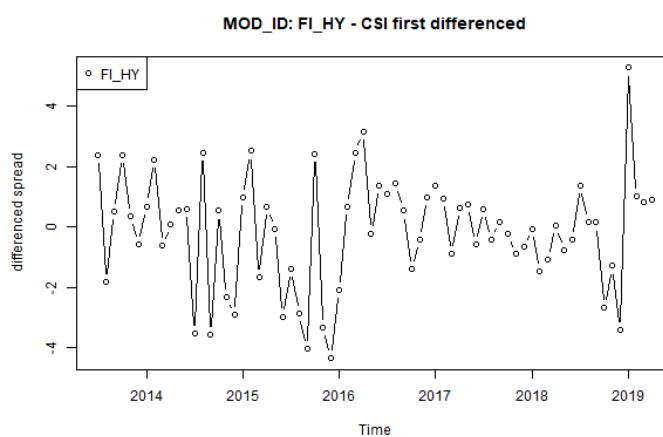
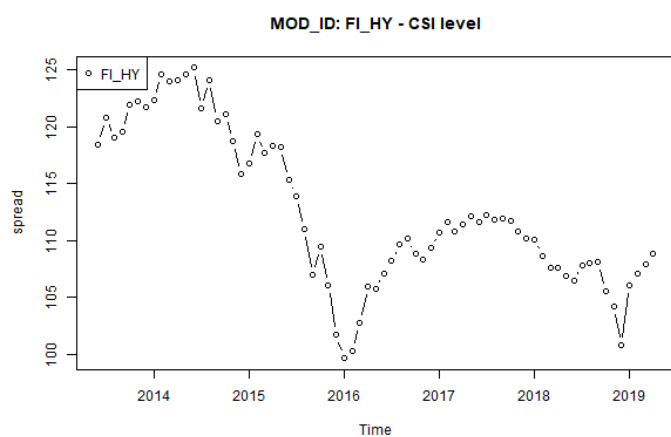
	EIGENVECTOR	MULTIPLIER
FI_IG_mid	1.00	1.00

Kendall Correlation

	FI_IG_mid	pc1
FI_IG_mid	1	1
pc1	1	1

Spearman Correlation

	FI_IG_mid	pc1
FI_IG_mid	1	1
pc1	1	1

3.6 Cluster: FI_{HY} 

Total Variance Explained By PC1 is 100%

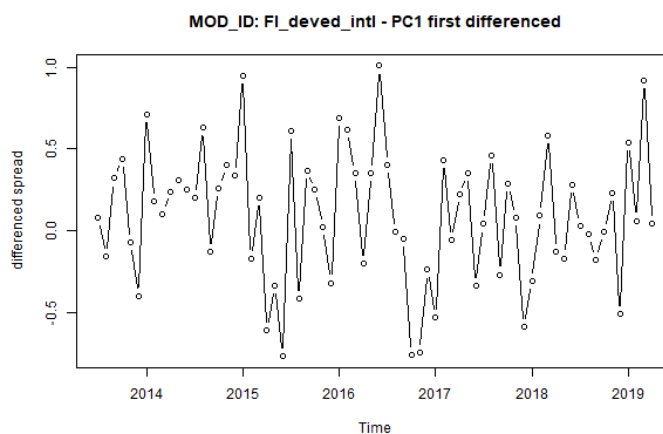
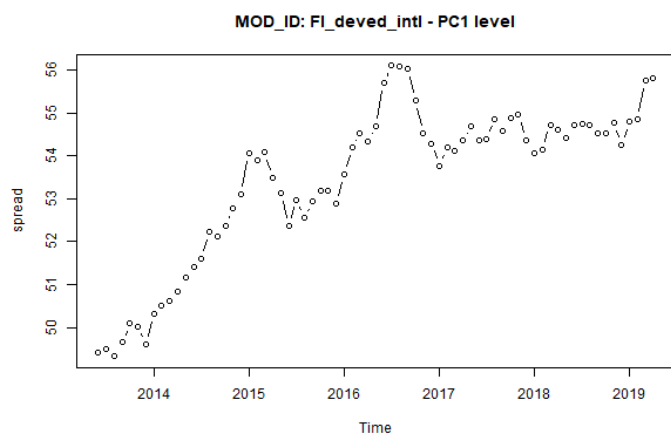
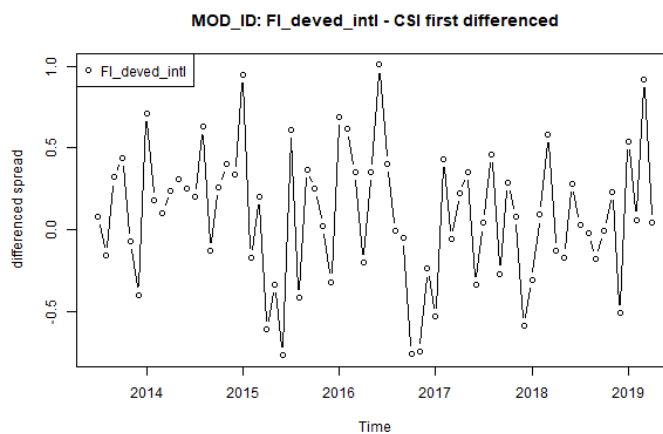
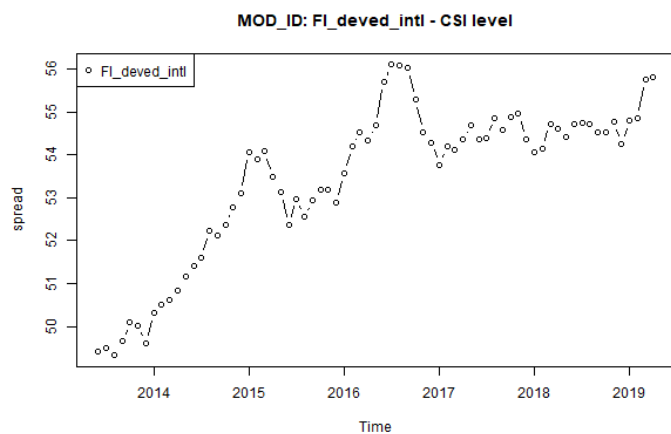
	EIGENVECTOR	MULTIPLIER
FI_HY	1.00	1.00

Kendall Correlation

	FL_HY	pc1
FL_HY	1	1
pc1	1	1

Spearman Correlation

	FL_HY	pc1
FL_HY	1	1
pc1	1	1

3.7 Cluster: FI_{deved_intl} 

Total Variance Explained By PC1 is 100%

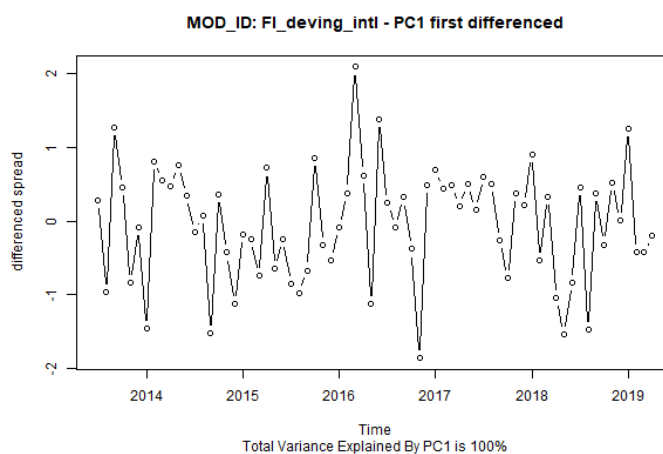
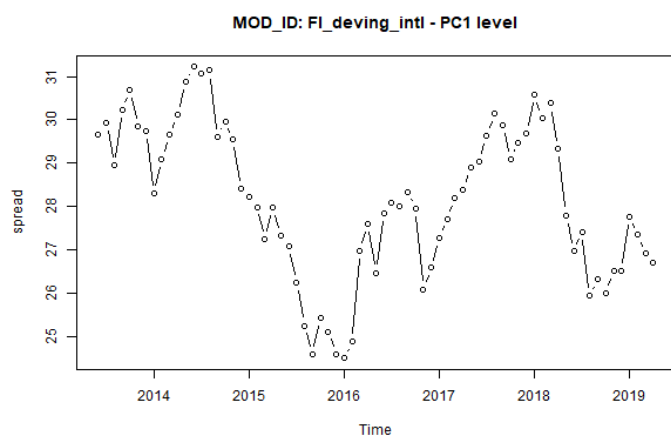
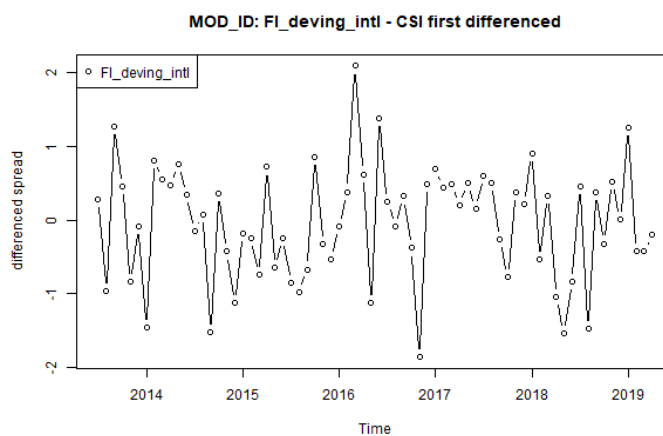
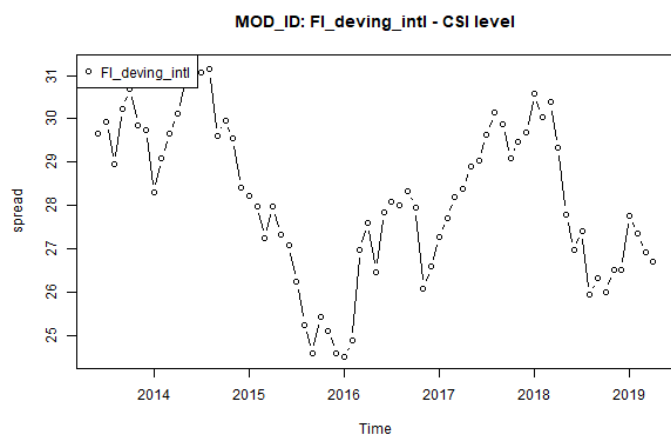
	EIGENVECTOR	MULTIPLIER
FI.deved.intl	1.00	1.00

Kendall Correlation

	FI_deved_intl	pc1
FI_deved_intl	1	1
pc1	1	1

Spearman Correlation

	FI_deved_intl	pc1
FI_deved_intl	1	1
pc1	1	1

3.8 Cluster: FI_{deving_intl} 

	EIGENVECTOR	MULTIPLIER
FI.deving_intl	1.00	1.00

Kendall Correlation

	FI_deving_intl	pc1
FI_deving_intl	1	1
pc1	1	1

Spearman Correlation

	FI_deving_intl	pc1
FI_deving_intl	1	1
pc1	1	1

4 Variable Selection

4.1 Single Factor Model for FI_{treas_hort}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG10YR_IndexMoMDifferenceL0	0.19	26.26	-0.78	0.00	0.07			
2	USGG10YR_IndexMoMPercentChangeL0	0.18	27.64	-1.63	0.00	0.05			
3	USGG2YR_IndexMoMPercentChangeL0	0.17	28.27	-0.88	0.00	0.04			
4	USGG30YR_IndexMoMDifferenceL0	0.16	29.16	-0.78	0.01	0.08			
5	USGG30YR_IndexMoMPercentChangeL0	0.14	30.49	-2.09	0.01	0.07			
6	USGG2YR_IndexMoMDifferenceL0	0.13	31.31	-0.99	0.00	0.12			

Table 3: Single Factor Analysis- FI_{treas_hort}

4.1.1 Model - 1

Backtesting on differenced values

FI_treas_short - Backtesting

$R^2: 0.19 / -0.7808 * USGG10YR_IndexMoMDifferenceL0$

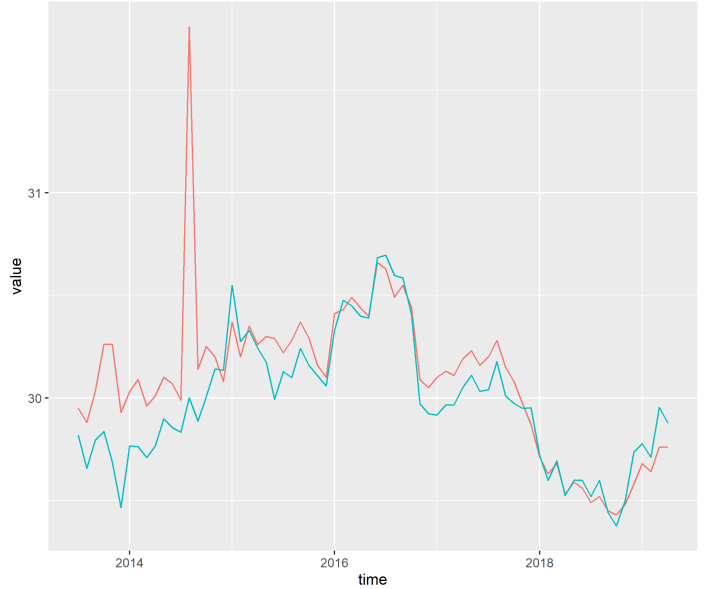


variable — FI_treas_short_History — FI_treas_short_FittedValue

Backtesting on level values

FI_treas_short - Level Value Backtesting

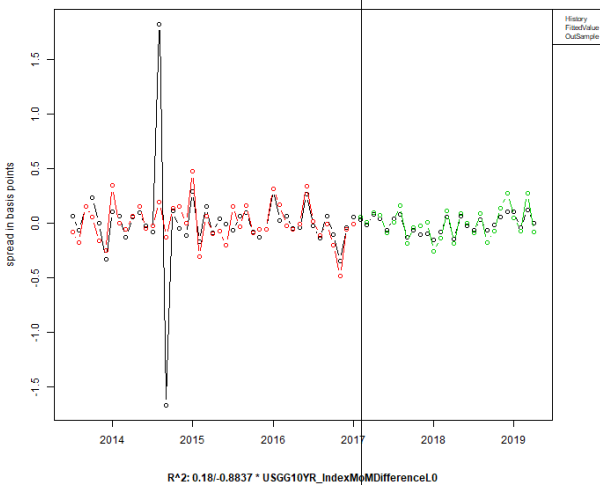
$R^2: 0.19 / -0.7808 * USGG10YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

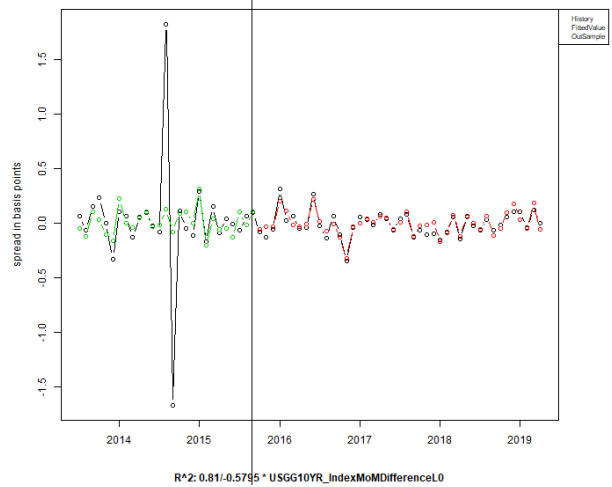
FI_treas_short Jul 2013-Apr 2019



$R^2: 0.18 / -0.8837 * USGG10YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



$R^2: 0.81 / -0.5795 * USGG10YR_IndexMoMDifferenceL0$

4.1.2 Model - 2

Backtesting on differenced values

FI_treas_short - Backtesting

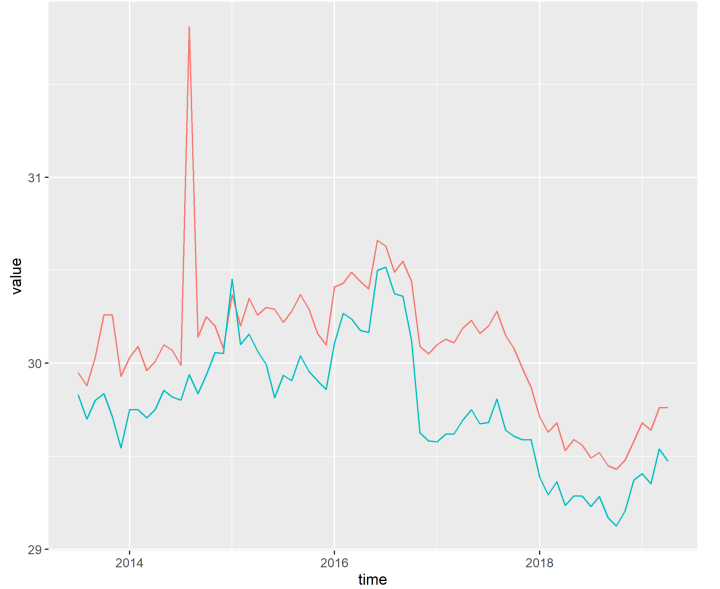
$R^2: 0.18 / -1.6332 * USGG10YR_IndexMoMPercentChangeL0$



Backtesting on level values

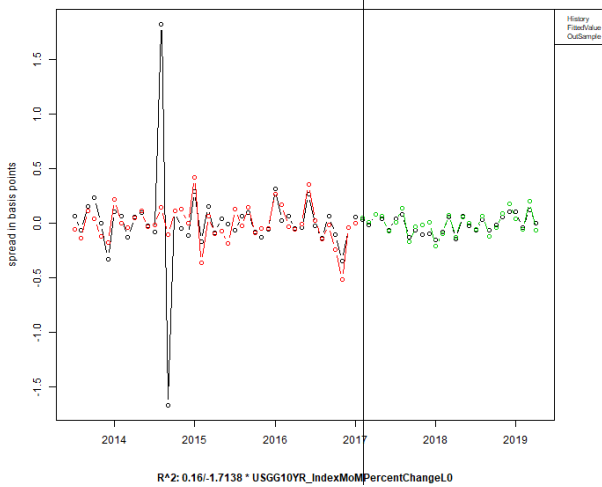
FI_treas_short - Level Value Backtesting

$R^2: 0.18 / -1.6332 * USGG10YR_IndexMoMPercentChangeL0$



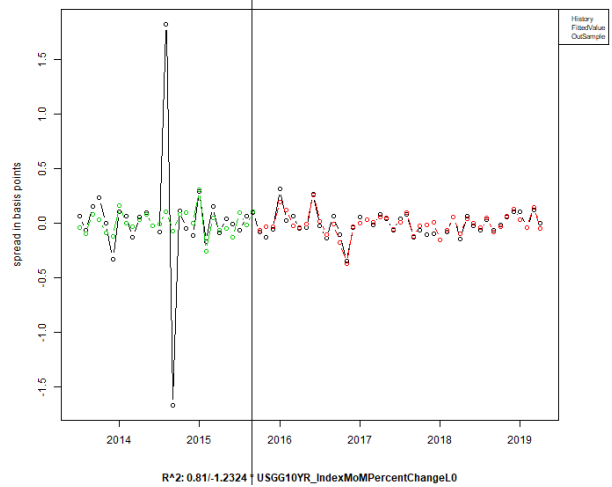
Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



4.1.3 Model - 3

Backtesting on differenced values

FI_treas_short - Backtesting

$R^2: 0.17 / -0.8816 * USGG2YR_IndexMoMPercentChangeL0$

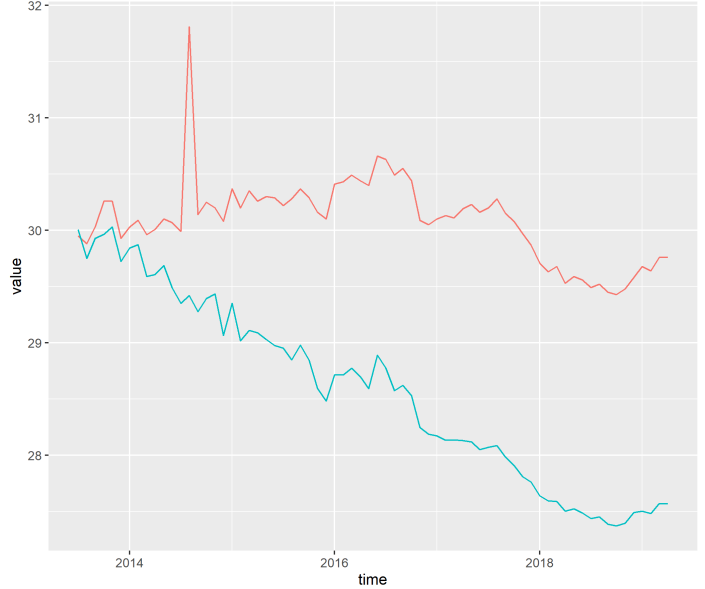


variable — FI_treas_short_History — FI_treas_short_FittedValue

Backtesting on level values

FI_treas_short - Level Value Backtesting

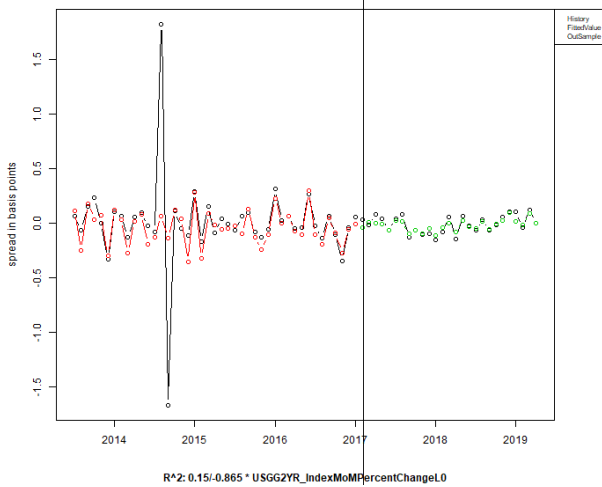
$R^2: 0.17 / -0.8816 * USGG2YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

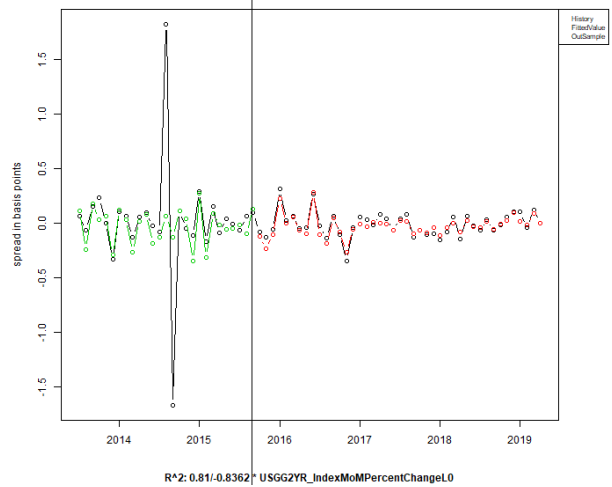
FI_treas_short Jul 2013-Apr 2019



$R^2: 0.15 / -0.865 * USGG2YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



$R^2: 0.81 / -0.8362 * USGG2YR_IndexMoMPercentChangeL0$

4.1.4 Model - 4

Backtesting on differenced values

FI_treas_short - Backtesting

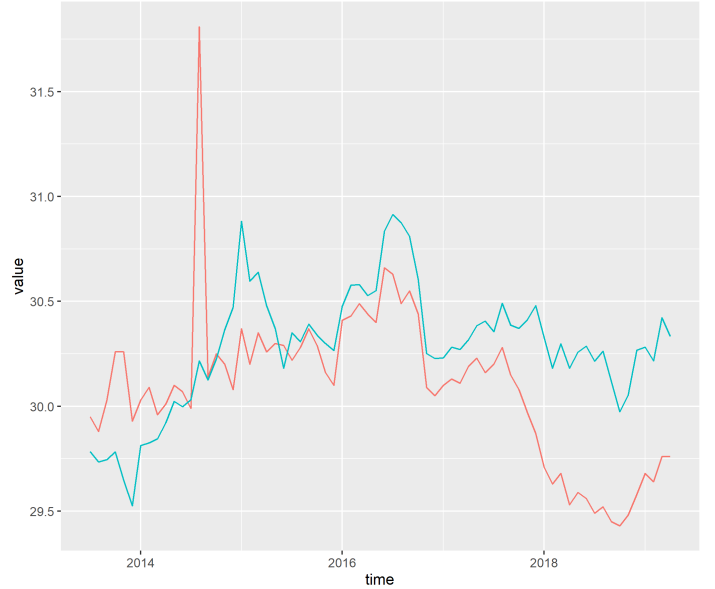
$R^2: 0.16 / -0.7773 * USGG30YR_IndexMoMDifferenceL0$



Backtesting on level values

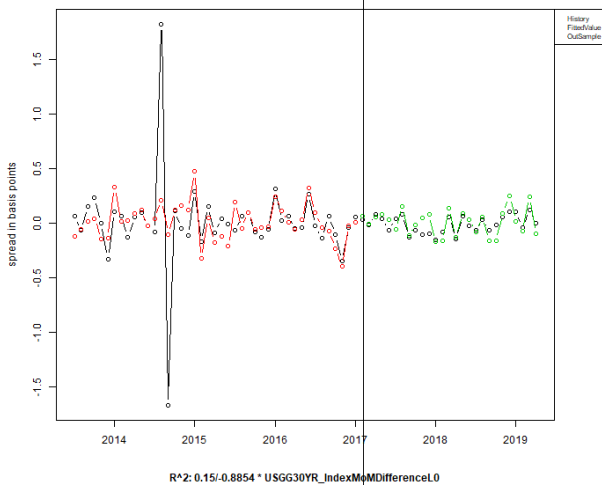
FI_treas_short - Level Value Backtesting

$R^2: 0.16 / -0.7773 * USGG30YR_IndexMoMDifferenceL0$



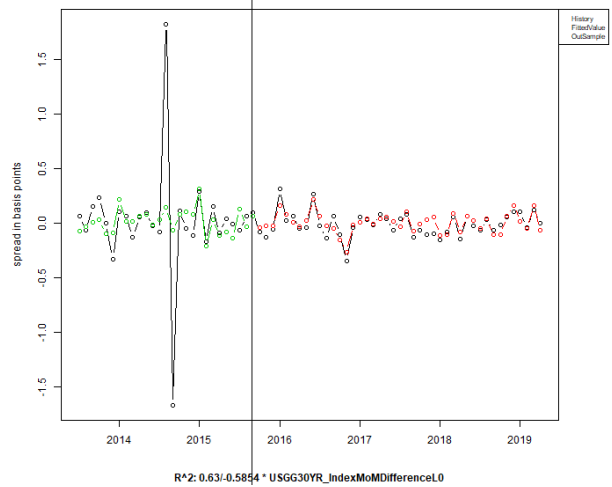
Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



4.1.5 Model - 5

Backtesting on differenced values

FI_treas_short - Backtesting

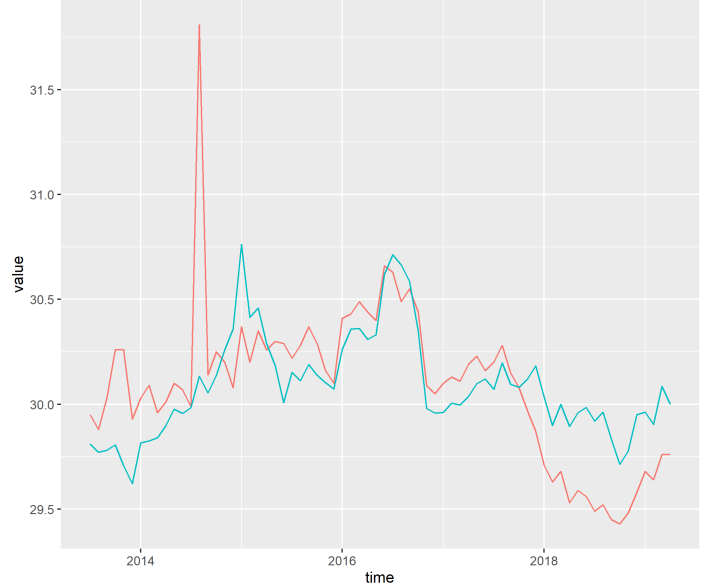
$R^2: 0.14 / -2.091 * USGG30YR_IndexMoMPercentChangeL0$



Backtesting on level values

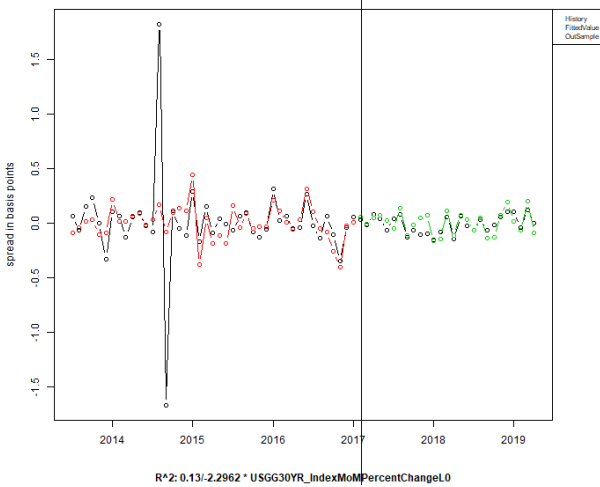
FI_treas_short - Level Value Backtesting

$R^2: 0.14 / -2.091 * USGG30YR_IndexMoMPercentChangeL0$



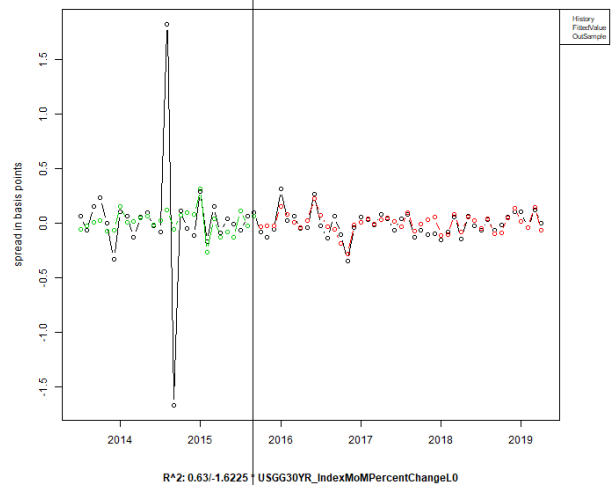
Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



4.1.6 Model - 6

Backtesting on differenced values

FI_treas_short - Backtesting

$R^2: 0.13 / -0.9943 * USGG2YR_IndexMoMDifferenceL0$

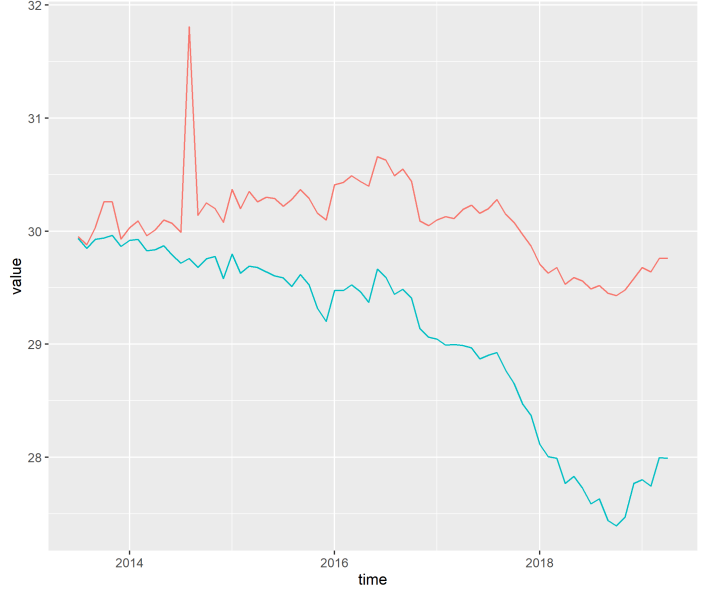


variable — FI_treas_short_History — FI_treas_short_FittedValue

Backtesting on level values

FI_treas_short - Level Value Backtesting

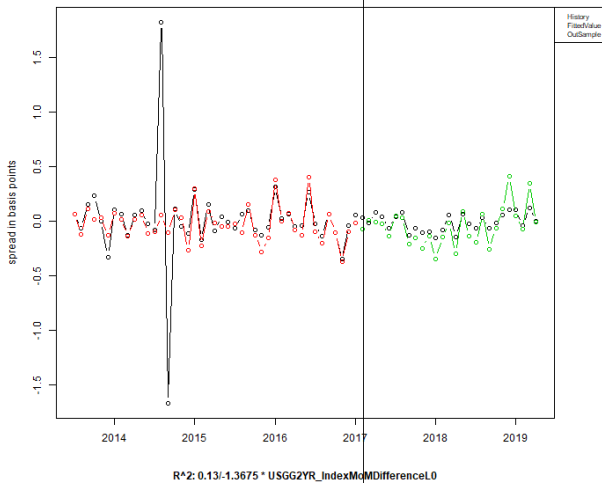
$R^2: 0.13 / -0.9943 * USGG2YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

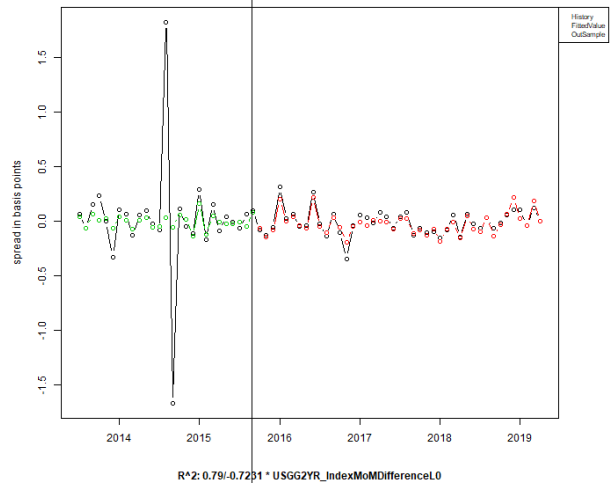
FI_treas_short Jul 2013-Apr 2019



$R^2: 0.13 / -1.3675 * USGG2YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_treas_short Jul 2013-Apr 2019



$R^2: 0.79 / -0.7231 * USGG2YR_IndexMoMDifferenceL0$

4.2 Two Factor Model for $FI_{t^{reashort}}$

4.3 Single Factor Model for $FI_{t_{reasoning}}$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG30YR_IndexMoMDifferenceL0	0.98	-83.38	-6.06	0.00	0.10			
2	USGG30YR_IndexMoMPercentChangeL0	0.98	-72.47	-17.09	0.00	0.13			
3	USGG10YR_IndexMoMPercentChangeL0	0.89	50.80	-11.53	0.00	0.34			
4	USGG10YR_IndexMoMDifferenceL0	0.89	51.58	-5.29	0.00	0.29			
5	USGG2YR_IndexMoMDifferenceL0	0.43	165.75	-5.64	0.00	0.64			
6	USGG2YR_IndexMoMPercentChangeL0	0.31	180.07	-3.78	0.00	0.70			
7	BAA10YB_IndexMoMPercentChangeL0	0.27	183.25	5.47	0.00	0.79			
8	BAA10YB_IndexMoMDifferenceL0	0.26	184.93	0.04	0.00	0.76			
9	UNRATEMoMPercentChangeL0	0.12	196.38	14.20	0.00	0.74			
10	UNRATEMoMDifferenceL0	0.10	197.90	2.53	0.01	0.75			

Table 4: Single Factor Analysis- $FI_{t_{reasoning}}$

4.3.1 Model - 1

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.98 / -6.0572 * USGG30YR_IndexMoMDifferenceL0$

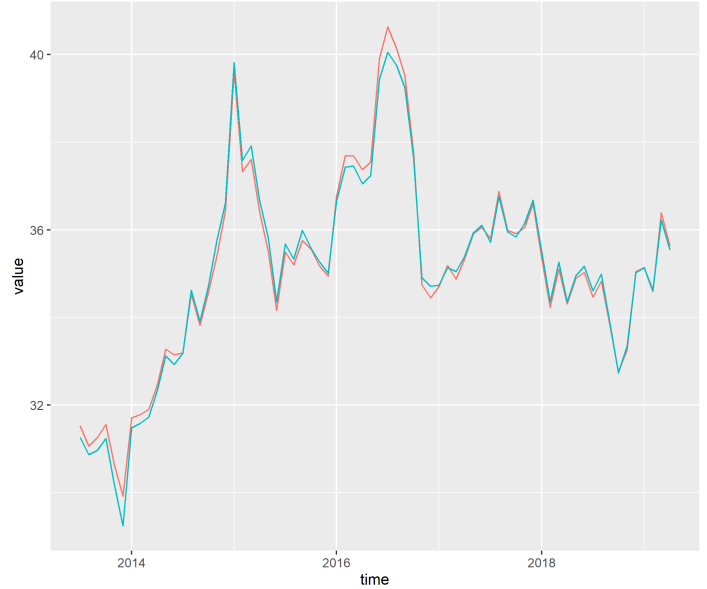


variable — FI_treas_long_History — FI_treas_long_FittedValue

Backtesting on level values

FI_treas_long - Level Value Backtesting

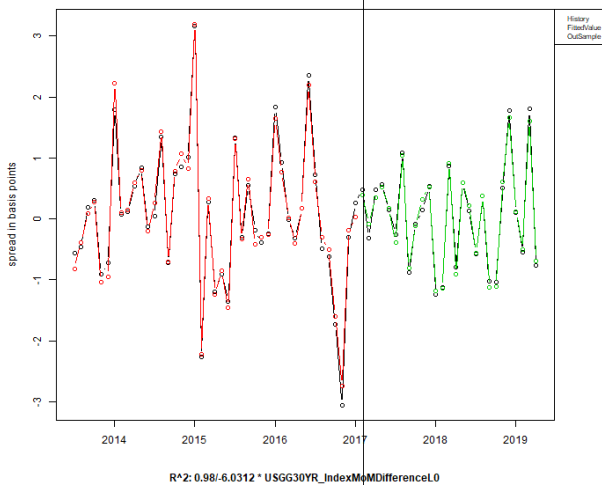
$R^2: 0.98 / -6.0572 * USGG30YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

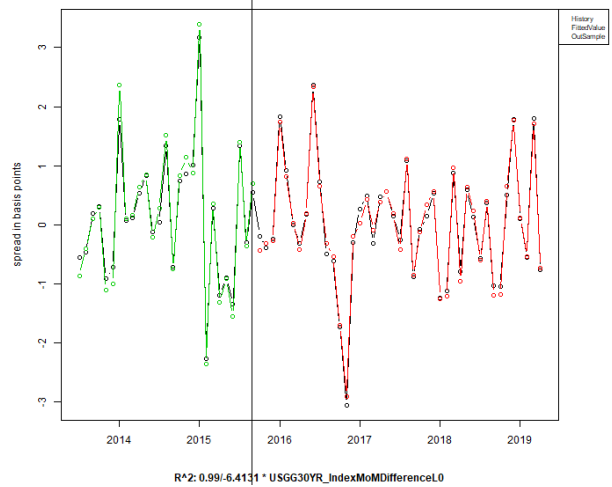
FI_treas_long Jul 2013-Apr 2019



$R^2: 0.98 / -6.0312 * USGG30YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



$R^2: 0.99 / -6.4131 * USGG30YR_IndexMoMDifferenceL0$

4.3.2 Model - 2

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.98 / -17.091 * USGG30YR_IndexMoMPercentChangeL0$

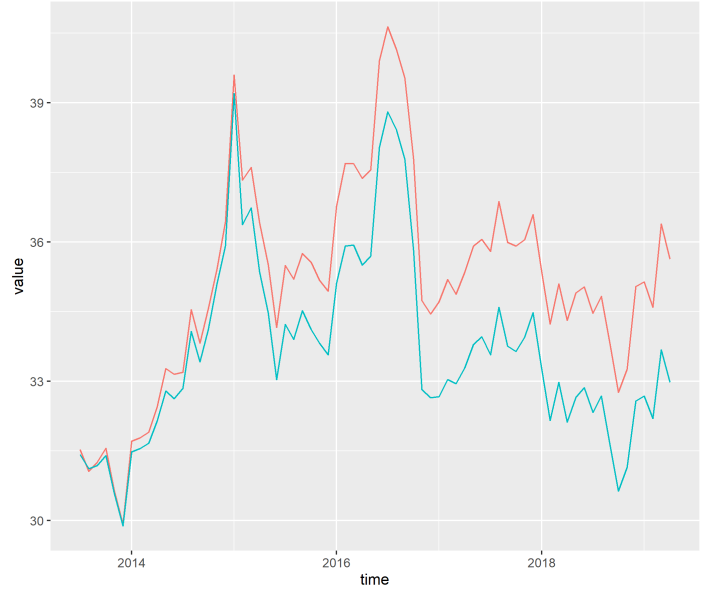


variable — FI_treas_long_History — FI_treas_long_FittedValue

Backtesting on level values

FI_treas_long - Level Value Backtesting

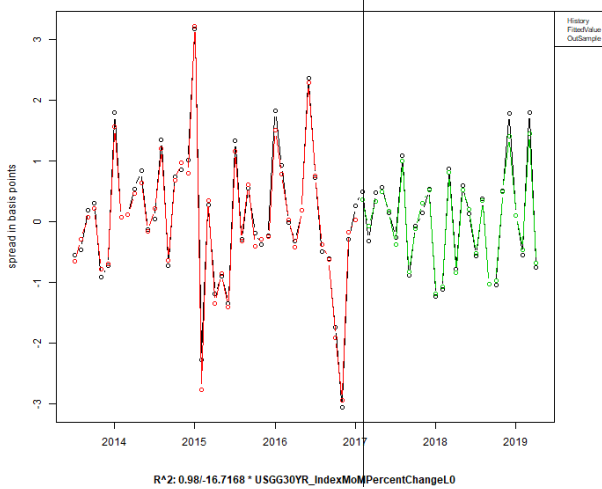
$R^2: 0.98 / -17.091 * USGG30YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

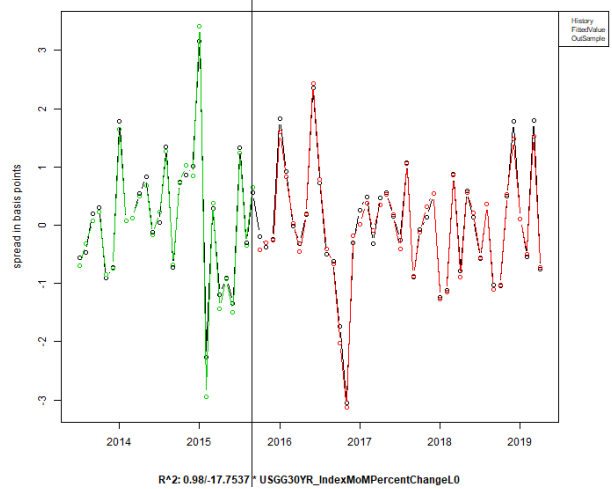
FI_treas_long Jul 2013-Apr 2019



$R^2: 0.98 / -16.7168 * USGG30YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



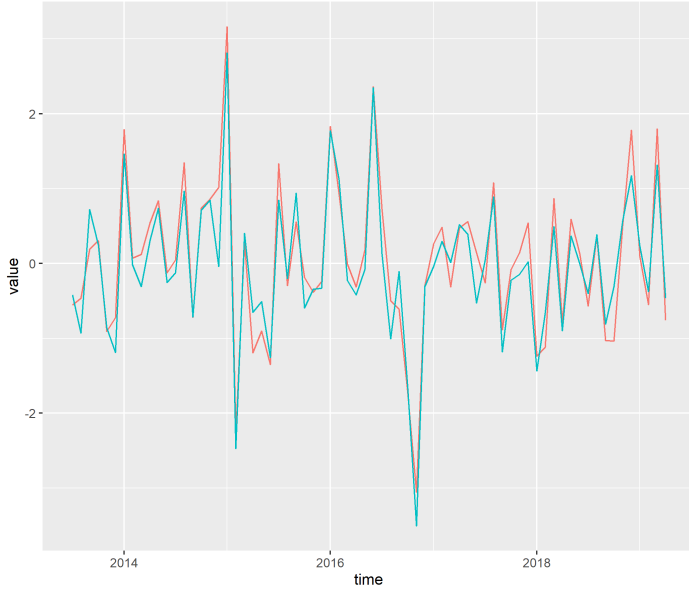
$R^2: 0.98 / -17.7537 * USGG30YR_IndexMoMPercentChangeL0$

4.3.3 Model - 3

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.89 / -11.5251 * USGG10YR_IndexMoMPercentChangeL0$

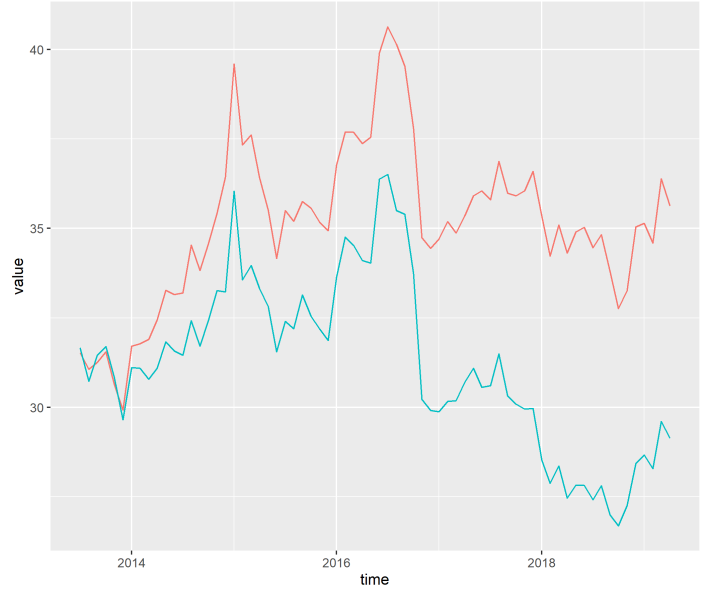


variable — FI_treas_long_History — FI_treas_long_FittedValue

Backtesting on level values

FI_treas_long - Level Value Backtesting

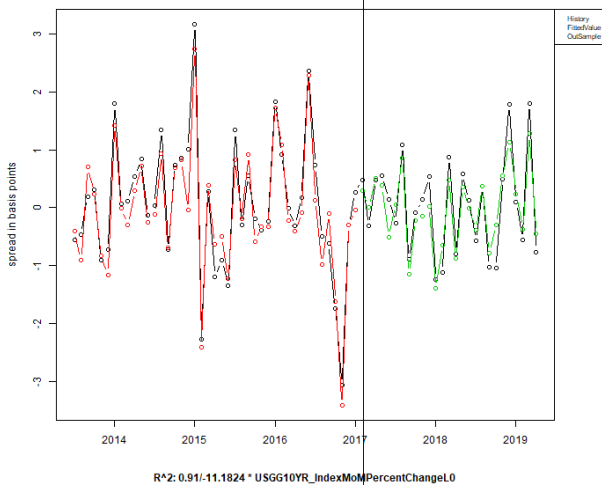
$R^2: 0.89 / -11.5251 * USGG10YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

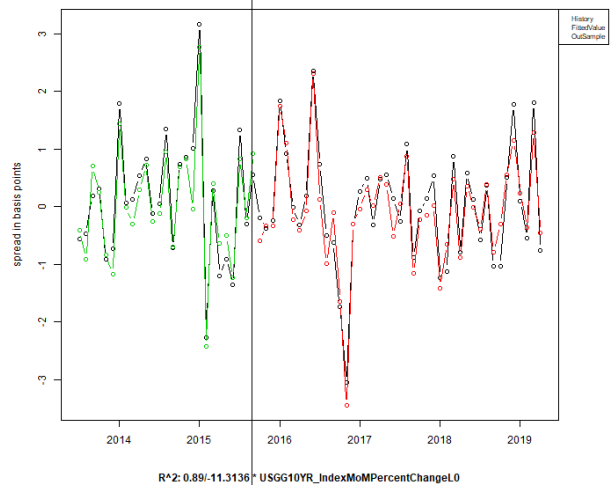
FI_treas_long Jul 2013-Apr 2019



$R^2: 0.91 / -11.1824 * USGG10YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



$R^2: 0.89 / -11.3136 * USGG10YR_IndexMoMPercentChangeL0$

4.3.4 Model - 4

Backtesting on differenced values

FI_treas_long - Backtesting

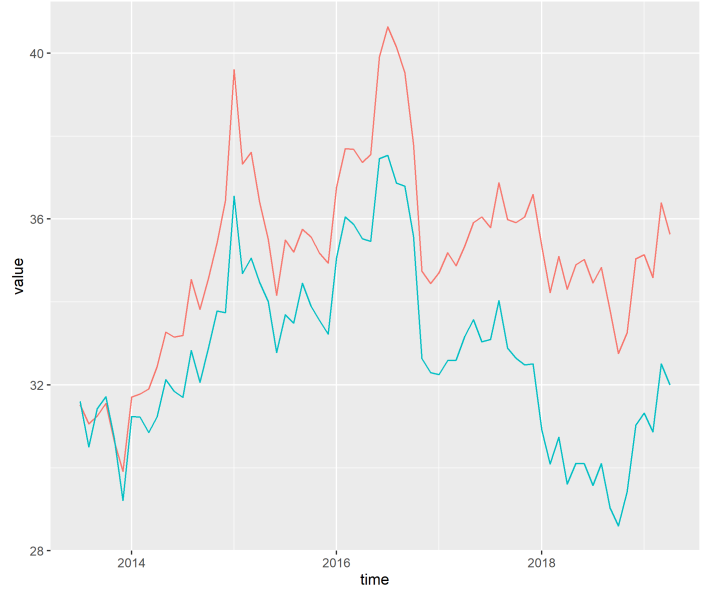
$R^2: 0.89 / -5.2878 * USGG10YR_IndexMoMDifferenceL0$



Backtesting on level values

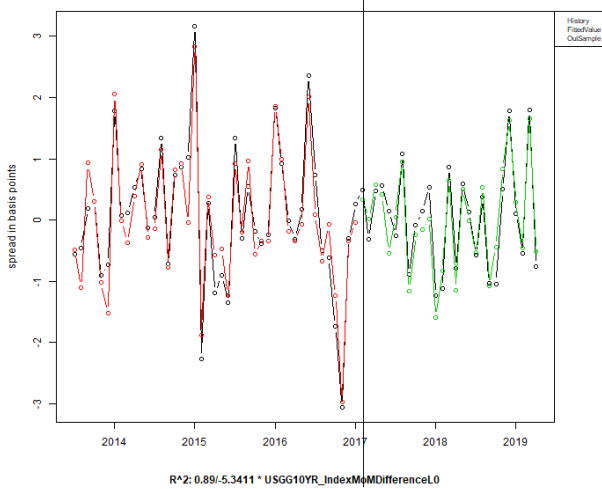
FI_treas_long - Level Value Backtesting

$R^2: 0.89 / -5.2878 * USGG10YR_IndexMoMDifferenceL0$



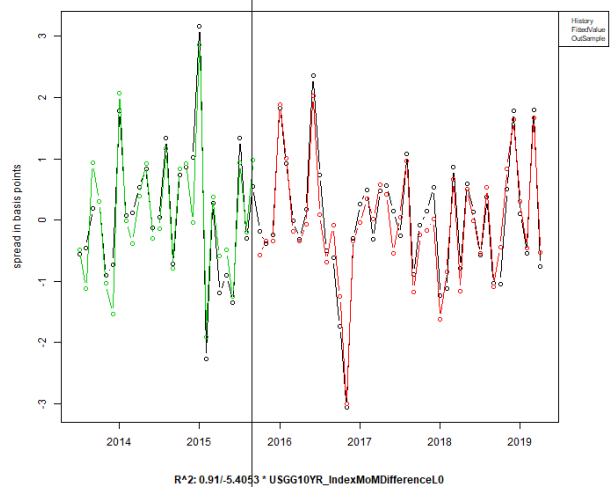
Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019

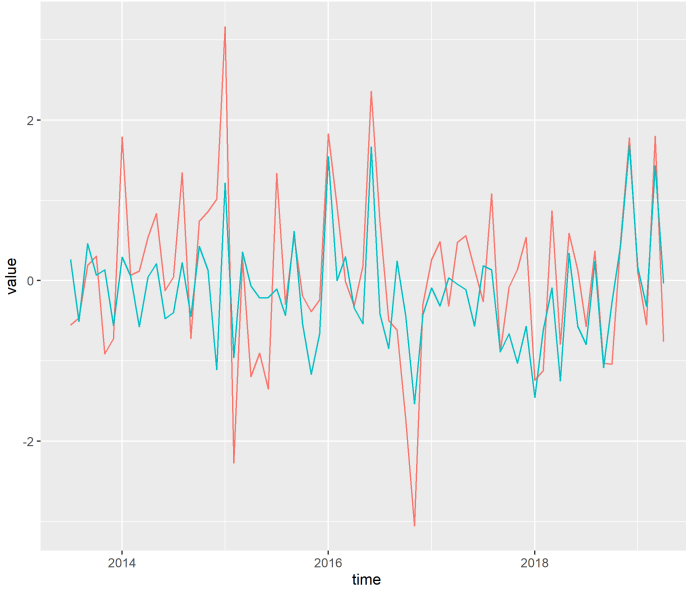


4.3.5 Model - 5

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.43 / -5.6421 * USGG2YR_IndexMoMDifferenceL0$

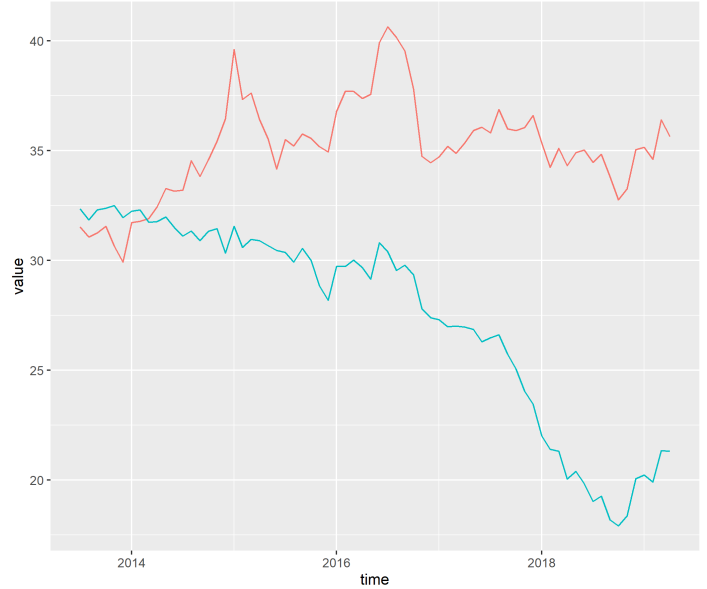


variable — FI_treas_long_History — FI_treas_long_FittedValue

Backtesting on level values

FI_treas_long - Level Value Backtesting

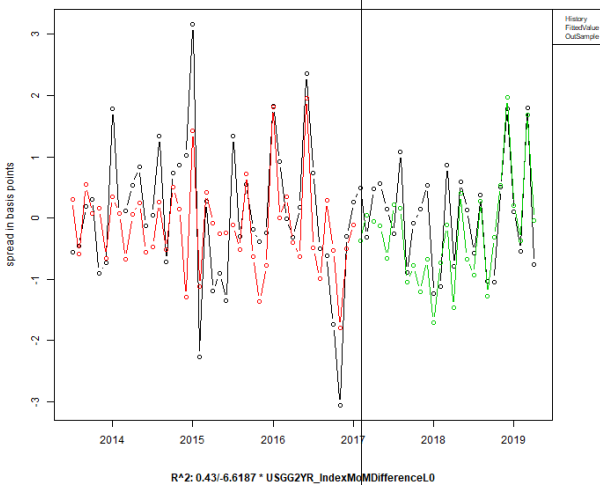
$R^2: 0.43 / -5.6421 * USGG2YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

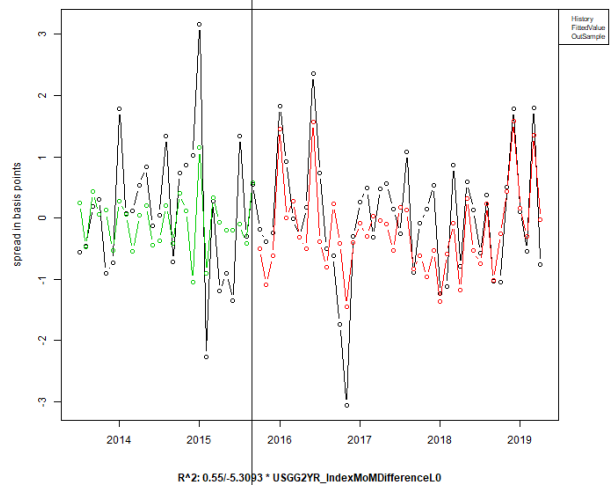
FI_treas_long Jul 2013-Apr 2019



$R^2: 0.43 / -5.6187 * USGG2YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



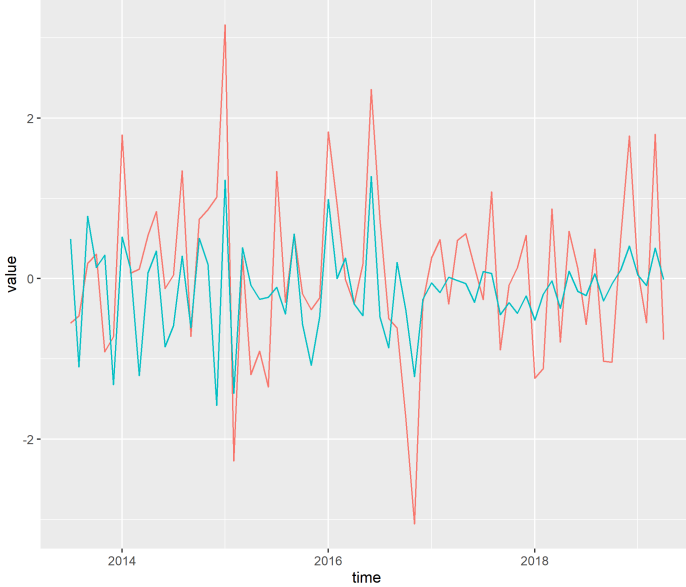
$R^2: 0.55 / -5.3093 * USGG2YR_IndexMoMDifferenceL0$

4.3.6 Model - 6

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.31 / -3.777 * USGG2YR_IndexMoMPercentChangeL0$

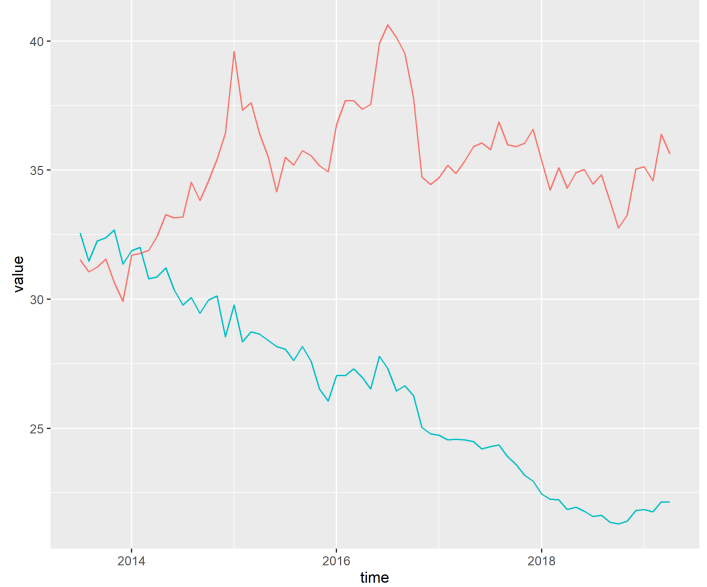


variable — FI_treas_long_History — FI_treas_long_FittedValue

Backtesting on level values

FI_treas_long - Level Value Backtesting

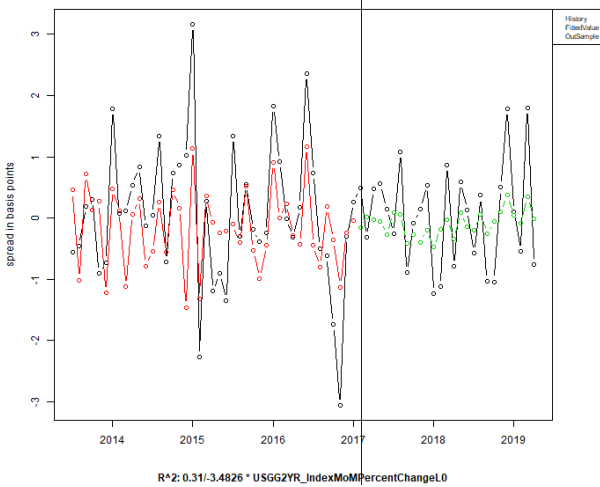
$R^2: 0.31 / -3.777 * USGG2YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

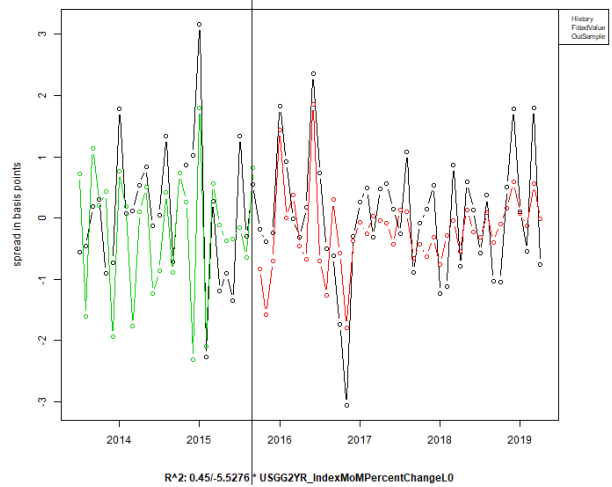
FI_treas_long Jul 2013-Apr 2019



$R^2: 0.31 / -3.4826 * USGG2YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



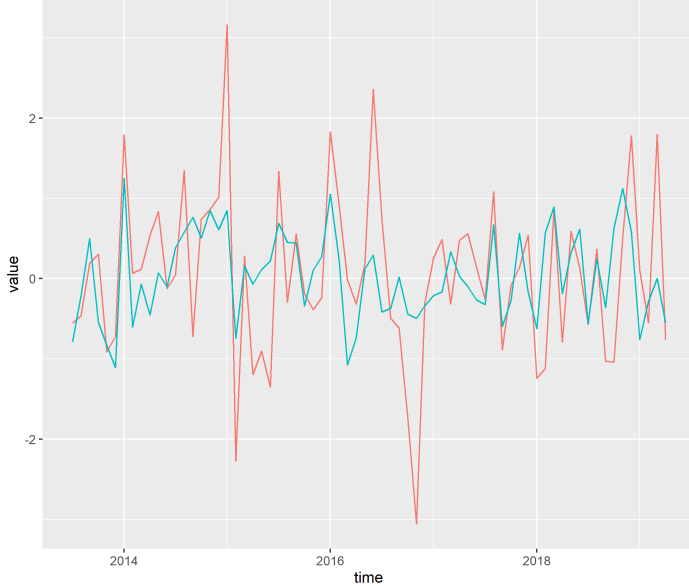
$R^2: 0.45 / -5.5276 * USGG2YR_IndexMoMPercentChangeL0$

4.3.7 Model - 7

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.27 / 5.4676 * BAA10YB_IndexMoMPercentChangeL0$

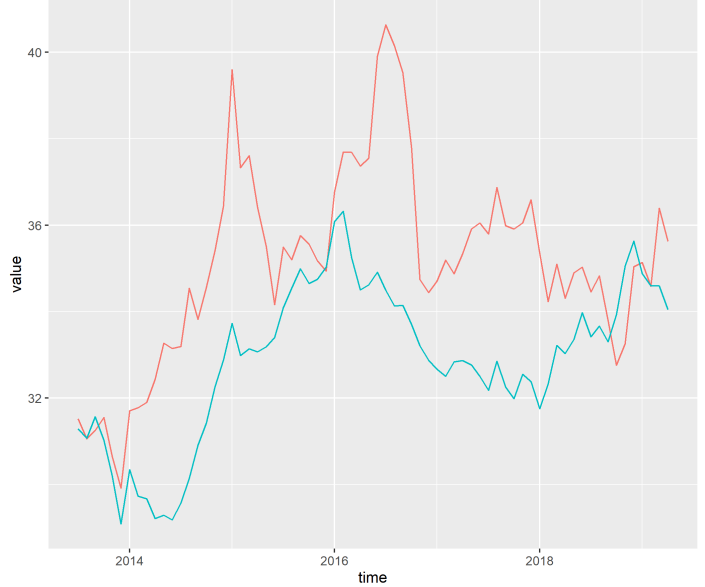


variable — FI_treas_long_History — FI_treas_long_FittedValue

Backtesting on level values

FI_treas_long - Level Value Backtesting

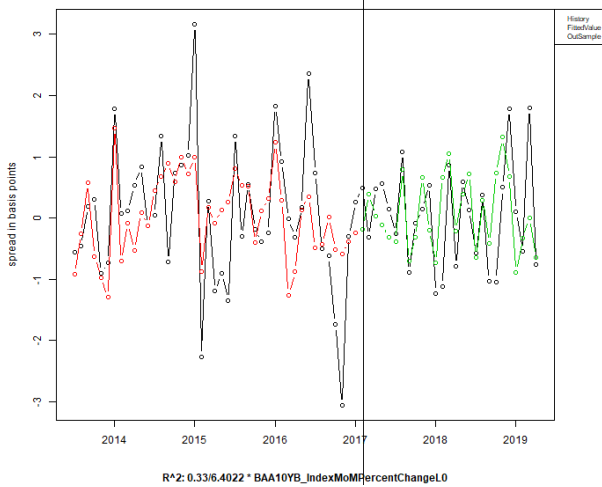
$R^2: 0.27 / 5.4676 * BAA10YB_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

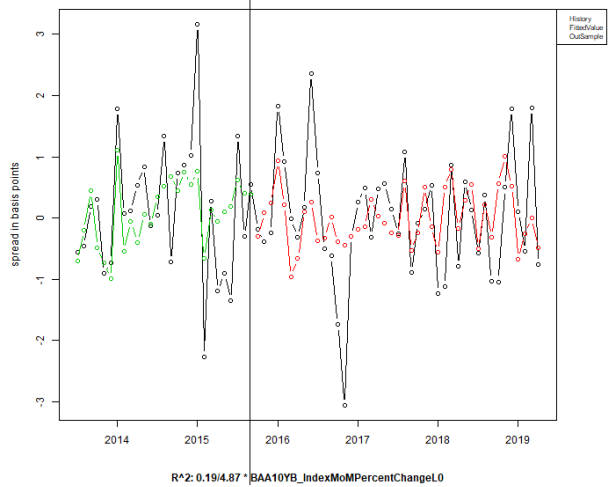
FI_treas_long Jul 2013-Apr 2019



$R^2: 0.33 / 6.4022 * BAA10YB_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



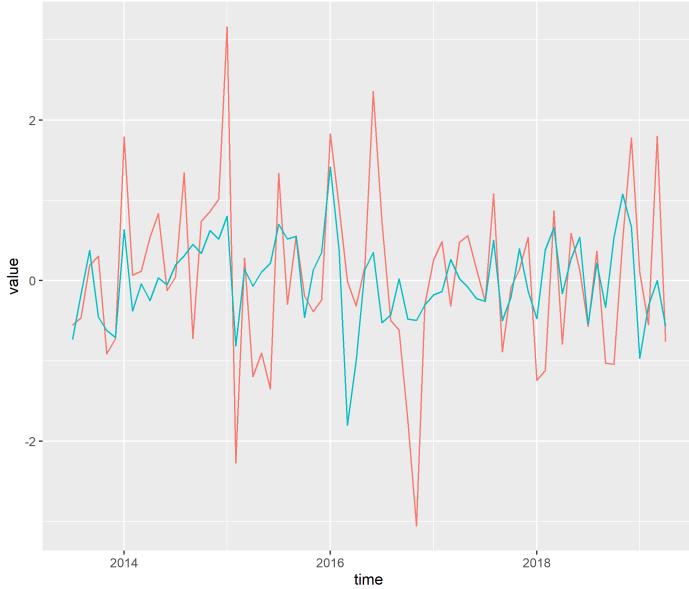
$R^2: 0.19 / 4.87 * BAA10YB_IndexMoMPercentChangeL0$

4.3.8 Model - 8

Backtesting on differenced values

FI_treas_long - Backtesting

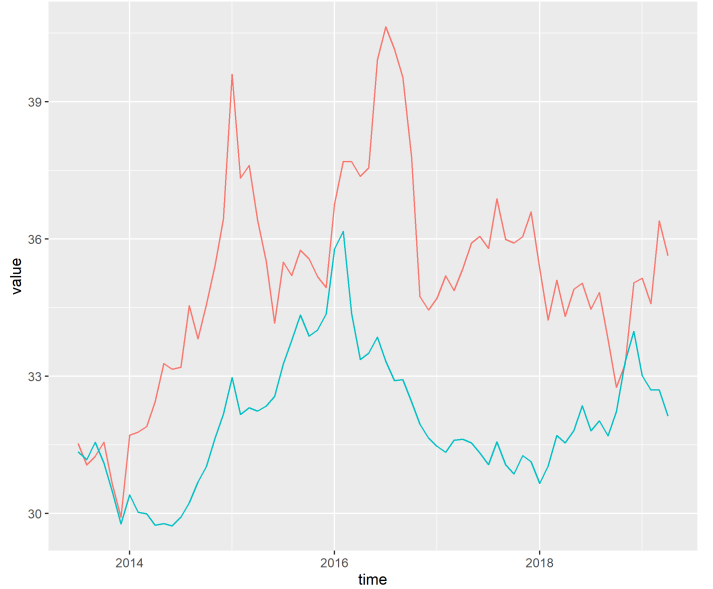
$R^2: 0.26 / 0.0365 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

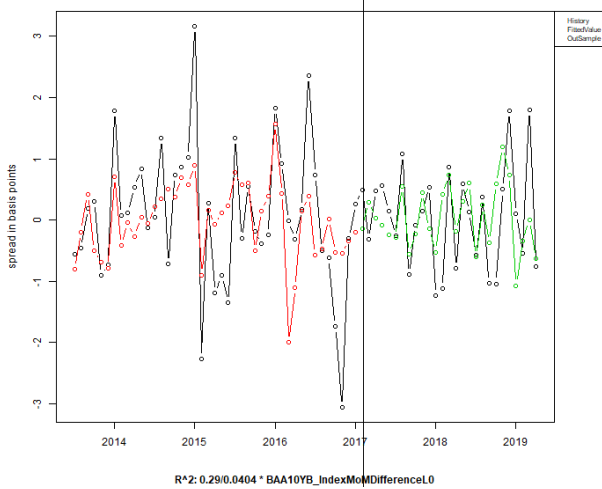
FI_treas_long - Level Value Backtesting

$R^2: 0.26 / 0.0365 * BAA10YB_IndexMoMDifferenceL0$



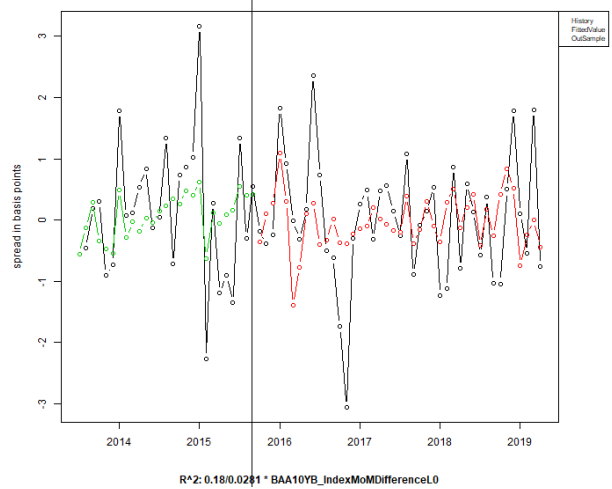
Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019

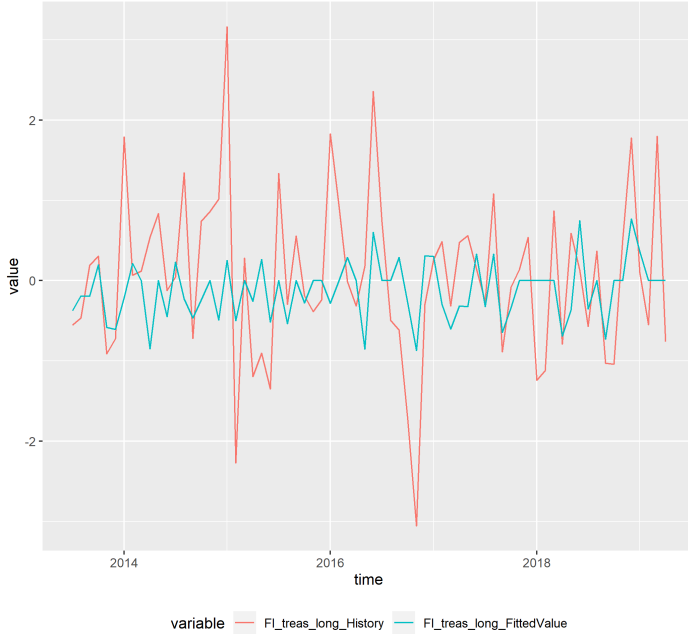


4.3.9 Model - 9

Backtesting on differenced values

FI_treas_long - Backtesting

$R^2: 0.12 / 14.1965 * \text{UNRATEMoMPercentChangeL0}$



Backtesting on level values

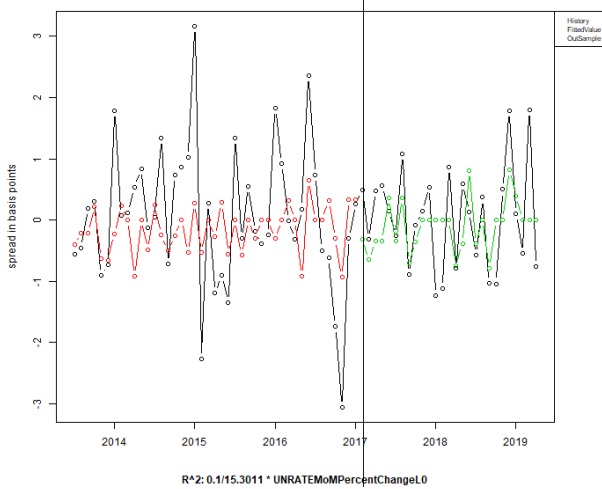
FI_treas_long - Level Value Backtesting

$R^2: 0.12 / 14.1965 * \text{UNRATEMoMPercentChangeL0}$



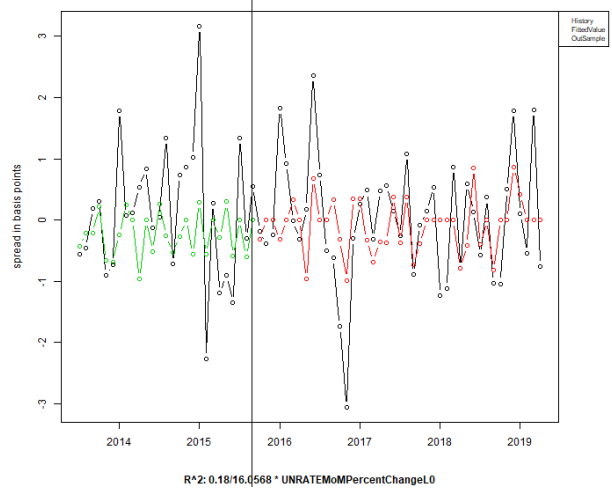
Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



Out-of-Sample testing

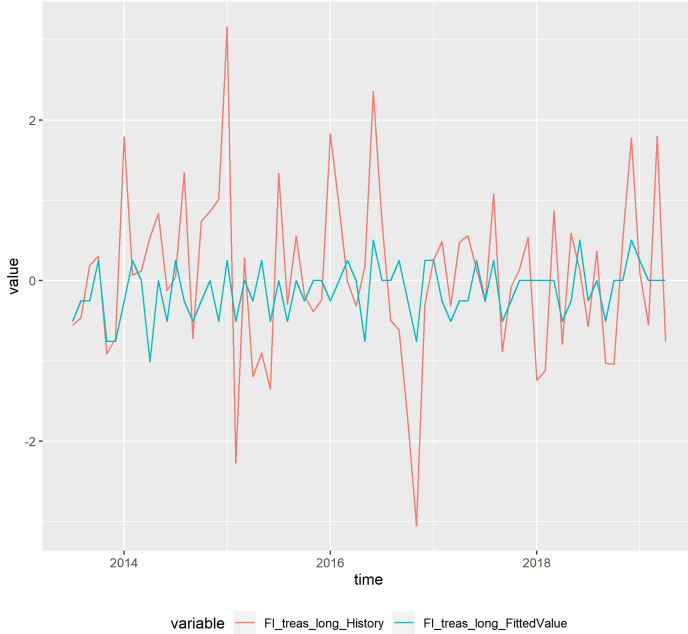
FI_treas_long Jul 2013-Apr 2019



4.3.10 Model - 10

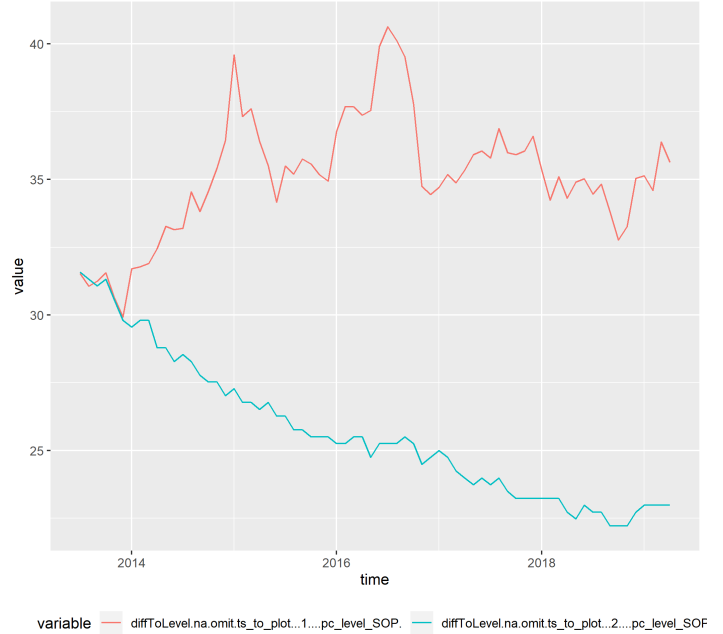
Backtesting on differenced values

FI_treas_long - Backtesting
R²: 0.1/ 2.5283 * UNRATEMoMDifferenceL0



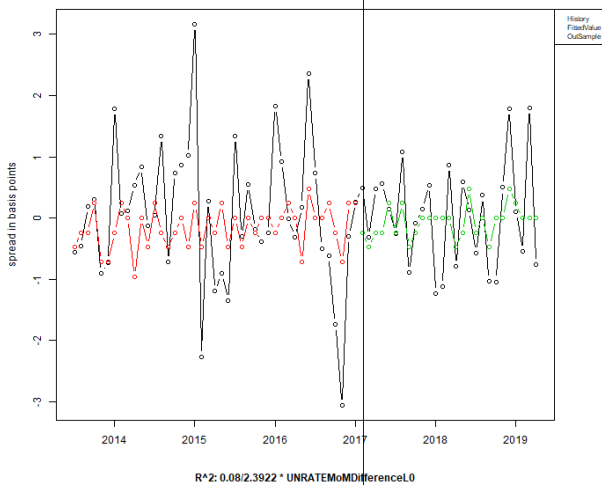
Backtesting on level values

FI_treas_long - Level Value Backtesting
R²: 0.1/ 2.5283 * UNRATEMoMDifferenceL0



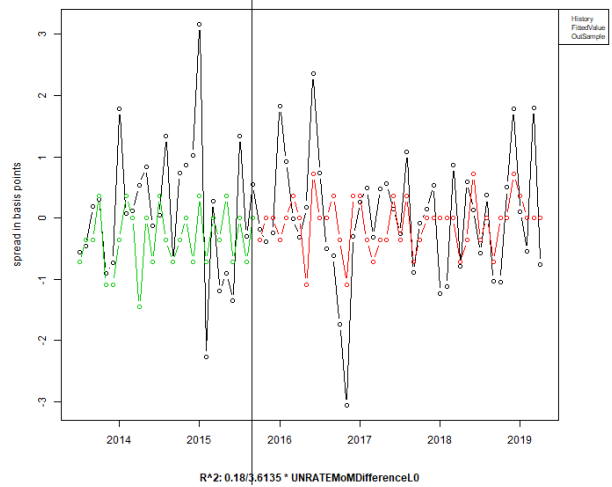
Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



Out-of-Sample testing

FI_treas_long Jul 2013-Apr 2019



4.4 Two Factor Model for $FI_{t^{reasoning}}$

4.5 Single Factor Model for $FI_I G_{short}$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG2YR_IndexMoMDifferenceL0	0.37	-176.07	-0.43	0.00	0.07			
2	USGG2YR_IndexMoMPercentChangeL0	0.36	-175.17	-0.33	0.00	0.07			
3	USGG10YR_IndexMoMDifferenceL0	0.29	-168.53	-0.25	0.00	0.07			
4	USGG10YR_IndexMoMPercentChangeL0	0.27	-166.22	-0.53	0.00	0.07			
5	VIX_IndexMoMDifferenceL0	0.21	-160.49	-0.01	0.00	0.07			
6	VIX_IndexMoMPercentChangeL0	0.19	-159.32	-0.14	0.00	0.07			
7	USGG30YR_IndexMoMDifferenceL0	0.18	-157.62	-0.22	0.00	0.08			
8	USGG30YR_IndexMoMPercentChangeL0	0.17	-157.09	-0.60	0.00	0.08			
9	SPX_Index_30DRSIMoMPercentChangeL0	0.16	-156.18	0.24	0.00	0.07			
10	SPX_Index_30DRSIMoMDifferenceL0	0.15	-155.18	0.00	0.00	0.08			
11	SPX_Index_14DRSIMoMPercentChangeL0	0.12	-153.12	0.11	0.01	0.08			
12	SPX_Index_14DRSIMoMDifferenceL0	0.12	-152.94	0.00	0.00	0.08			

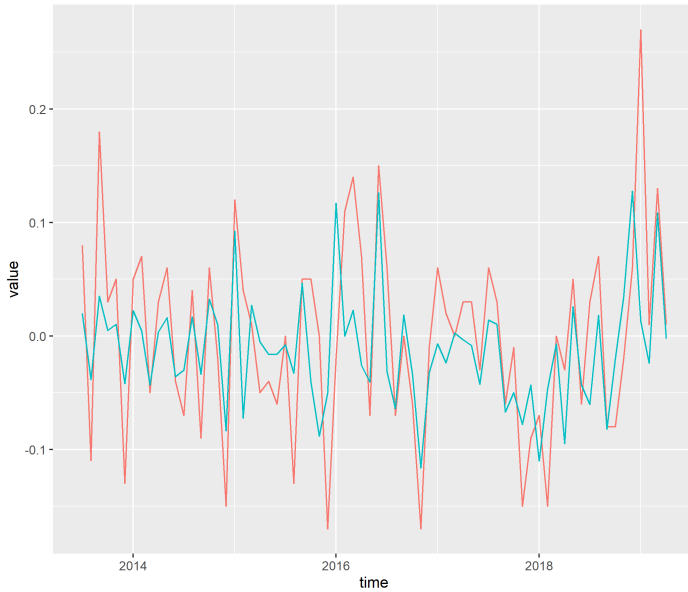
Table 5: Single Factor Analysis- $FI_I G_{short}$

4.5.1 Model - 1

Backtesting on differenced values

FI_IG_short - Backtesting

$R^2: 0.37 / -0.4272 * USGG2YR_IndexMoMDifferenceL0$

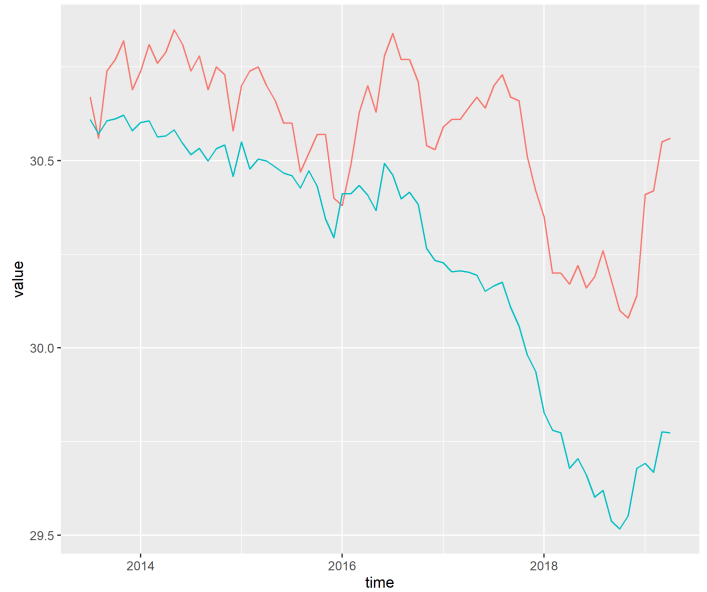


variable — FI_IG_short_History — FI_IG_short_FittedValue

Backtesting on level values

FI_IG_short - Level Value Backtesting

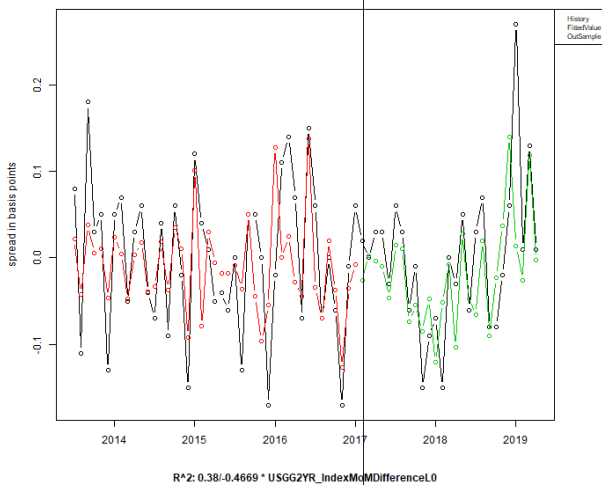
$R^2: 0.37 / -0.4272 * USGG2YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

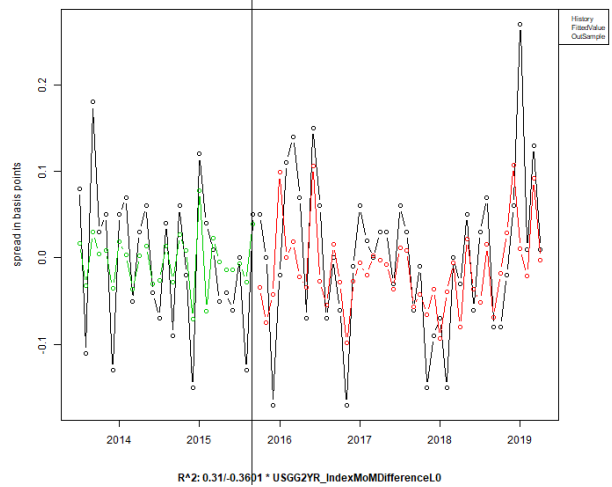
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

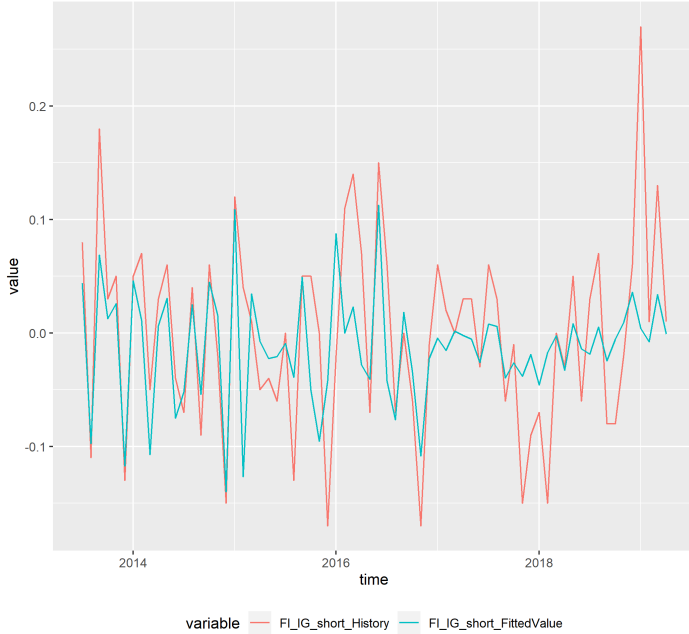


4.5.2 Model - 2

Backtesting on differenced values

FI_IG_short - Backtesting

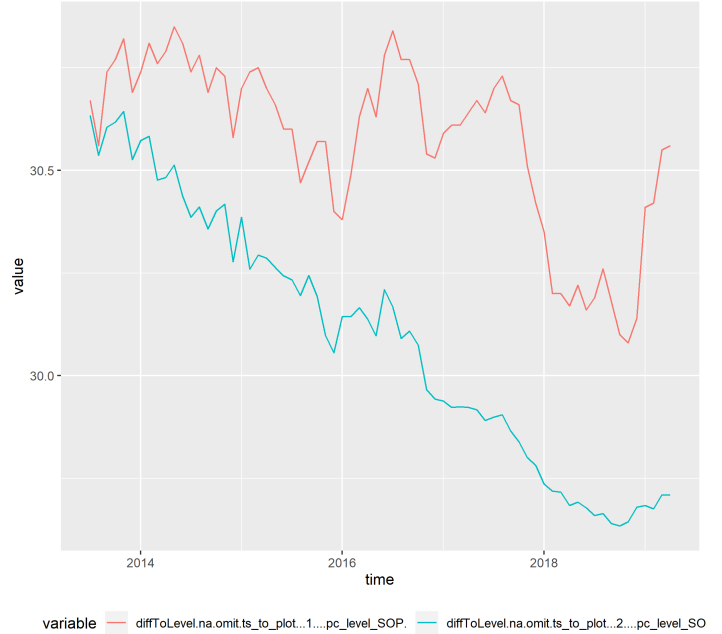
$R^2: 0.36 / -0.3349 * USGG2YR_IndexMoMPercentChangeL0$



Backtesting on level values

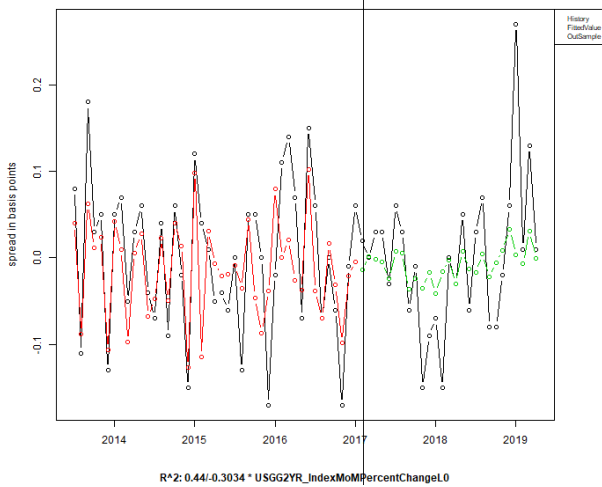
FI_IG_short - Level Value Backtesting

$R^2: 0.36 / -0.3349 * USGG2YR_IndexMoMPercentChangeL0$



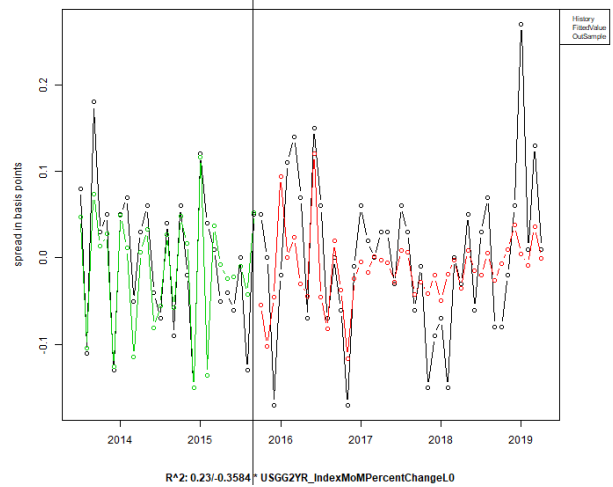
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

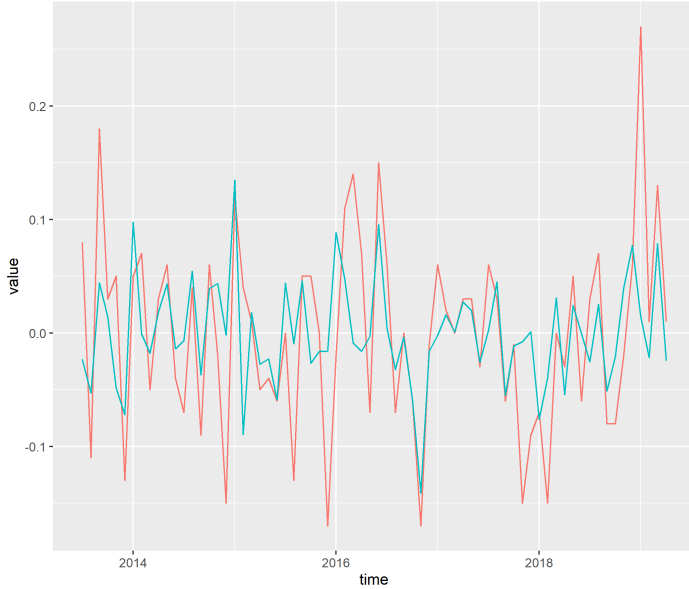


4.5.3 Model - 3

Backtesting on differenced values

FI_IG_short - Backtesting

$R^2: 0.29 / -0.2541 * USGG10YR_IndexMoMDifferenceL0$

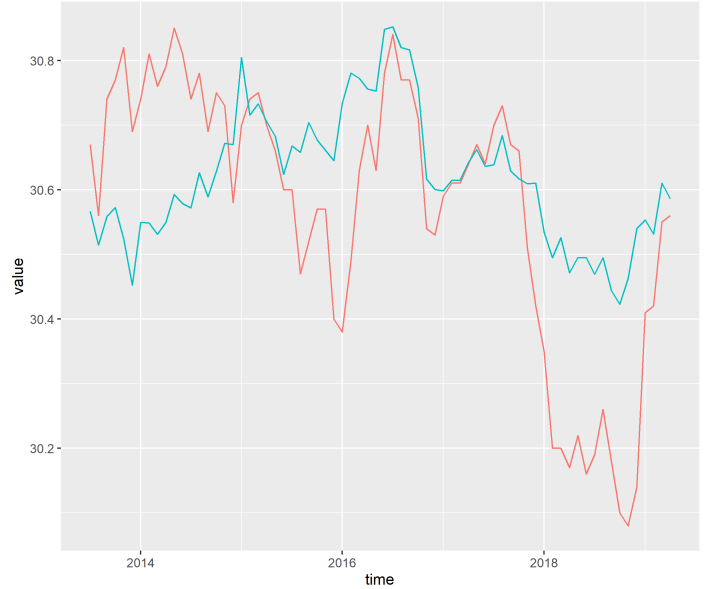


variable — FI_IG_short_History — FI_IG_short_FittedValue

Backtesting on level values

FI_IG_short - Level Value Backtesting

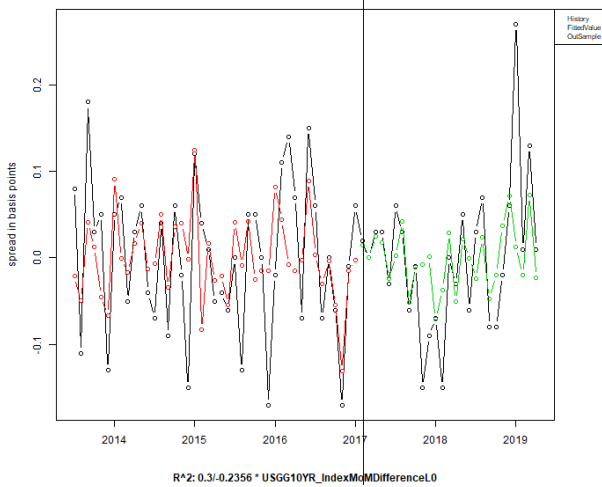
$R^2: 0.29 / -0.2541 * USGG10YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_1....pc_level_SOP — diffToLevel.na.omit.ts_2....pc_level_SO

Out-of-Sample testing

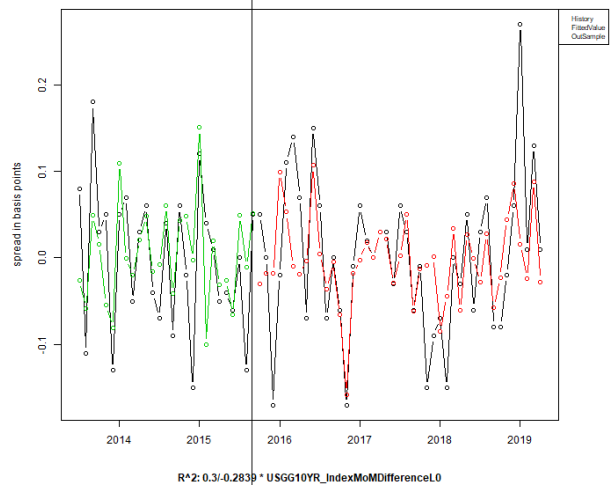
FI_IG_short Jul 2013-Apr 2019



$R^2: 0.3 / -0.2356 * USGG10YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



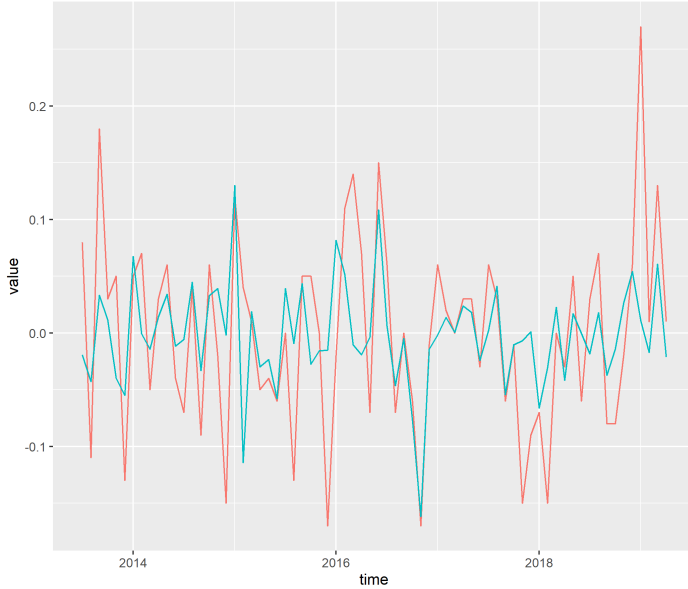
$R^2: 0.3 / -0.2839 * USGG10YR_IndexMoMDifferenceL0$

4.5.4 Model - 4

Backtesting on differenced values

FI_IG_short - Backtesting

$R^2: 0.27 / -0.5317 * USGG10YR_IndexMoMPercentChangeL0$

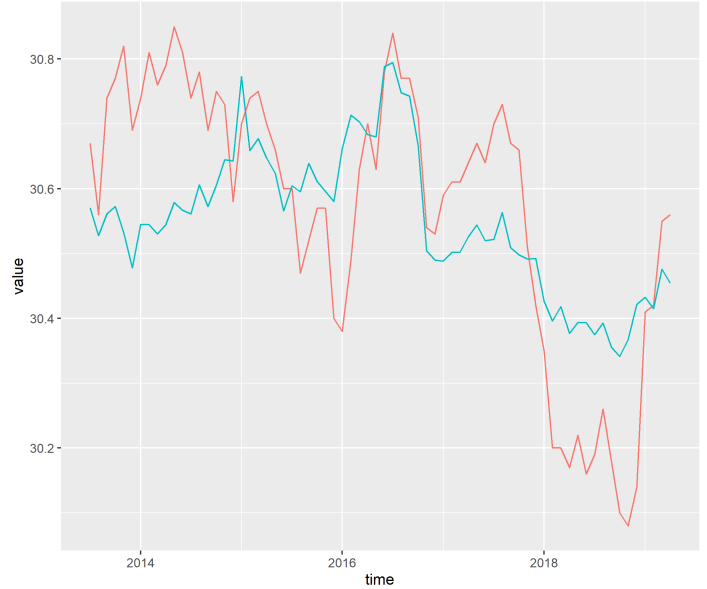


variable — FI_IG_short_History — FI_IG_short_FittedValue

Backtesting on level values

FI_IG_short - Level Value Backtesting

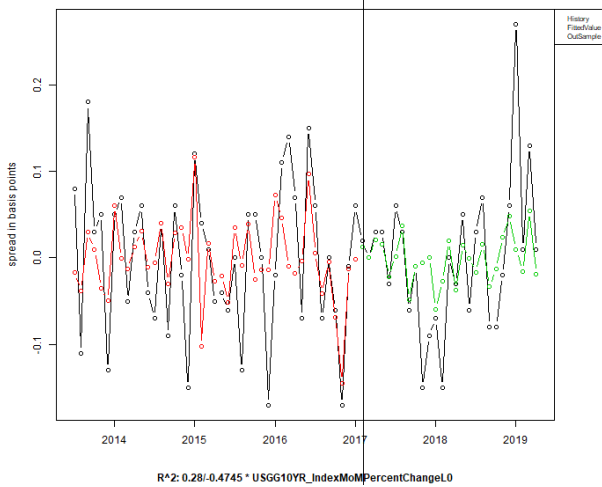
$R^2: 0.27 / -0.5317 * USGG10YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

Out-of-Sample testing

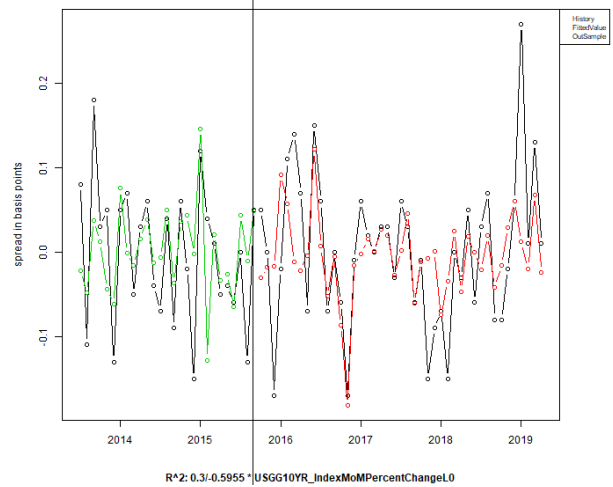
FI_IG_short Jul 2013-Apr 2019



$R^2: 0.28 / -0.4745 * USGG10YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

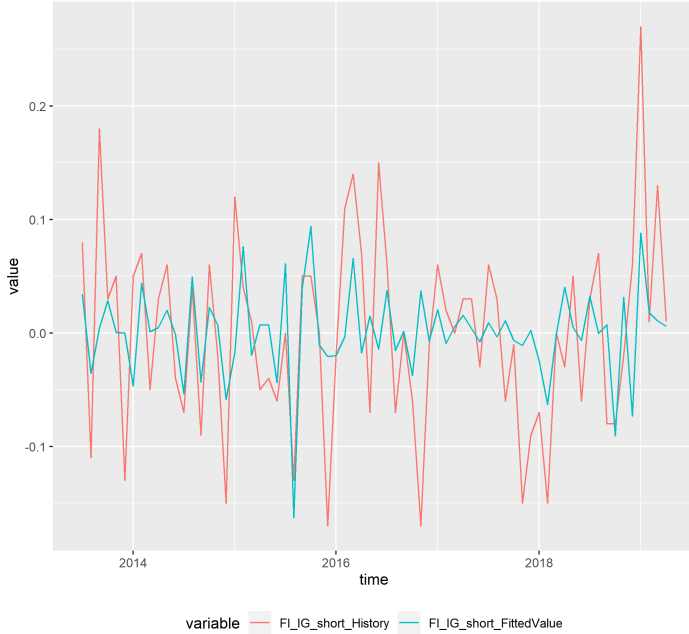


$R^2: 0.3 / -0.5955 * USGG10YR_IndexMoMPercentChangeL0$

4.5.5 Model - 5

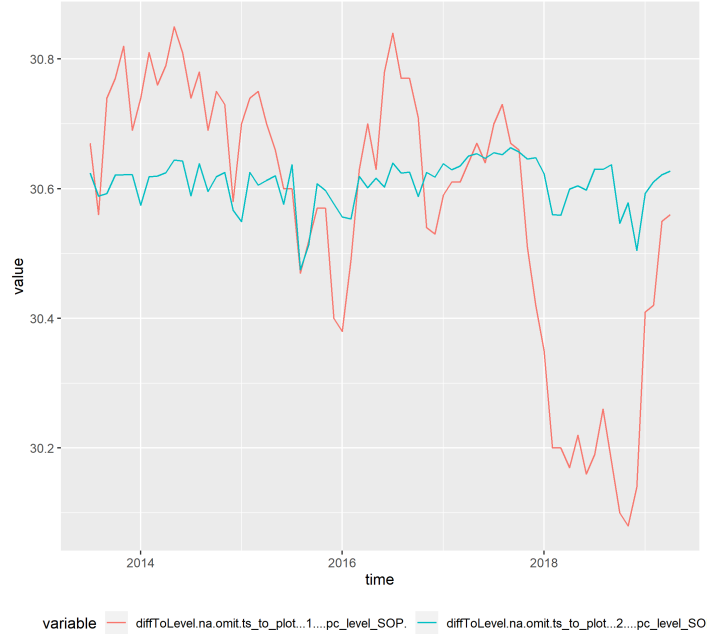
Backtesting on differenced values

FI_IG_short - Backtesting
R²: 0.21/ -0.01 * VIX_IndexMoMDifferenceL0



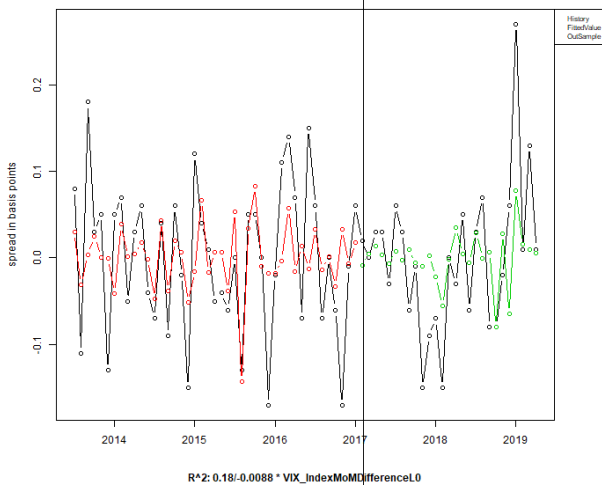
Backtesting on level values

FI_IG_short - Level Value Backtesting
R²: 0.21/ -0.01 * VIX_IndexMoMDifferenceL0



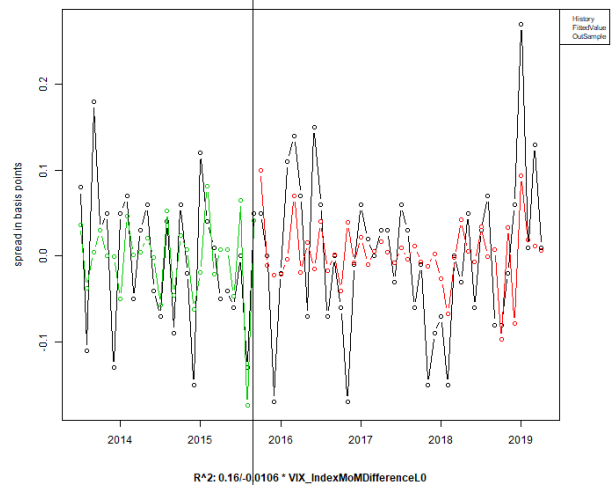
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

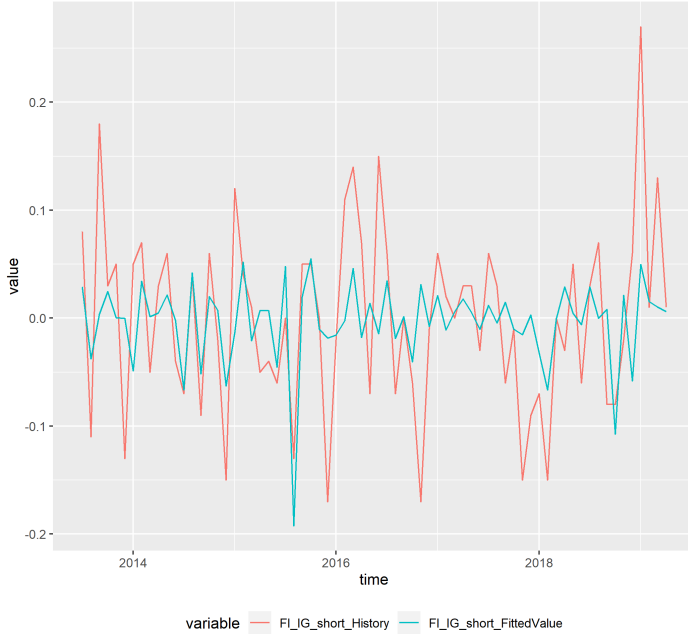


4.5.6 Model - 6

Backtesting on differenced values

FI_IG_short - Backtesting

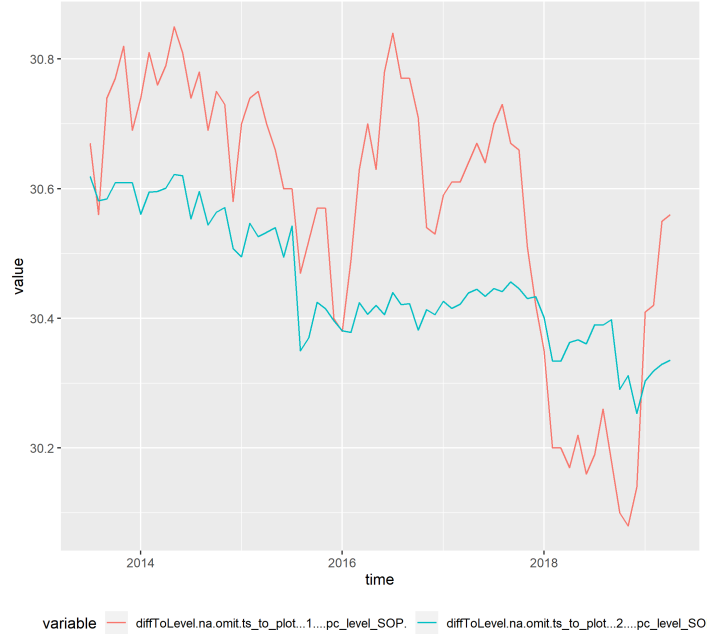
$R^2: 0.19 / -0.1429 * VIX_IndexMoMPercentChangeL0$



Backtesting on level values

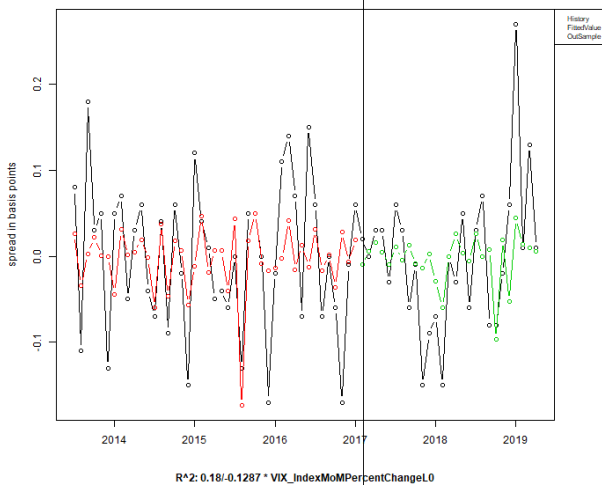
FI_IG_short - Level Value Backtesting

$R^2: 0.19 / -0.1429 * VIX_IndexMoMPercentChangeL0$



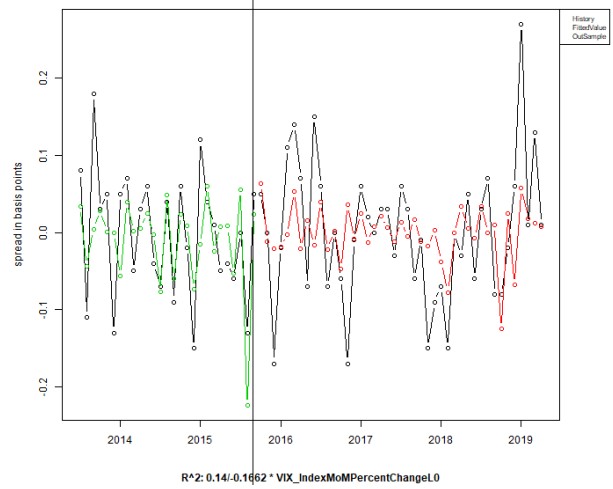
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



Out-of-Sample testing

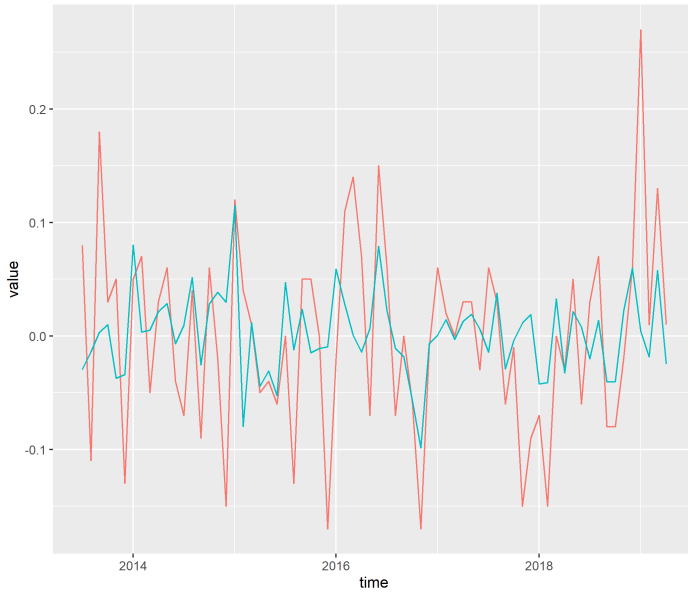
FI_IG_short Jul 2013-Apr 2019



4.5.7 Model - 7

Backtesting on differenced values

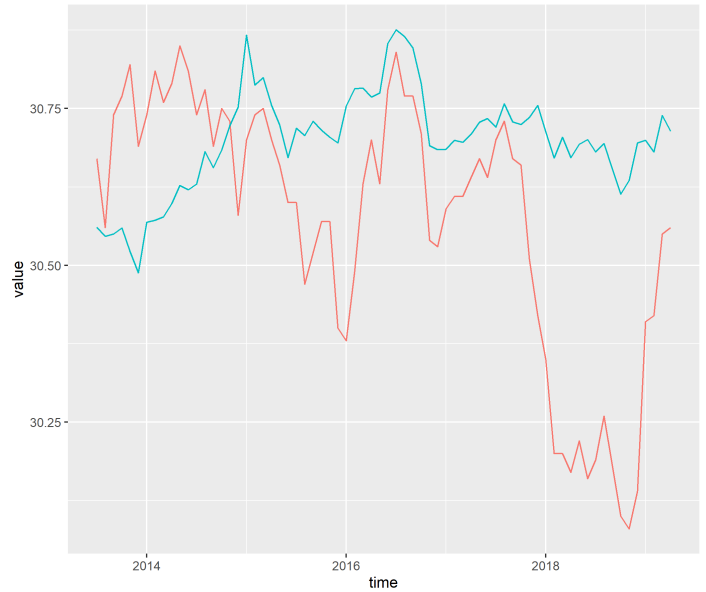
FI_IG_short - Backtesting

 $R^2: 0.18 / -0.2171 * USGG30YR_IndexMoMDifferenceL0$ 

variable — FI_IG_short_History — FI_IG_short_FittedValue

Backtesting on level values

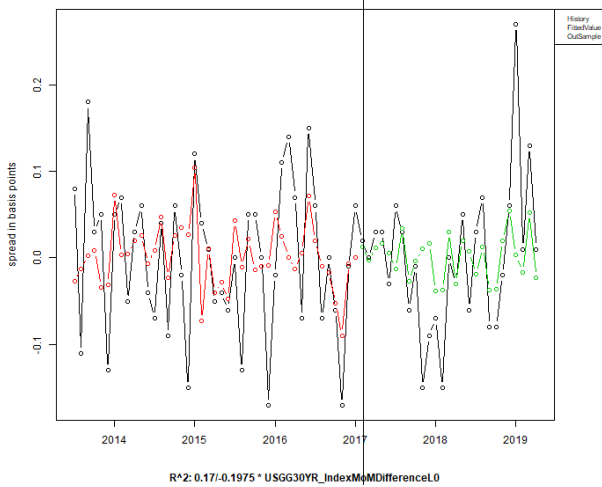
FI_IG_short - Level Value Backtesting

 $R^2: 0.18 / -0.2171 * USGG30YR_IndexMoMDifferenceL0$ 

variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SC

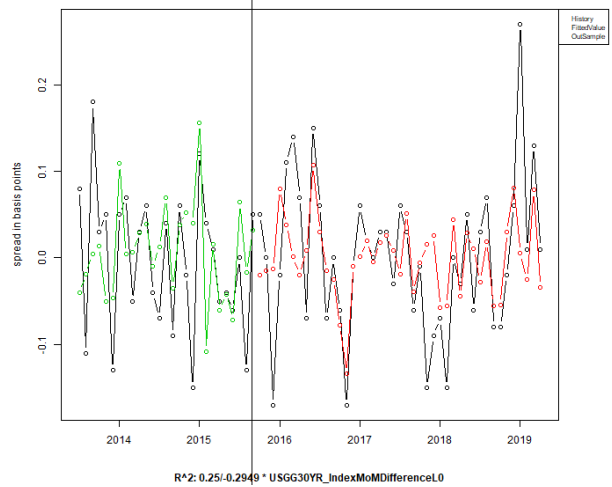
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

 $R^2: 0.17 / -0.1975 * USGG30YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

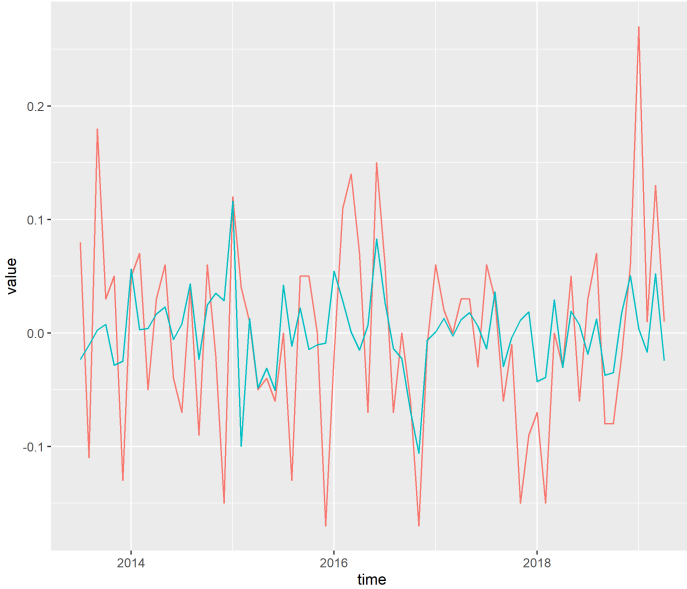
 $R^2: 0.25 / -0.2949 * USGG30YR_IndexMoMDifferenceL0$

4.5.8 Model - 8

Backtesting on differenced values

FI_IG_short - Backtesting

$R^2: 0.17 / -0.6029 \cdot \text{USGG30YR_IndexMoMPercentChangeL0}$

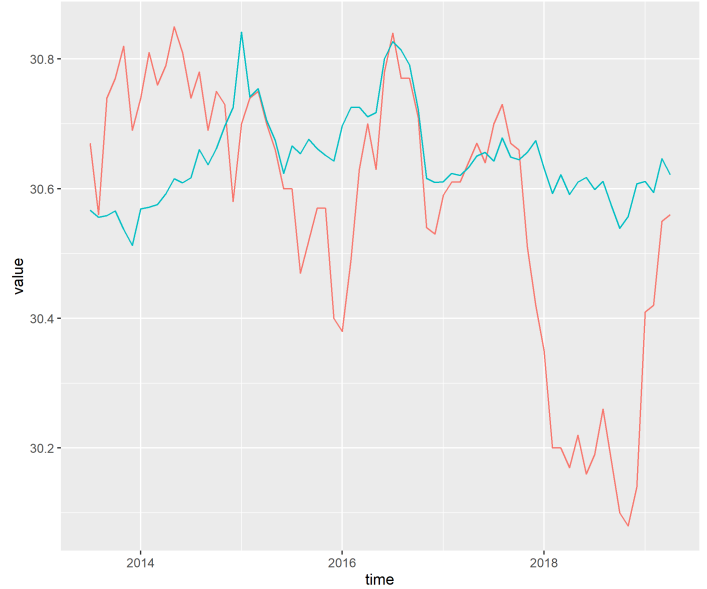


variable — FI_IG_short_History — FI_IG_short_FittedValue

Backtesting on level values

FI_IG_short - Level Value Backtesting

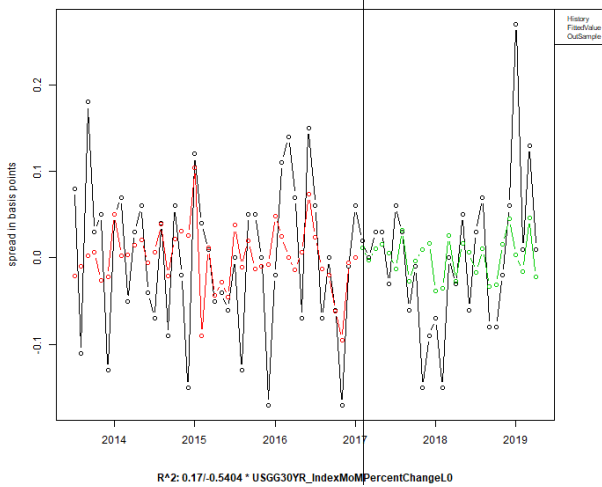
$R^2: 0.17 / -0.6029 \cdot \text{USGG30YR_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

Out-of-Sample testing

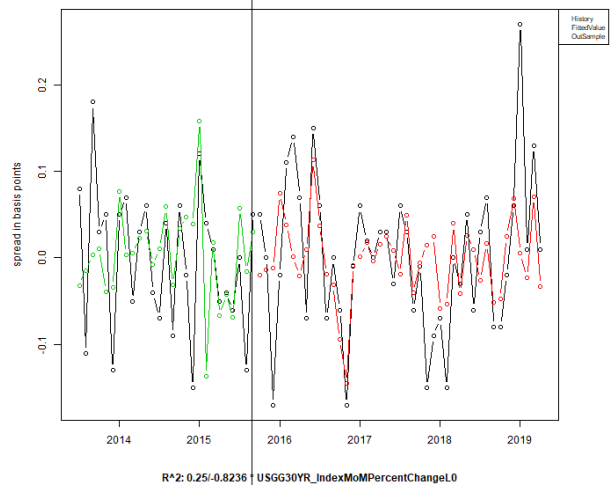
FI_IG_short Jul 2013-Apr 2019



$R^2: 0.17 / -0.5404 \cdot \text{USGG30YR_IndexMoMPercentChangeL0}$

Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



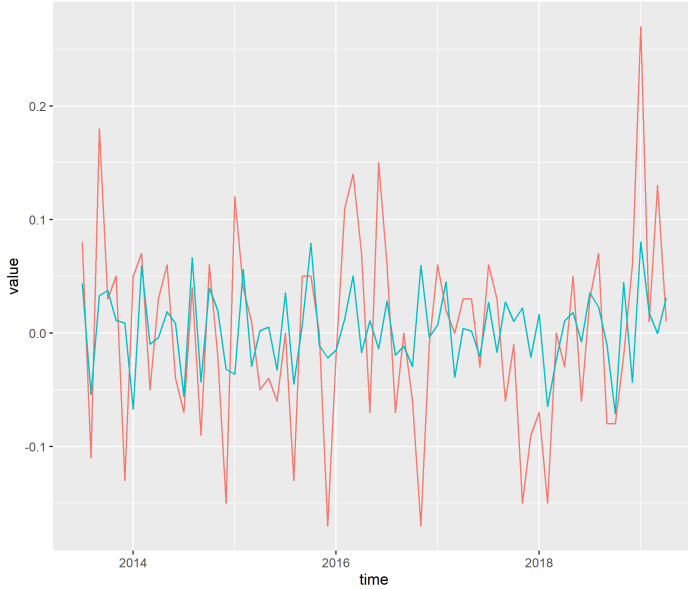
$R^2: 0.25 / -0.8236 \cdot \text{USGG30YR_IndexMoMPercentChangeL0}$

4.5.9 Model - 9

Backtesting on differenced values

FI_IG_short - Backtesting

$R^2: 0.16 / 0.2436 * SPX_Index_30DRSIMoMPercentChangeL0$

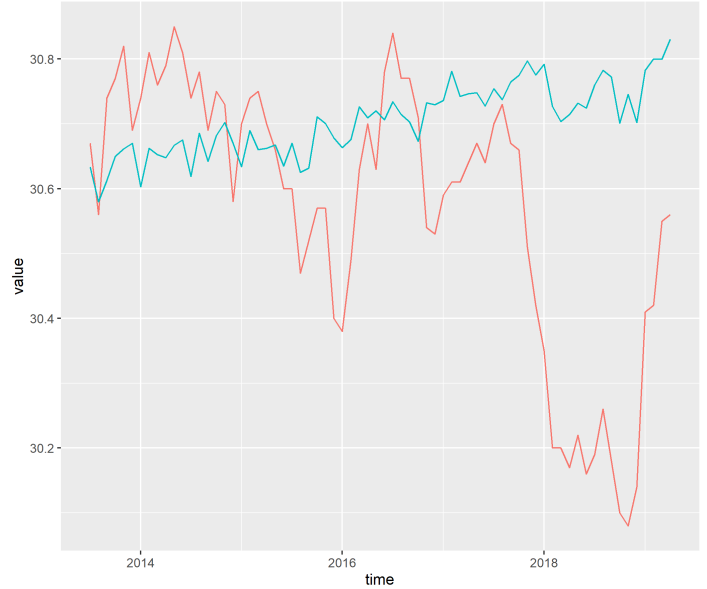


variable FI_IG_short_History FI_IG_short_FittedValue

Backtesting on level values

FI_IG_short - Level Value Backtesting

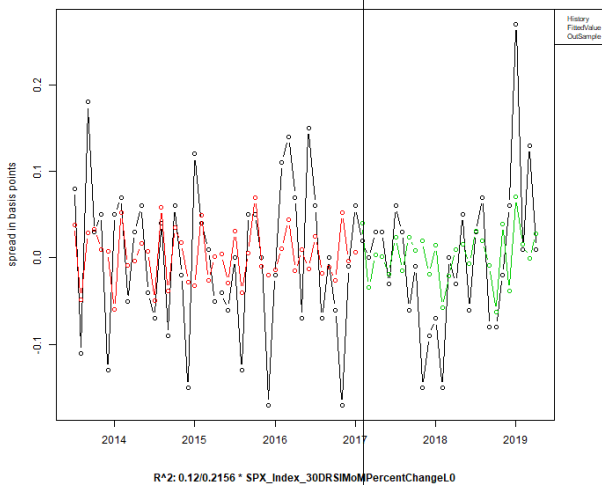
$R^2: 0.16 / 0.2436 * SPX_Index_30DRSIMoMPercentChangeL0$



variable diffToLevel.na.omit.ts_1....pc_level_SOP diffToLevel.na.omit.ts_2....pc_level_SO

Out-of-Sample testing

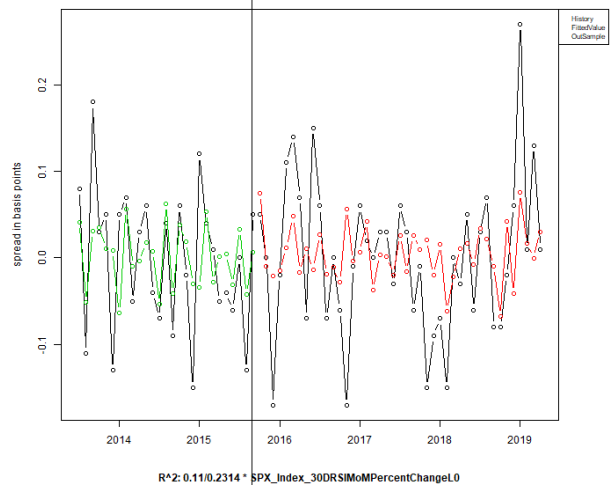
FI_IG_short Jul 2013-Apr 2019



$R^2: 0.12 / 0.2156 * SPX_Index_30DRSIMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



$R^2: 0.11 / 0.2314 * SPX_Index_30DRSIMoMPercentChangeL0$

4.5.10 Model - 10

Backtesting on differenced values

FI_IG_short - Backtesting

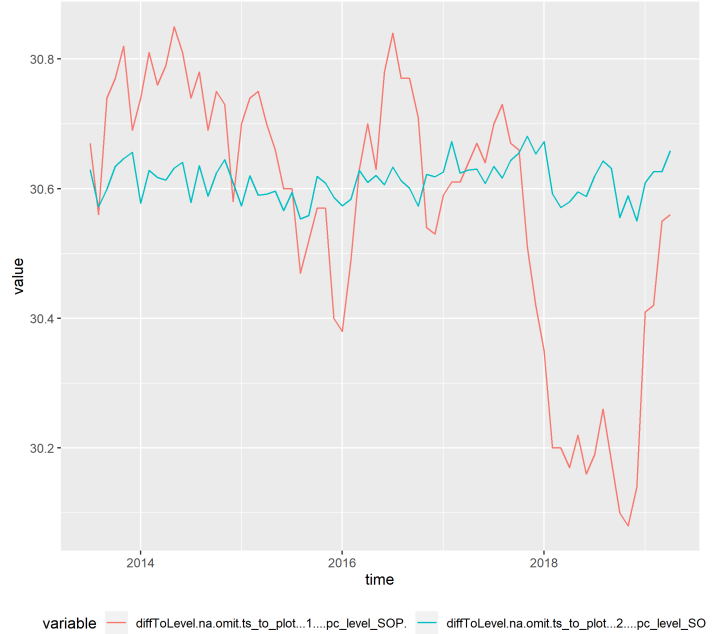
R^2 : 0.15/ 0.0044 * SPX_Index_30DRSIMoMDifferenceL0



Backtesting on level values

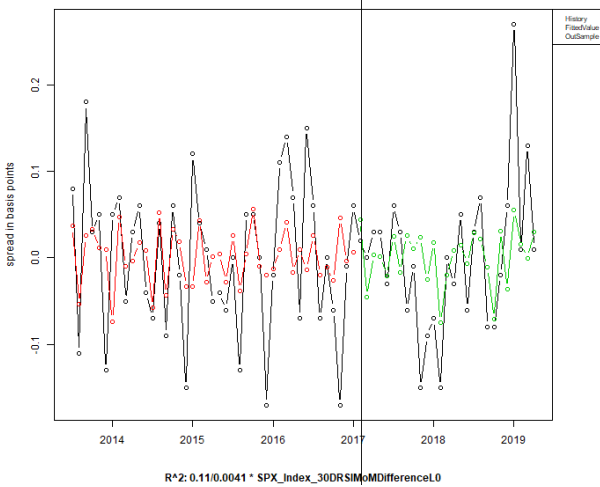
FI_IG_short - Level Value Backtesting

R^2 : 0.15/ 0.0044 * SPX_Index_30DRSIMoMDifferenceL0



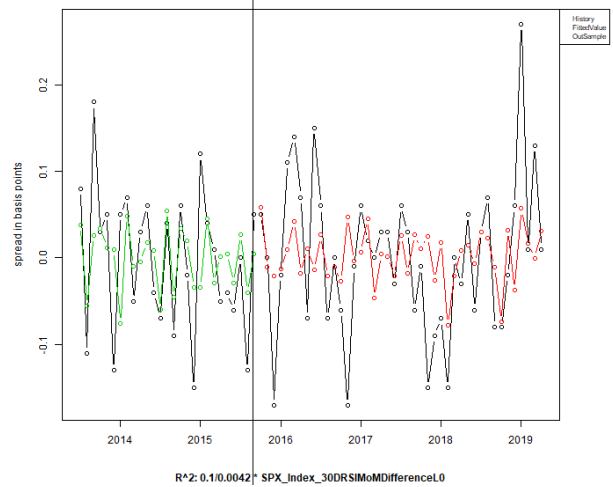
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019

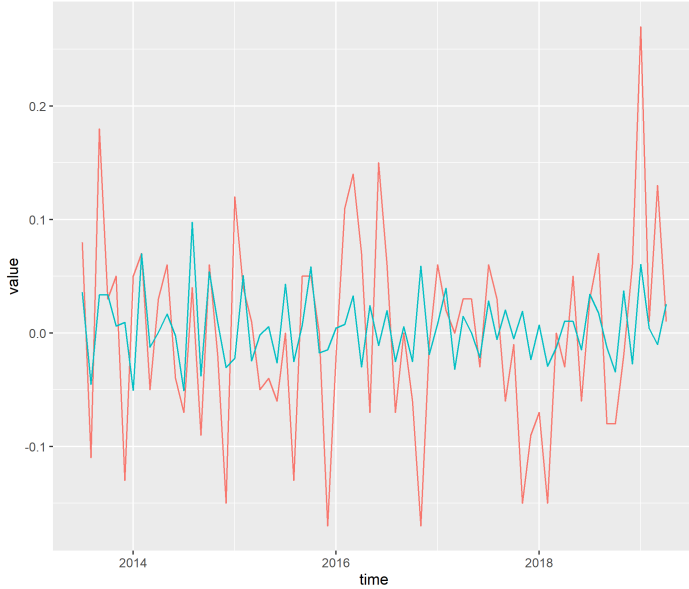


4.5.11 Model - 11

Backtesting on differenced values

FI_IG_short - Backtesting

$R^2: 0.12 / 0.1148 * SPX_Index_14DRSIMoMPercentChangeL0$

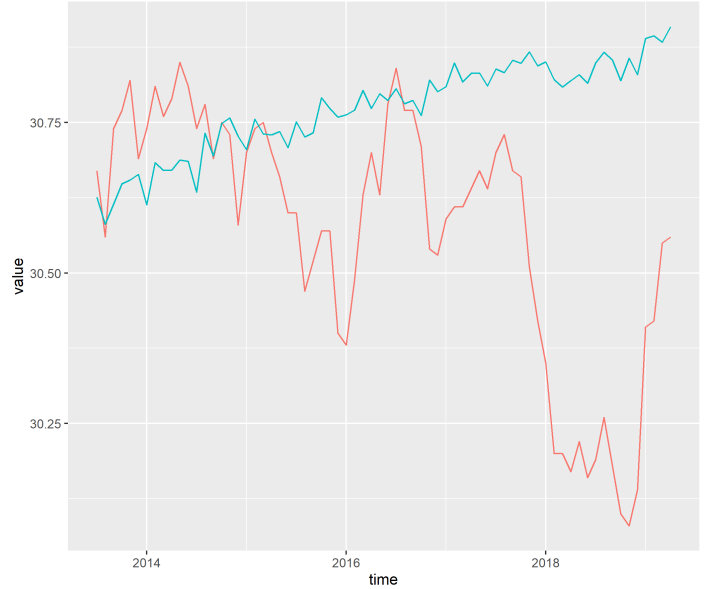


variable FI_IG_short_History FI_IG_short_FittedValue

Backtesting on level values

FI_IG_short - Level Value Backtesting

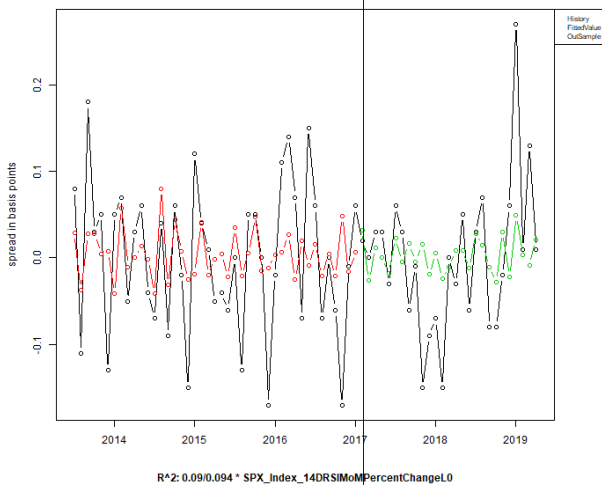
$R^2: 0.12 / 0.1148 * SPX_Index_14DRSIMoMPercentChangeL0$



variable diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP diffToLevel.na.omit.ts_to_plot...2....pc_level_SC

Out-of-Sample testing

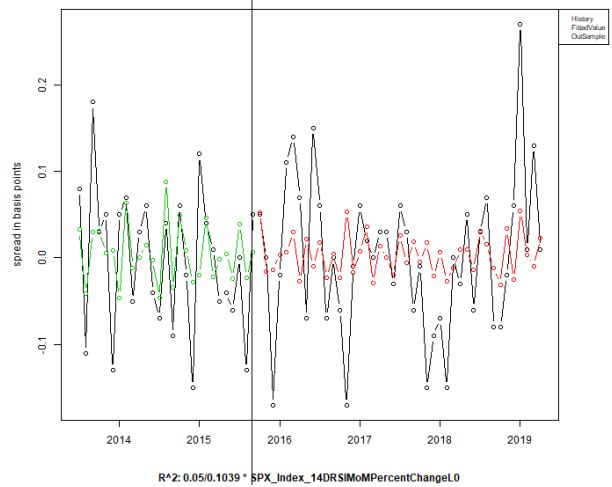
FI_IG_short Jul 2013-Apr 2019



$R^2: 0.09 / 0.094 * SPX_Index_14DRSIMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



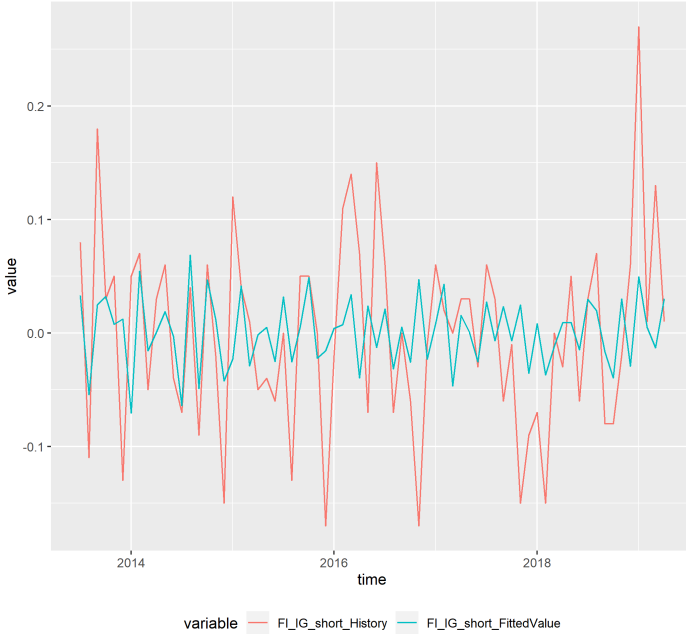
$R^2: 0.05 / 0.1039 * SPX_Index_14DRSIMoMPercentChangeL0$

4.5.12 Model - 12

Backtesting on differenced values

FI_IG_short - Backtesting

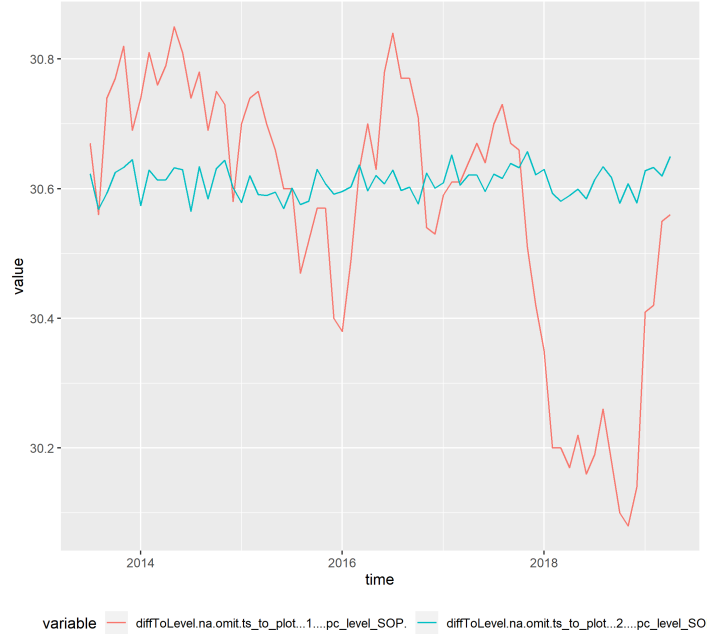
$R^2: 0.12 / 0.0022 * SPX_Index_14DRSIMoMDifferenceL0$



Backtesting on level values

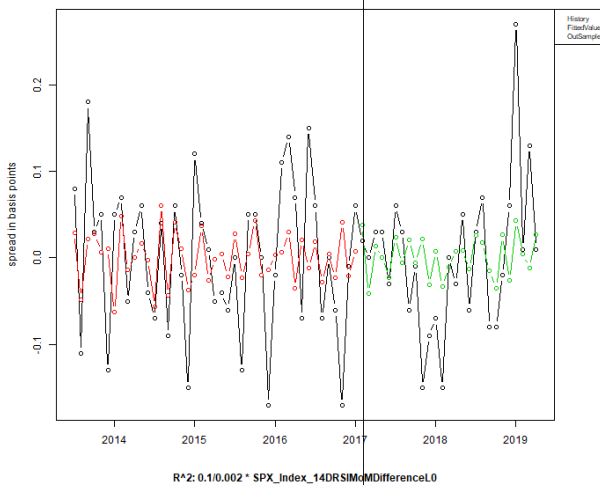
FI_IG_short - Level Value Backtesting

$R^2: 0.12 / 0.0022 * SPX_Index_14DRSIMoMDifferenceL0$



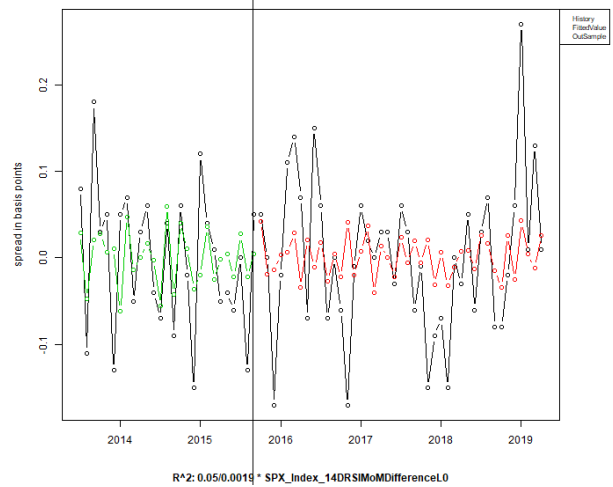
Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_short Jul 2013-Apr 2019



4.6 Two Factor Model for $\text{FI}_I G_{short}$

4.7 Single Factor Model for $FI_I G_{long}$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG30YR_IndexMoMPercentChangeL0	0.59	69.76	-7.83	0.00	0.39			
2	USGG30YR_IndexMoMDifferenceL0	0.58	70.26	-2.77	0.00	0.39			
3	USGG10YR_IndexMoMPercentChangeL0	0.48	86.15	-5.01	0.00	0.42			
4	USGG10YR_IndexMoMDifferenceL0	0.47	86.90	-2.29	0.00	0.42			
5	USGG2YR_IndexMoMDifferenceL0	0.22	113.85	-2.43	0.00	0.49			
6	USGG2YR_IndexMoMPercentChangeL0	0.18	117.75	-1.74	0.00	0.48			
7	UNRATEMoMPercentChangeL0	0.11	123.50	7.96	0.01	0.56			
8	UNRATEMoMDifferenceL0	0.10	124.23	1.47	0.01	0.52			
9	IPMANSCSMoMPercentChangeL0	0.00	131.68	-5.31	0.19	0.51			
10	UEMP27OVMoMPercentChangeL0	0.00	131.68	1.37	0.34	0.57			

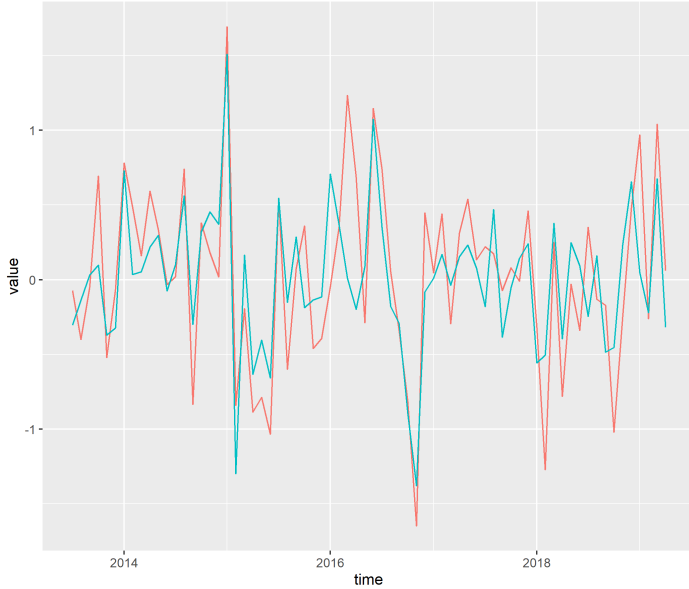
Table 6: Single Factor Analysis- $FI_I G_{long}$

4.7.1 Model - 1

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0.59 / -7.8342 * USGG30YR_IndexMoMPercentChangeL0$

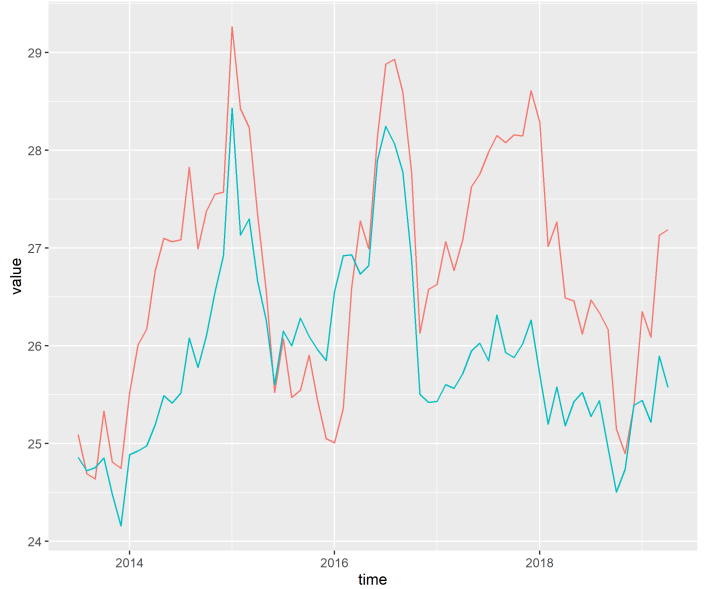


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

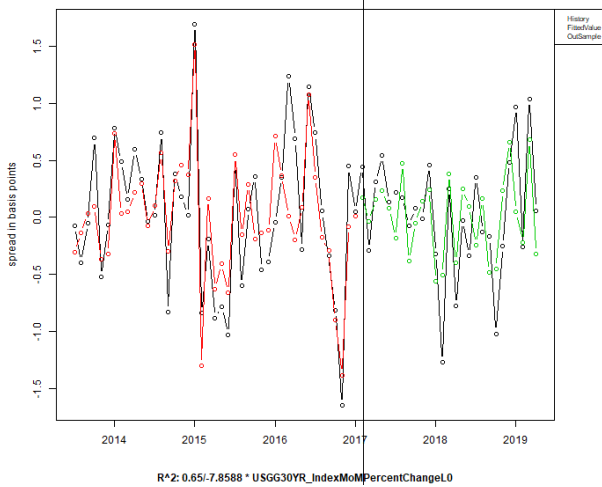
$R^2: 0.59 / -7.8342 * USGG30YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

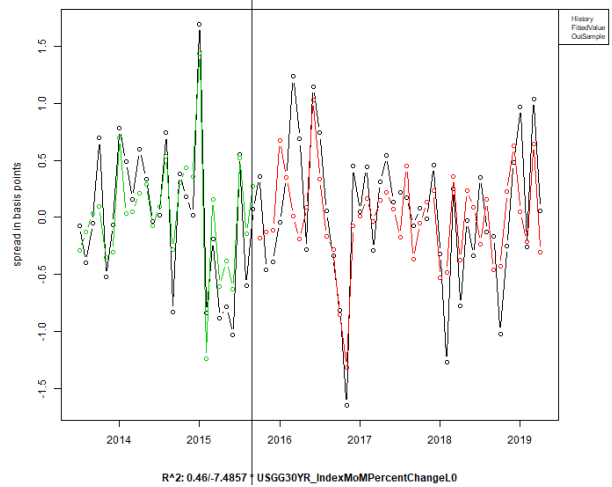
FI_IG_long Jul 2013-Apr 2019



$R^2: 0.65 / -7.8588 * USGG30YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



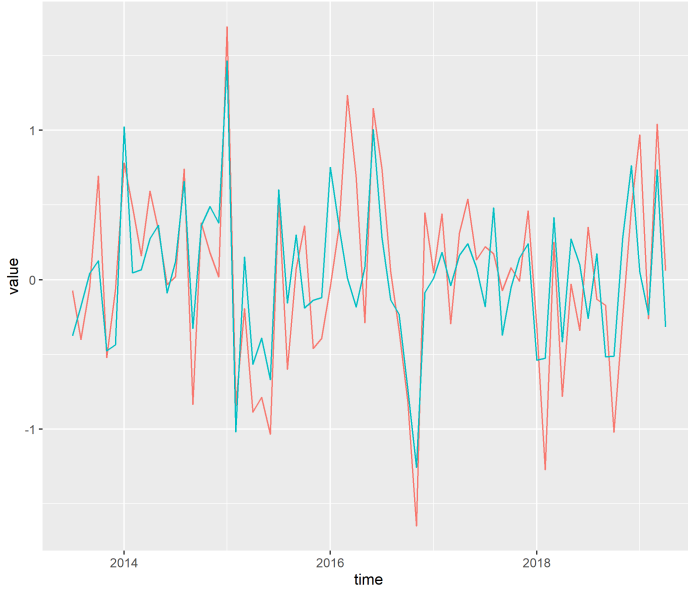
$R^2: 0.46 / -7.4857 * USGG30YR_IndexMoMPercentChangeL0$

4.7.2 Model - 2

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0.58 / -2.766 * USGG30YR_IndexMoMDifferenceL0$

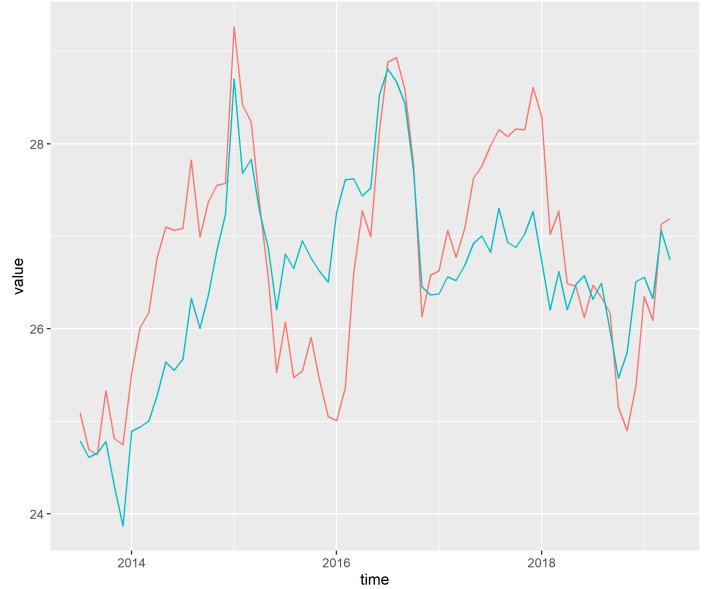


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

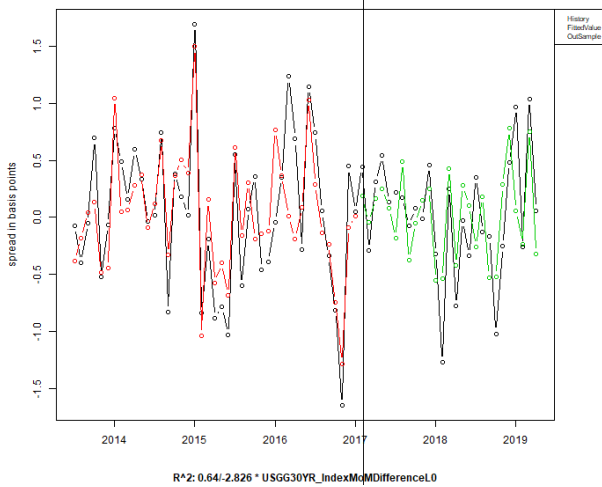
$R^2: 0.58 / -2.766 * USGG30YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

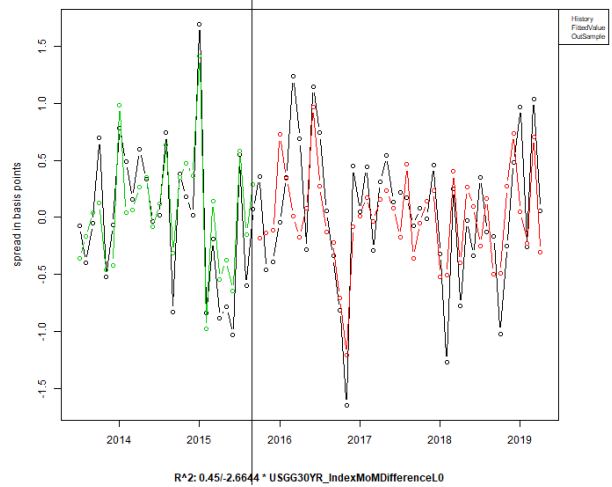
FI_IG_long Jul 2013-Apr 2019



$R^2: 0.64 / -2.826 * USGG30YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



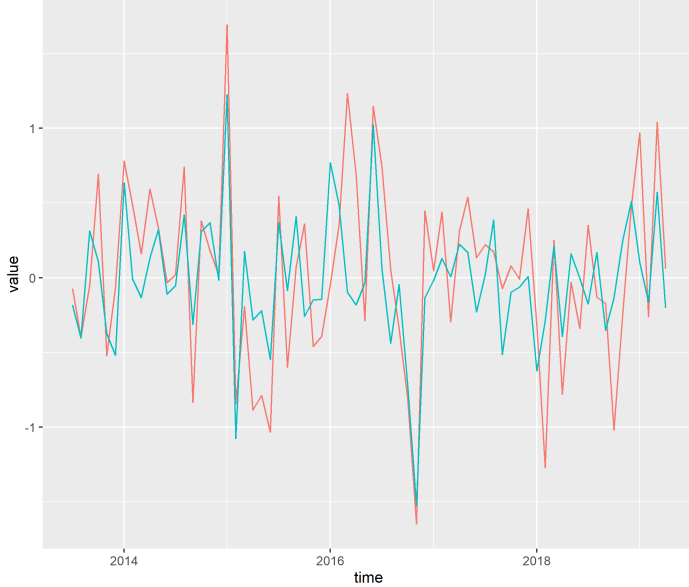
$R^2: 0.45 / -2.6644 * USGG30YR_IndexMoMDifferenceL0$

4.7.3 Model - 3

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0.48 / -5.0144 * \text{USGG10YR_IndexMoMPercentChangeL0}$

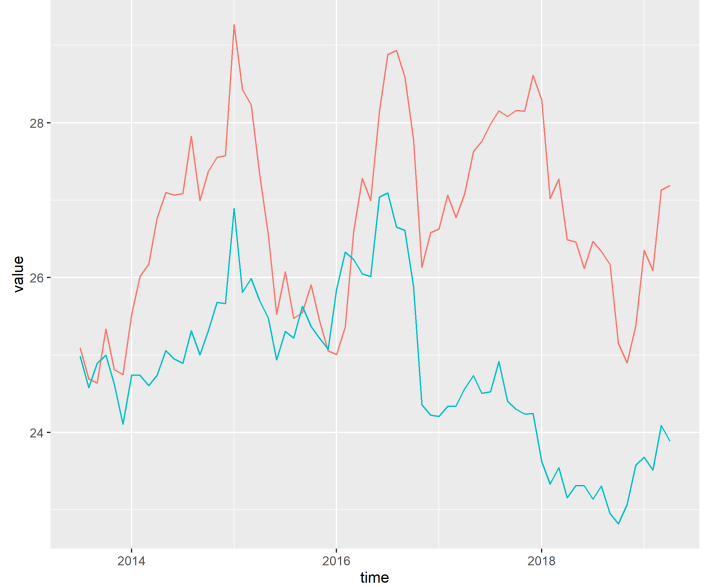


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

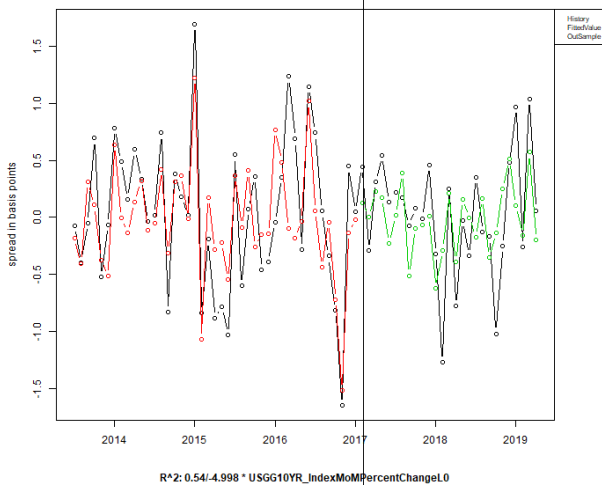
$R^2: 0.48 / -5.0144 * \text{USGG10YR_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

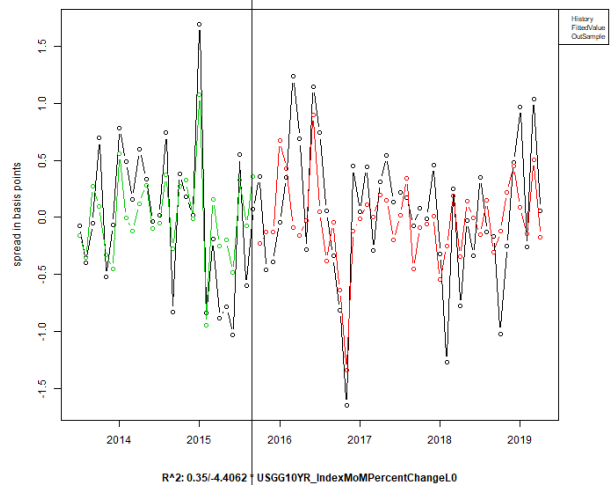
FI_IG_long Jul 2013-Apr 2019



$R^2: 0.54 / -4.998 * \text{USGG10YR_IndexMoMPercentChangeL0}$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



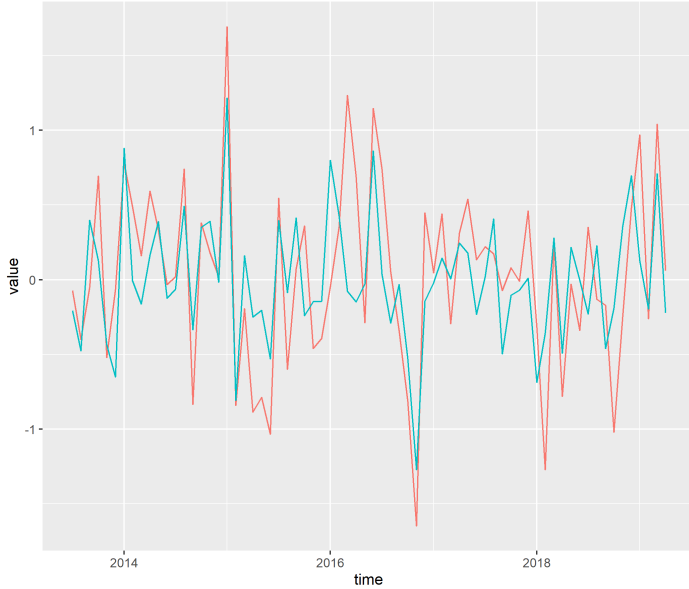
$R^2: 0.35 / -4.4062 * \text{USGG10YR_IndexMoMPercentChangeL0}$

4.7.4 Model - 4

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0.47 / -2.2891 * USGG10YR_IndexMoMDifferenceL0$

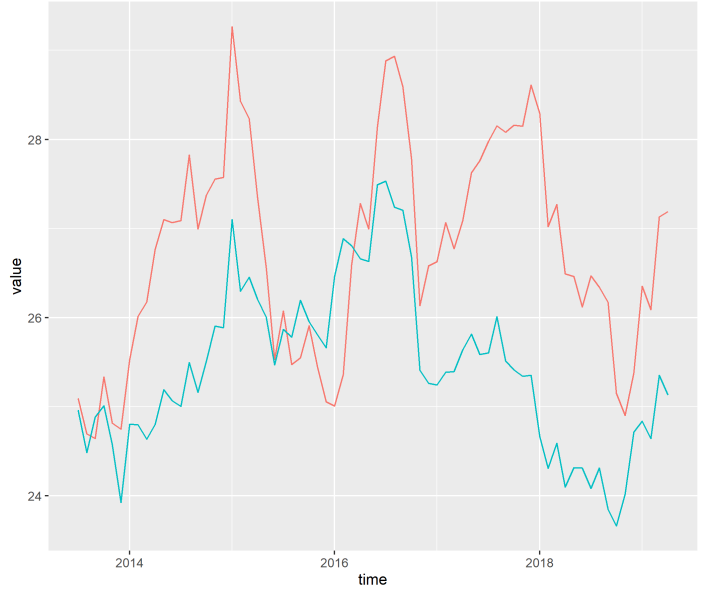


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

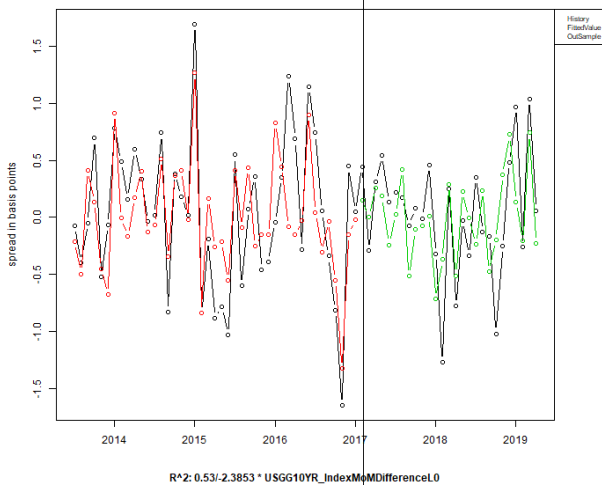
$R^2: 0.47 / -2.2891 * USGG10YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts...1....pc_level_SOP — diffToLevel.na.omit.ts...2....pc_level_SOF

Out-of-Sample testing

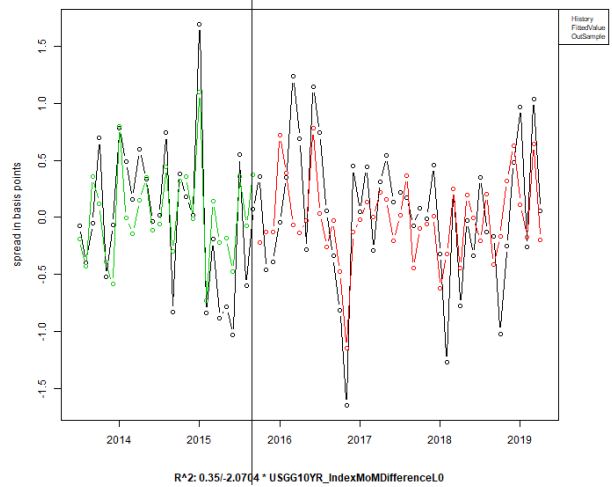
FI_IG_long Jul 2013-Apr 2019



$R^2: 0.53 / -2.3853 * USGG10YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



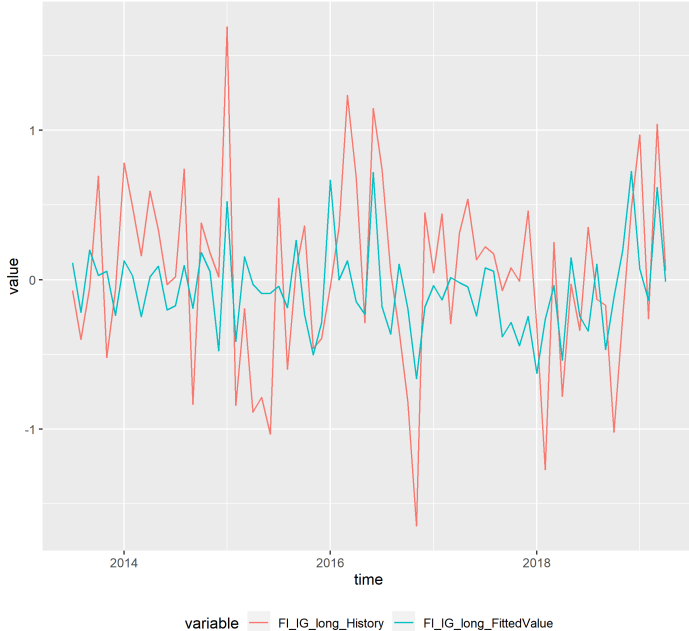
$R^2: 0.35 / -2.0704 * USGG10YR_IndexMoMDifferenceL0$

4.7.5 Model - 5

Backtesting on differenced values

FI_IG_long - Backtesting

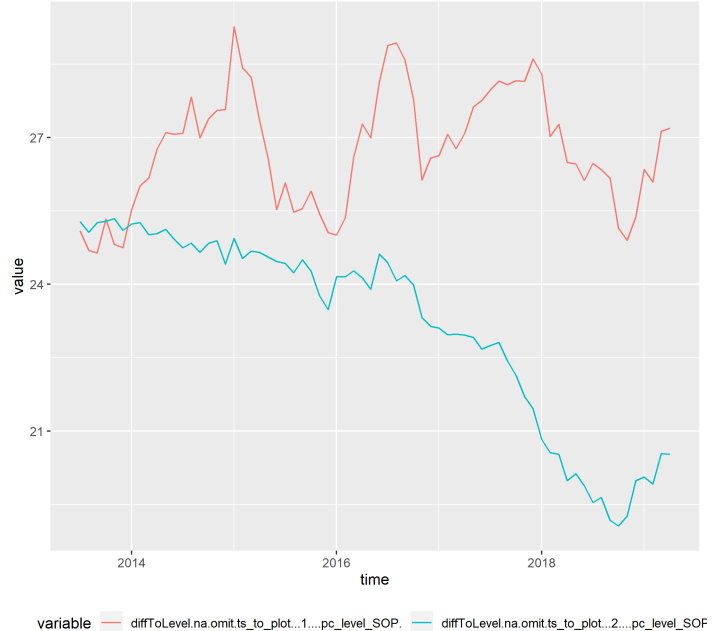
$R^2: 0.22 / -2.43 * USGG2YR_IndexMoMDifferenceL0$



Backtesting on level values

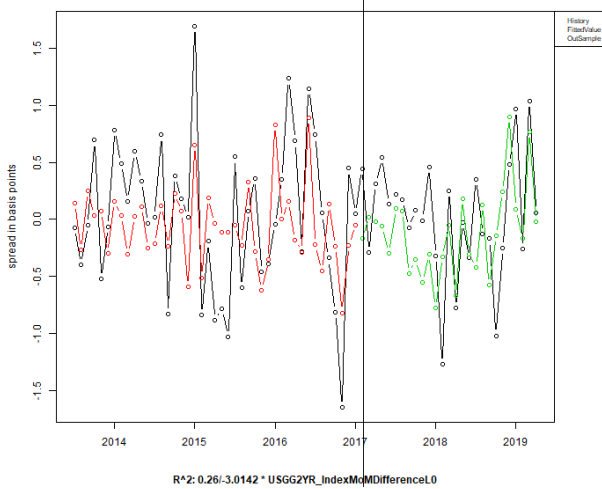
FI_IG_long - Level Value Backtesting

$R^2: 0.22 / -2.43 * USGG2YR_IndexMoMDifferenceL0$



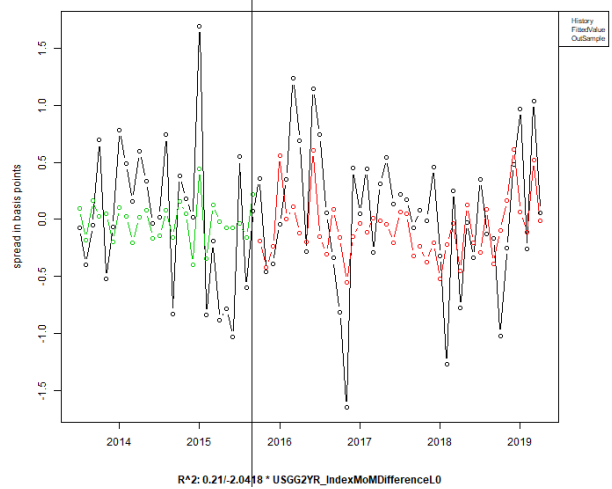
Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019

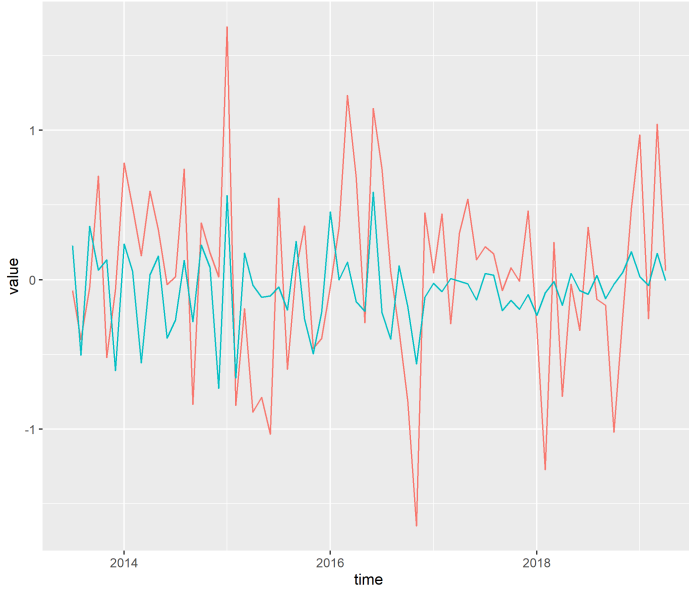


4.7.6 Model - 6

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0.18 / -1.7375 * USGG2YR_IndexMoMPercentChangeL0$

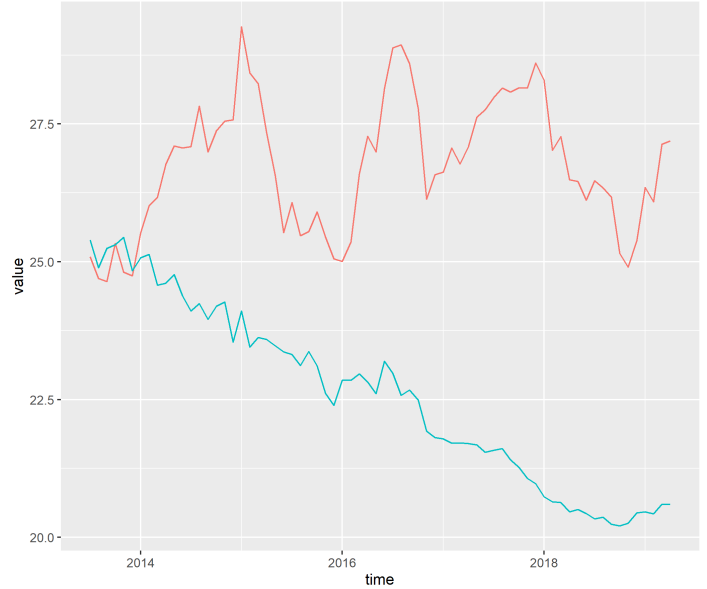


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

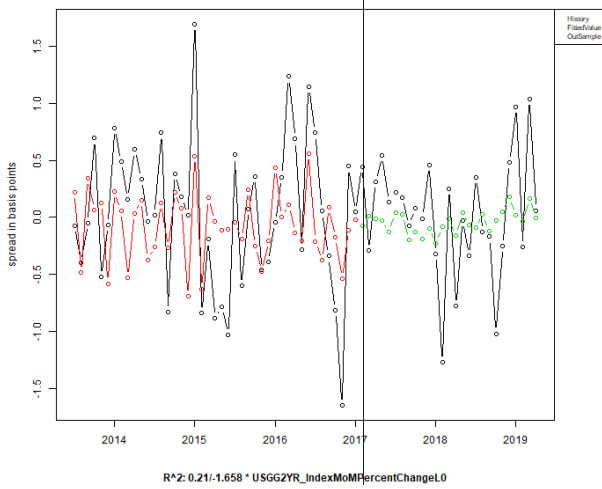
$R^2: 0.18 / -1.7375 * USGG2YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

Out-of-Sample testing

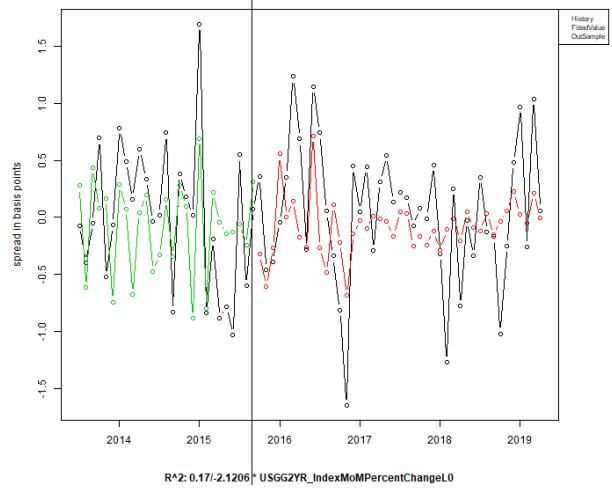
FI_IG_long Jul 2013-Apr 2019



$R^2: 0.21 / -1.658 * USGG2YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



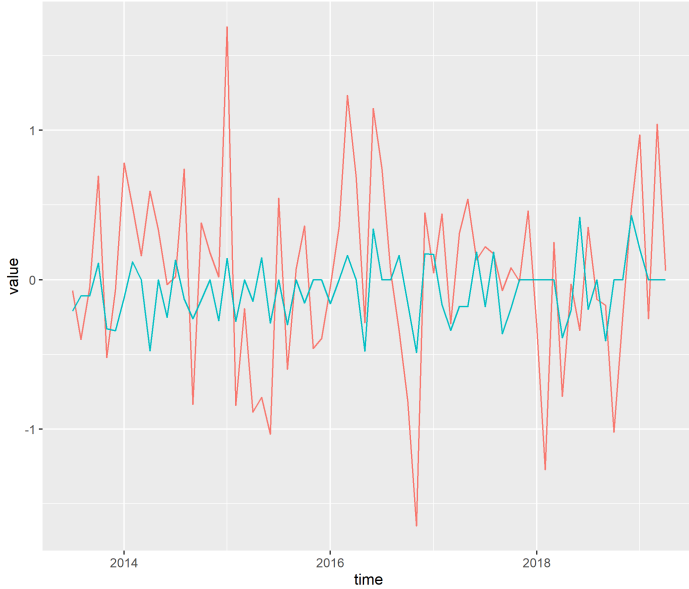
$R^2: 0.17 / -2.1206 * USGG2YR_IndexMoMPercentChangeL0$

4.7.7 Model - 7

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0.11 / 7.9583 * \text{UNRATEMoMPPercentChangeL0}$

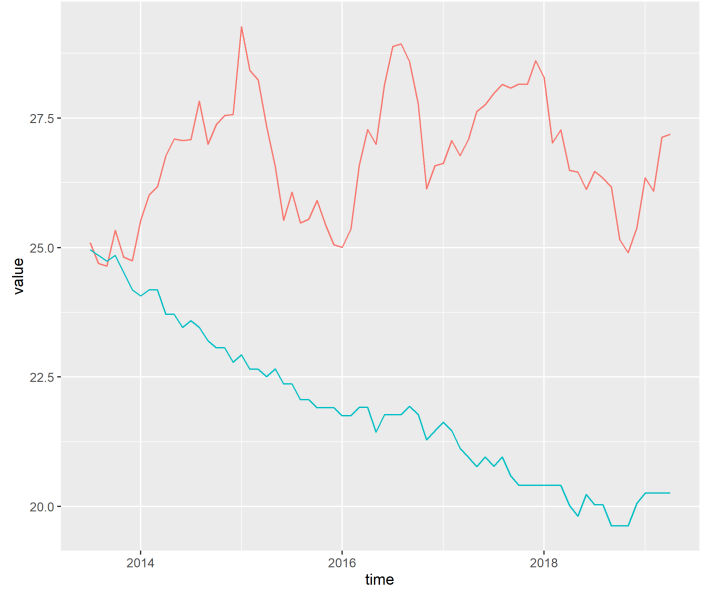


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

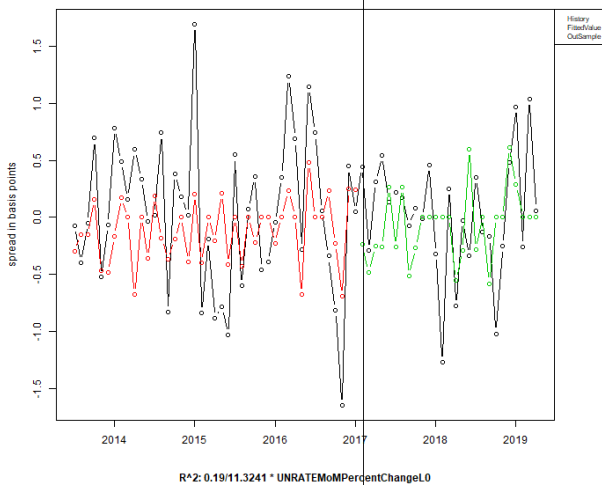
$R^2: 0.11 / 7.9583 * \text{UNRATEMoMPPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

Out-of-Sample testing

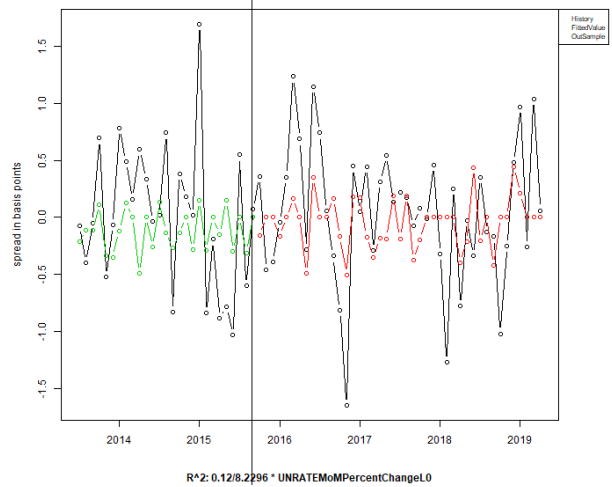
FI_IG_long Jul 2013-Apr 2019



$R^2: 0.19 / 11.3241 * \text{UNRATEMoMPPercentChangeL0}$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019

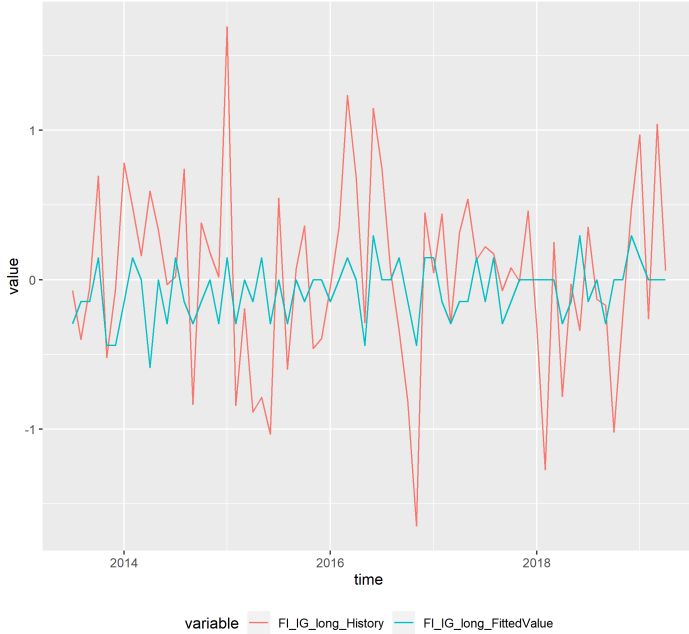


$R^2: 0.12 / 8.2296 * \text{UNRATEMoMPPercentChangeL0}$

4.7.8 Model - 8

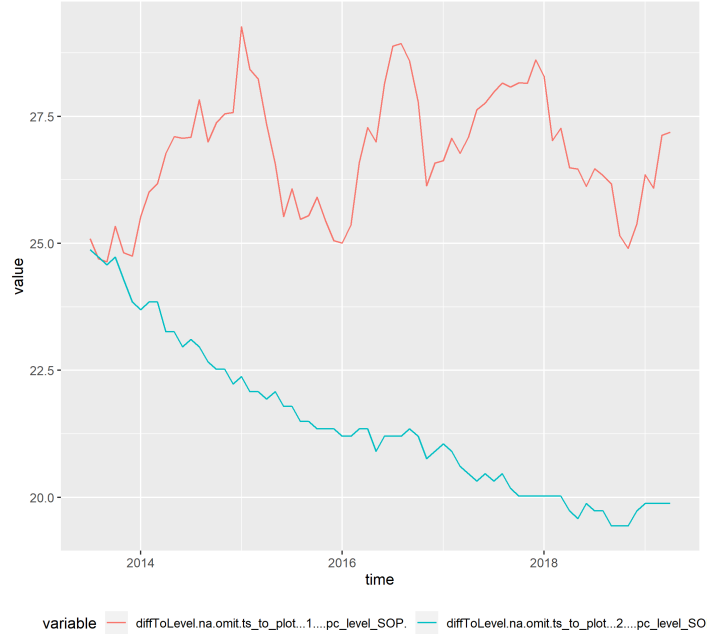
Backtesting on differenced values

FI_IG_long - Backtesting
R²: 0.1/ 1.4689 * UNRATEMoMDifferenceL0



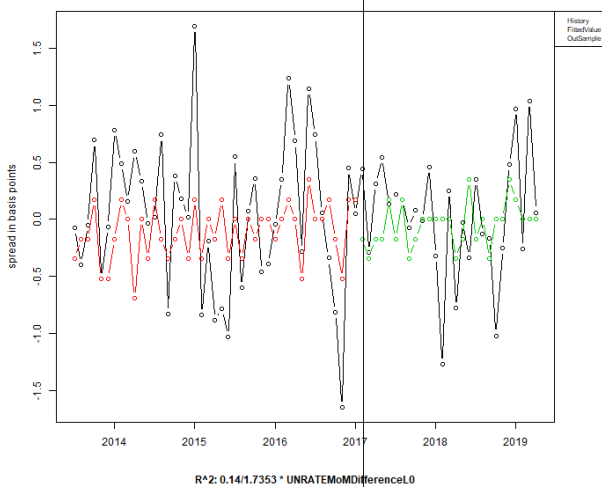
Backtesting on level values

FI_IG_long - Level Value Backtesting
R²: 0.1/ 1.4689 * UNRATEMoMDifferenceL0



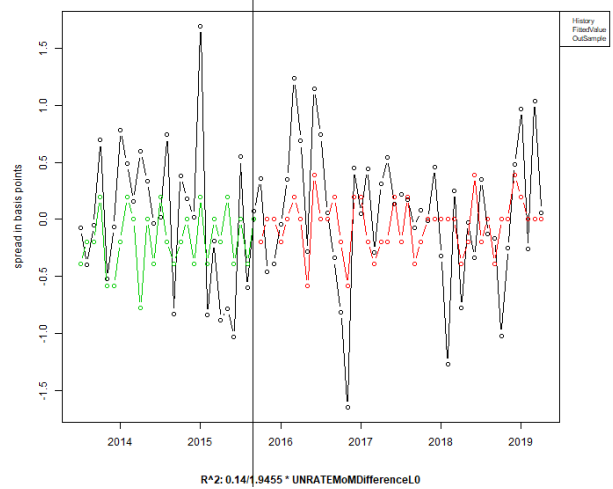
Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019

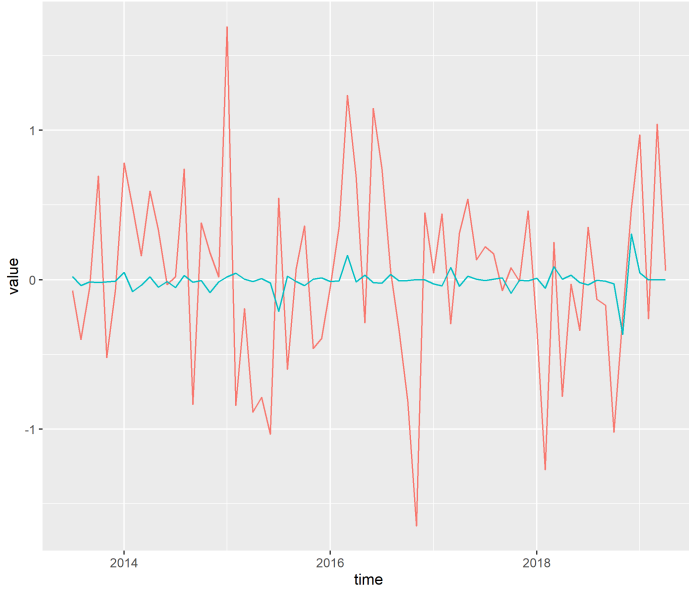


4.7.9 Model - 9

Backtesting on differenced values

FI_IG_long - Backtesting

$R^2: 0 / -5.3122 * IPMANSICSMoMPercentChangeL0$

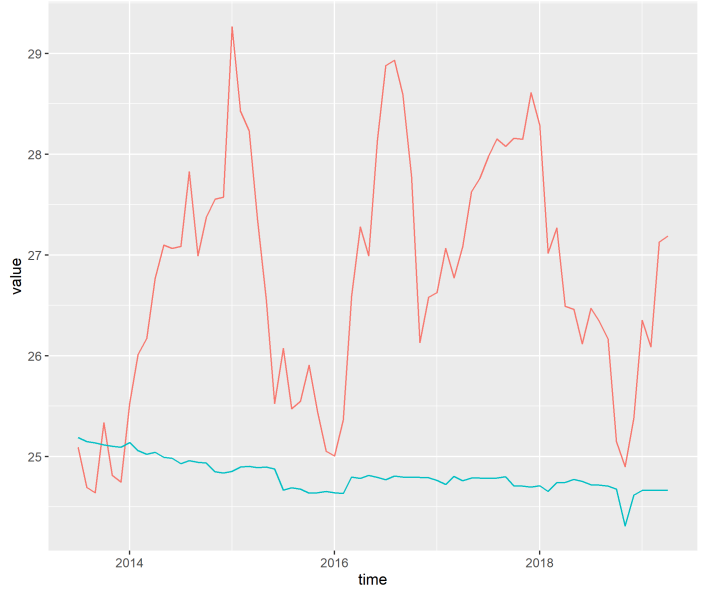


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

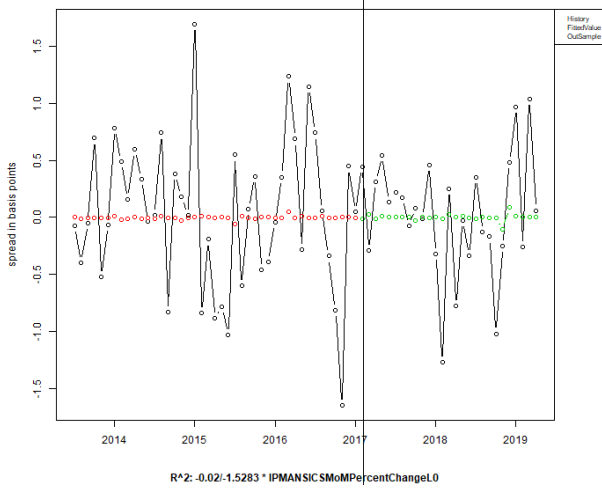
$R^2: 0 / -5.3122 * IPMANSICSMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

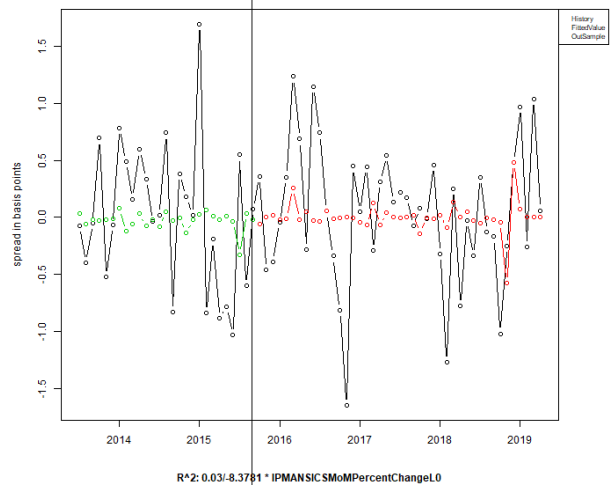
FI_IG_long Jul 2013-Apr 2019



$R^2: -0.02 / -1.5283 * IPMANSICSMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



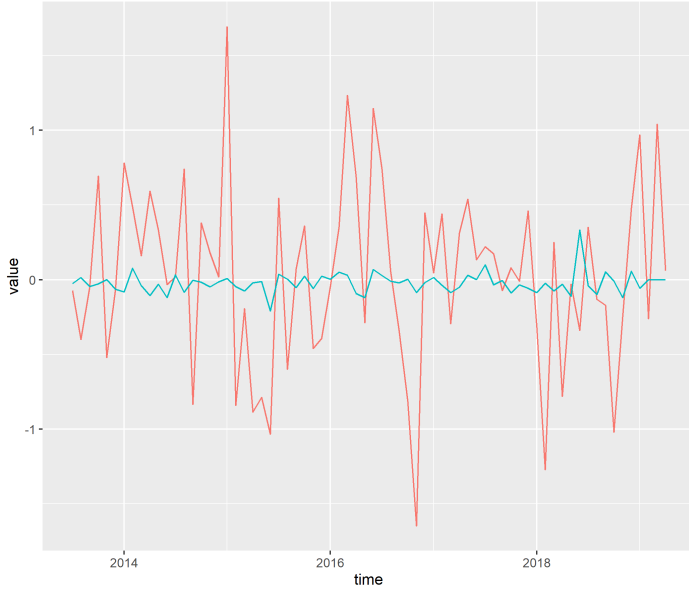
$R^2: 0.03 / -8.3781 * IPMANSICSMoMPercentChangeL0$

4.7.10 Model - 10

Backtesting on differenced values

FI_IG_long - Backtesting

R²: 0/ 1.3734 * UEMP27OVMoMPercentChangeL0

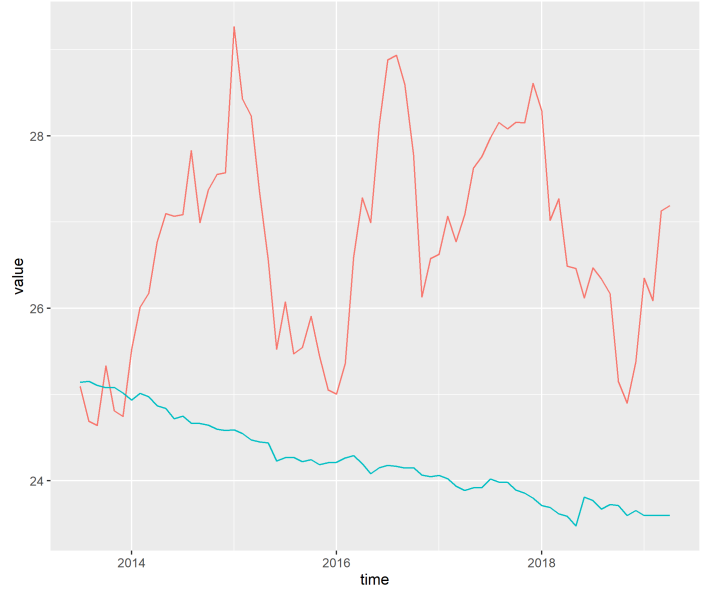


variable — FI_IG_long_History — FI_IG_long_FittedValue

Backtesting on level values

FI_IG_long - Level Value Backtesting

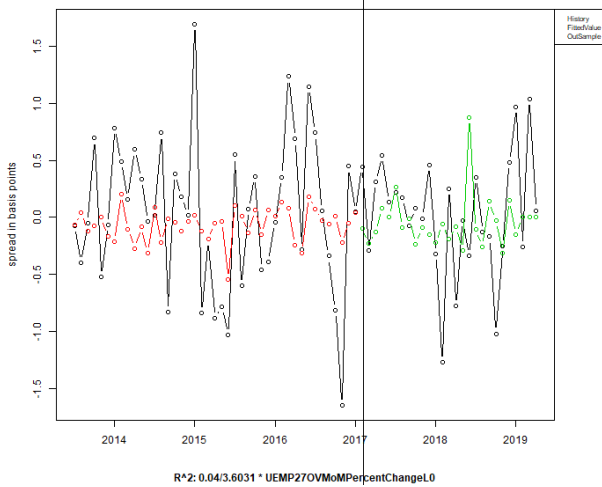
R²: 0/ 1.3734 * UEMP27OVMoMPercentChangeL0



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

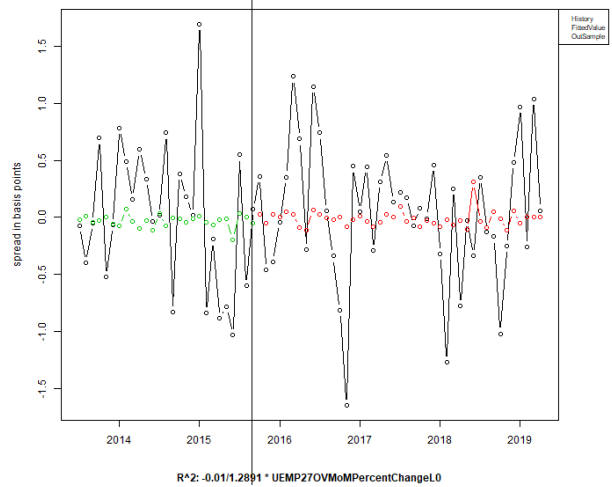
FI_IG_long Jul 2013-Apr 2019



R²: 0.04/3.6031 * UEMP27OVMoMPercentChangeL0

Out-of-Sample testing

FI_IG_long Jul 2013-Apr 2019



R²: -0.01/1.2891 * UEMP27OVMoMPercentChangeL0

4.8 Two Factor Model for $FI_I G_{long}$

4.9 Single Factor Model for $FI_I G_{mid}$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG10YR_IndexMoMDifferenceL0	0.60	-51.43	-1.10	0.00	0.18			
2	USGG10YR_IndexMoMPercentChangeL0	0.59	-49.88	-2.37	0.00	0.18			
3	USGG30YR_IndexMoMDifferenceL0	0.50	-36.30	-1.10	0.00	0.19			
4	USGG30YR_IndexMoMPercentChangeL0	0.49	-35.16	-3.08	0.00	0.19			
5	USGG2YR_IndexMoMDifferenceL0	0.46	-30.05	-1.46	0.00	0.22			
6	USGG2YR_IndexMoMPercentChangeL0	0.45	-28.98	-1.14	0.00	0.20			
7	UNRATEMoMPercentChangeL0	0.13	3.14	3.61	0.00	0.24			
8	UNRATEMoMDifferenceL0	0.12	4.00	0.67	0.00	0.23			

Table 7: Single Factor Analysis- $FI_I G_{mid}$

4.9.1 Model - 1

Backtesting on differenced values

FI_IG_mid - Backtesting

$R^2: 0.6 / -1.0976 * USGG10YR_IndexMoMDifferenceL0$



Backtesting on level values

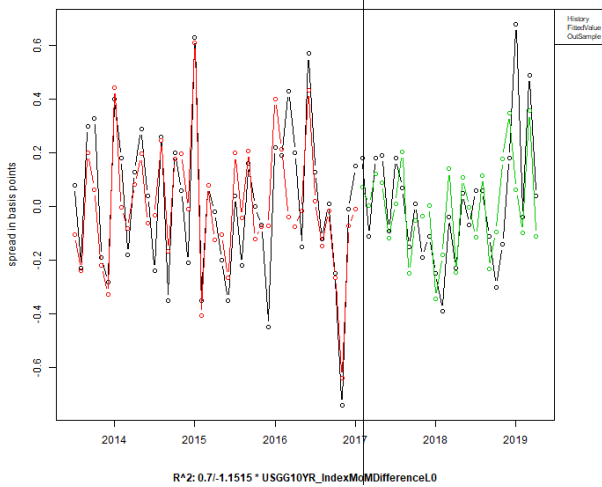
FI_IG_mid - Level Value Backtesting

$R^2: 0.6 / -1.0976 * USGG10YR_IndexMoMDifferenceL0$



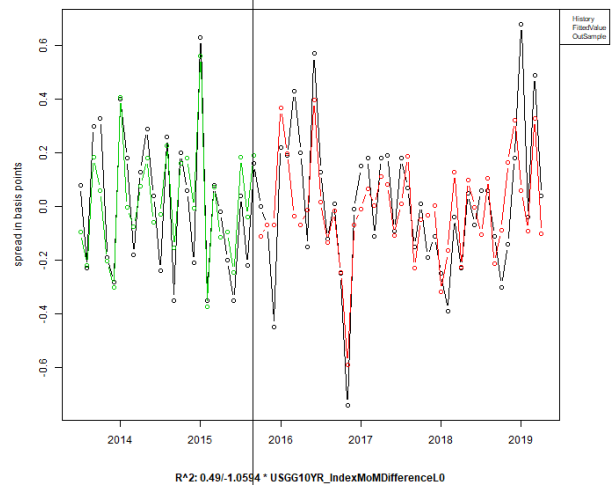
Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019

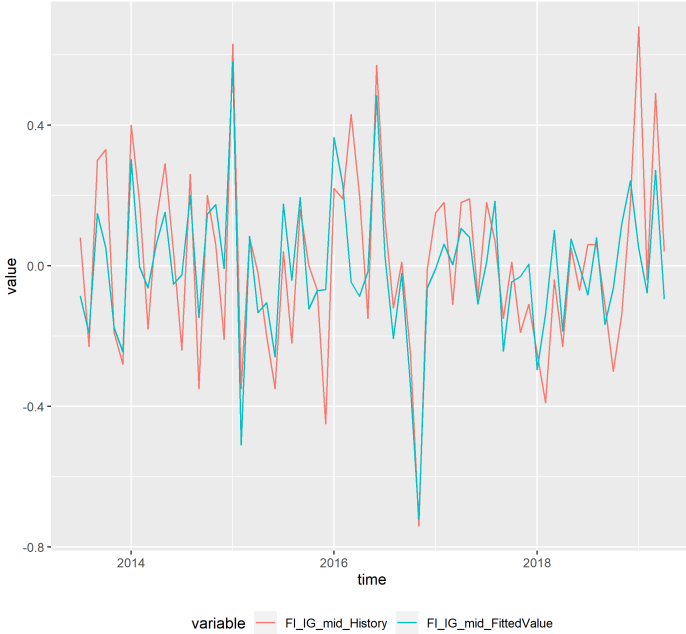


4.9.2 Model - 2

Backtesting on differenced values

FI_IG_mid - Backtesting

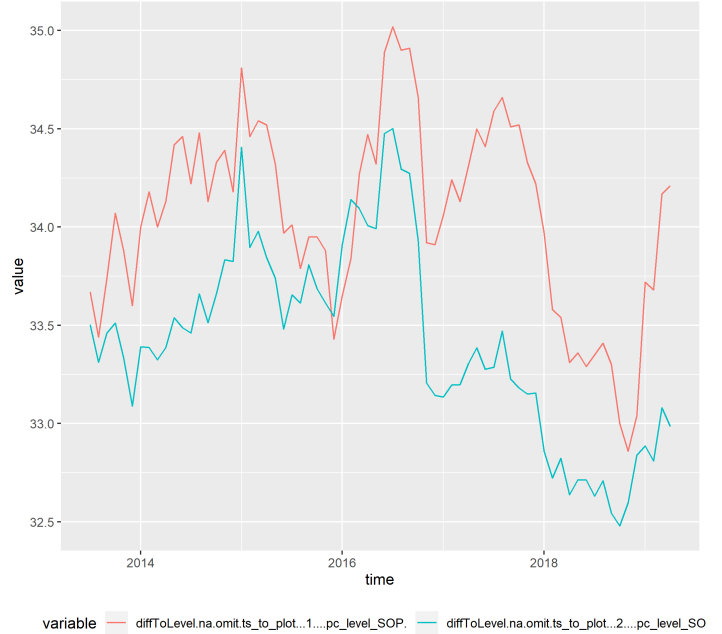
$R^2: 0.59 / -2.3731 * USGG10YR_IndexMoMPercentChangeL0$



Backtesting on level values

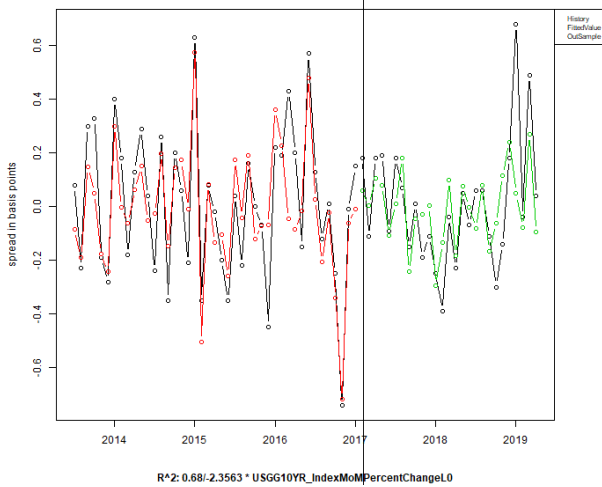
FI_IG_mid - Level Value Backtesting

$R^2: 0.59 / -2.3731 * USGG10YR_IndexMoMPercentChangeL0$



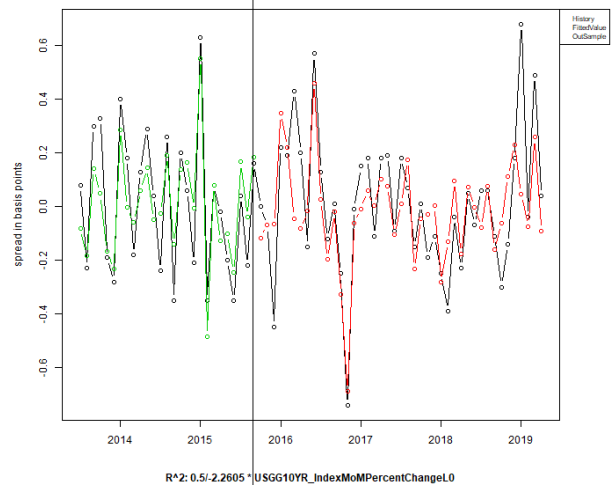
Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



4.9.3 Model - 3

Backtesting on differenced values

FI_IG_mid - Backtesting

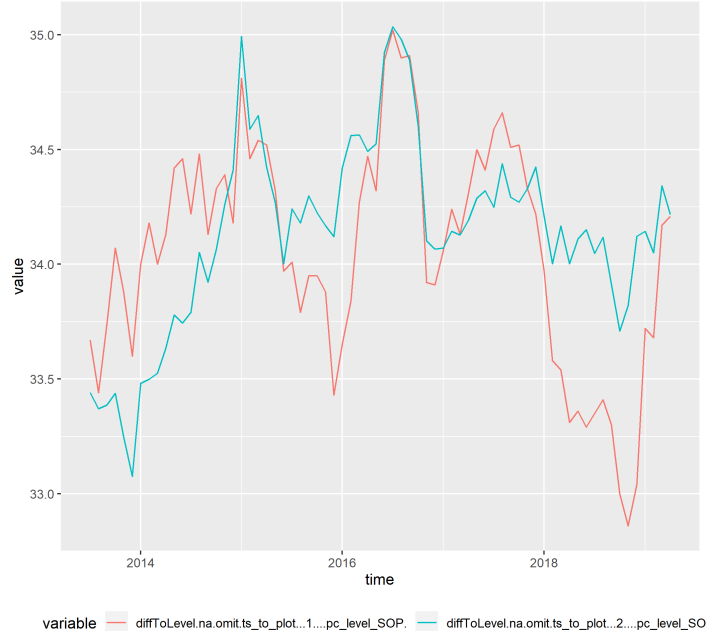
$R^2: 0.5 / -1.098 * USGG30YR_IndexMoMDifferenceL0$



Backtesting on level values

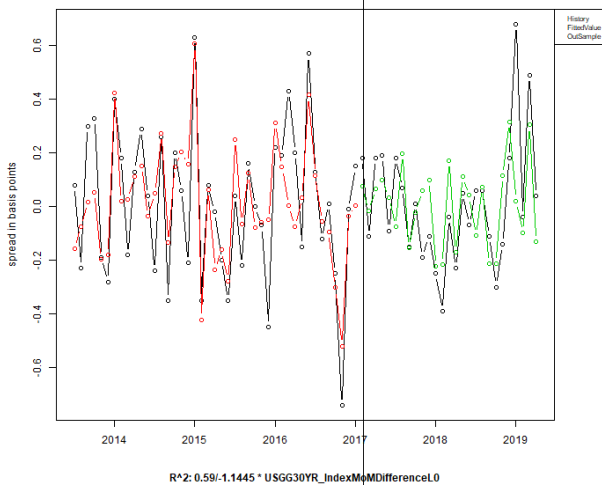
FI_IG_mid - Level Value Backtesting

$R^2: 0.5 / -1.098 * USGG30YR_IndexMoMDifferenceL0$



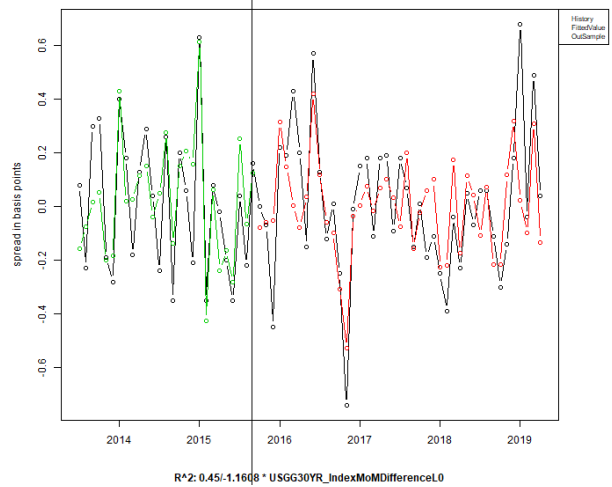
Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019

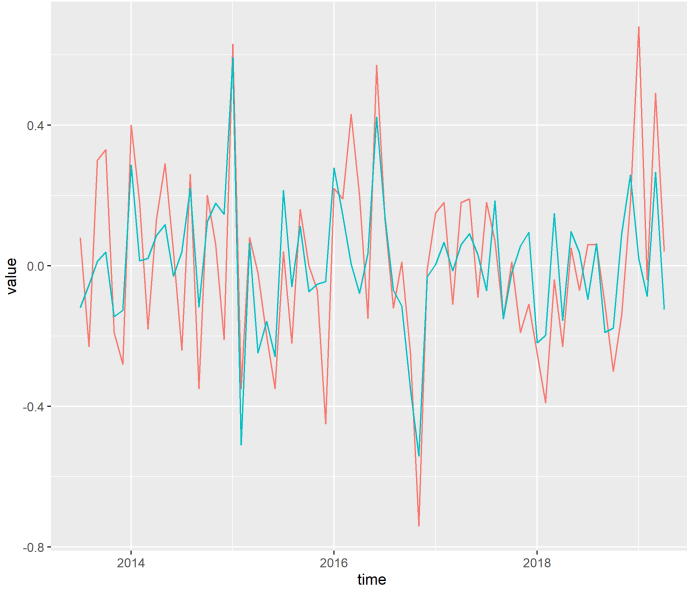


4.9.4 Model - 4

Backtesting on differenced values

FI_IG_mid - Backtesting

$R^2: 0.49 / -3.0777 * USGG30YR_IndexMoMPercentChangeL0$

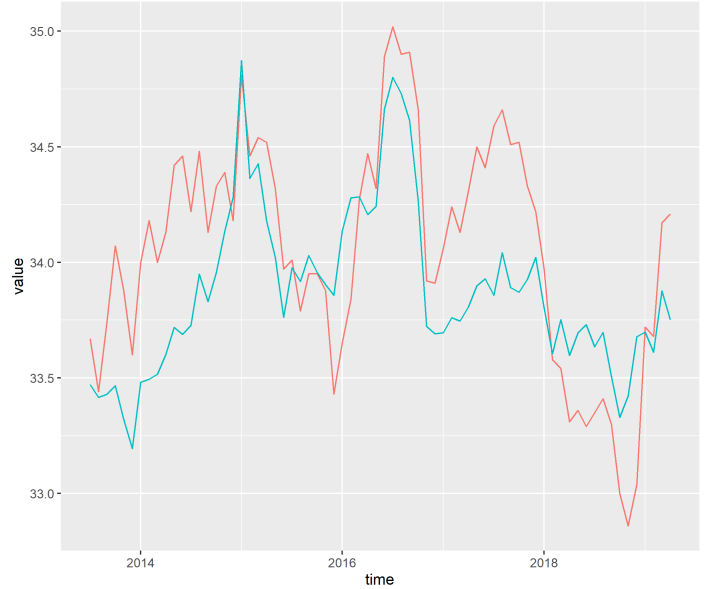


variable — FI_IG_mid_History — FI_IG_mid_FittedValue

Backtesting on level values

FI_IG_mid - Level Value Backtesting

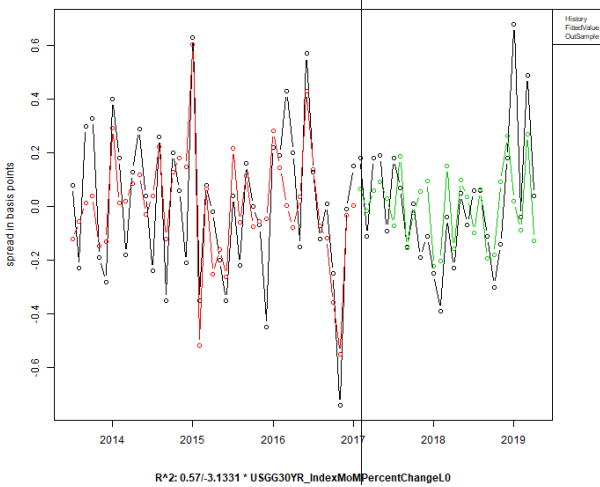
$R^2: 0.49 / -3.0777 * USGG30YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SO

Out-of-Sample testing

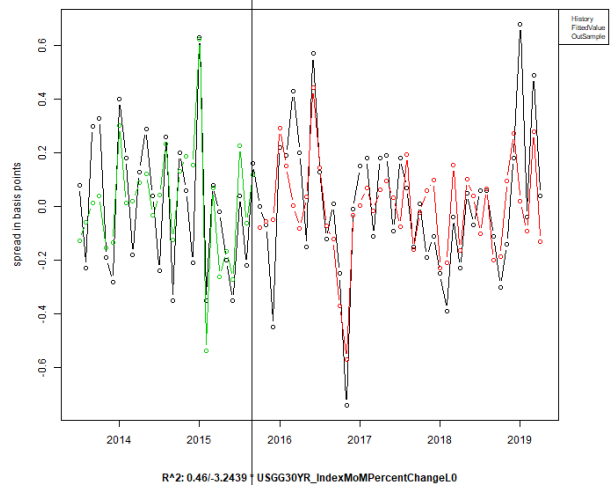
FI_IG_mid Jul 2013-Apr 2019



$R^2: 0.57 / -3.1331 * USGG30YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



$R^2: 0.46 / -3.2439 * USGG30YR_IndexMoMPercentChangeL0$

4.9.5 Model - 5

Backtesting on differenced values

FI_IG_mid - Backtesting

$R^2: 0.46 / -1.4552 * USGG2YR_IndexMoMDifferenceL0$

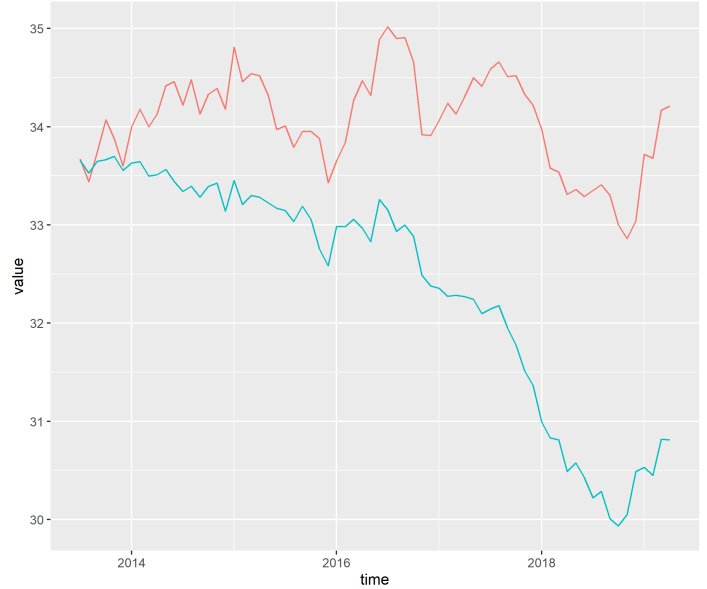


variable — FI_IG_mid_History — FI_IG_mid_FittedValue

Backtesting on level values

FI_IG_mid - Level Value Backtesting

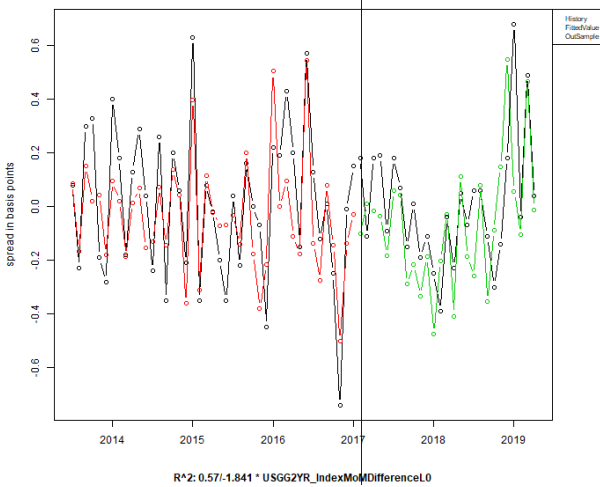
$R^2: 0.46 / -1.4552 * USGG2YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

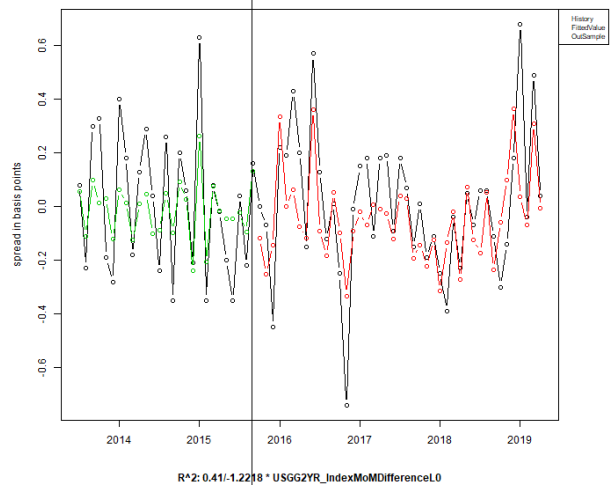
FI_IG_mid Jul 2013-Apr 2019



$R^2: 0.57 / -1.841 * USGG2YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



$R^2: 0.41 / -1.2218 * USGG2YR_IndexMoMDifferenceL0$

4.9.6 Model - 6

Backtesting on differenced values

FI_IG_mid - Backtesting

$R^2: 0.45 / -1.1427 * USGG2YR_IndexMoMPercentChangeL0$

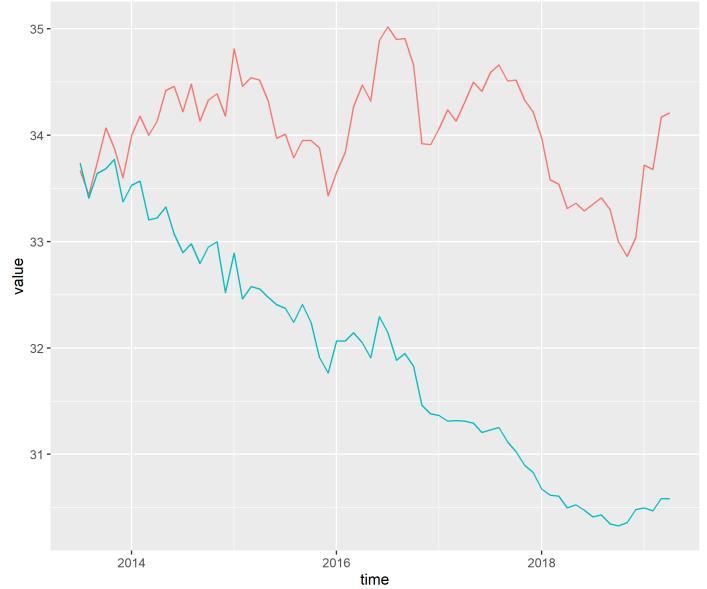


variable — FI_IG_mid_History — FI_IG_mid_FittedValue

Backtesting on level values

FI_IG_mid - Level Value Backtesting

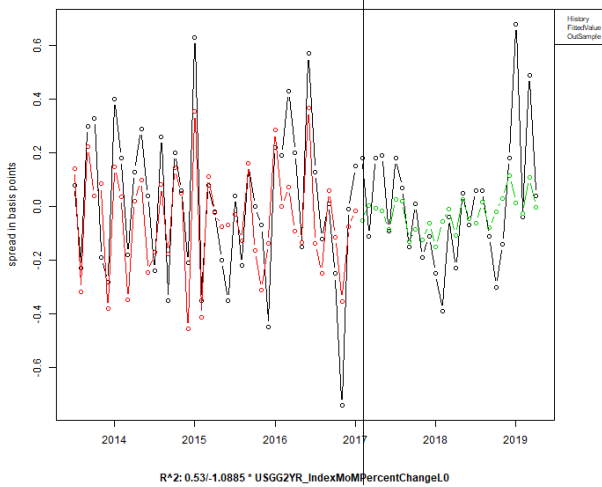
$R^2: 0.45 / -1.1427 * USGG2YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOF

Out-of-Sample testing

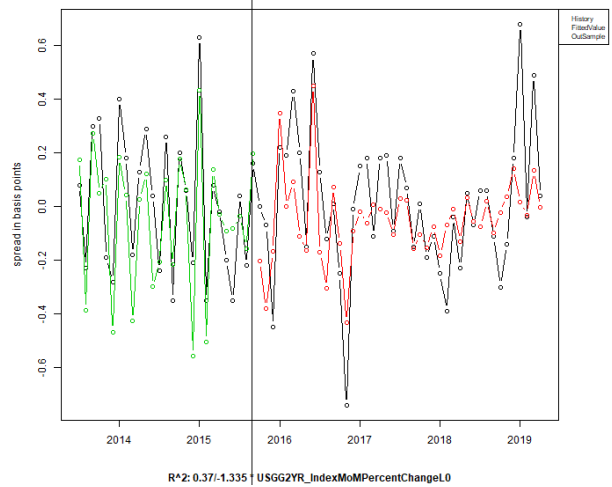
FI_IG_mid Jul 2013-Apr 2019



$R^2: 0.53 / -1.0885 * USGG2YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



$R^2: 0.37 / -1.335 * USGG2YR_IndexMoMPercentChangeL0$

4.9.7 Model - 7

Backtesting on differenced values

FI_IG_mid - Backtesting

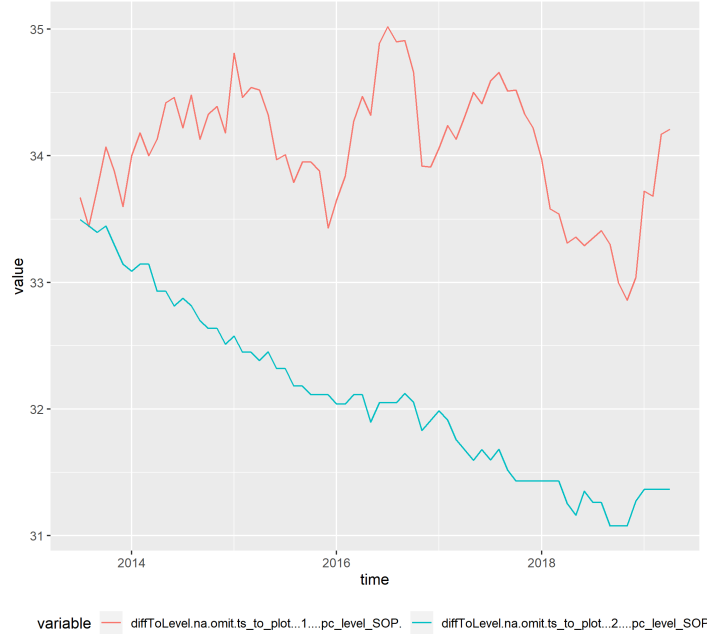
$R^2: 0.13 / 3.6087 * \text{UNRATEMoMPercentChangeL0}$



Backtesting on level values

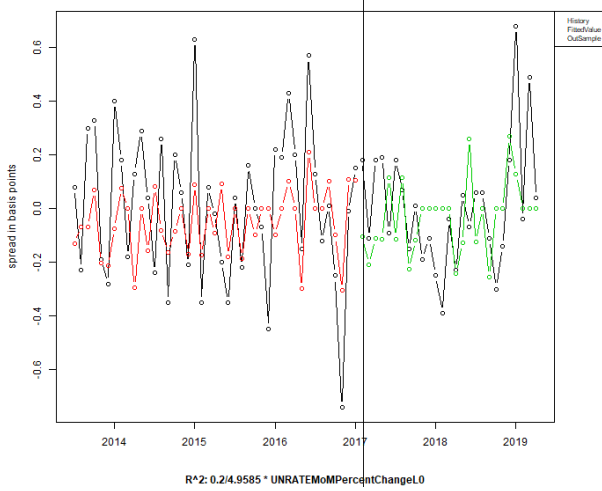
FI_IG_mid - Level Value Backtesting

$R^2: 0.13 / 3.6087 * \text{UNRATEMoMPercentChangeL0}$



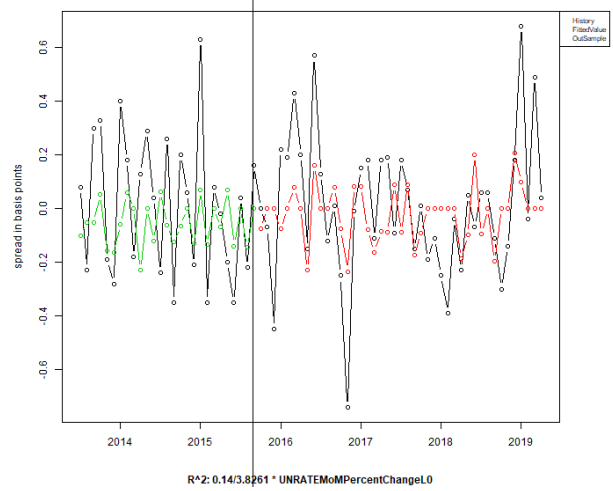
Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



4.9.8 Model - 8

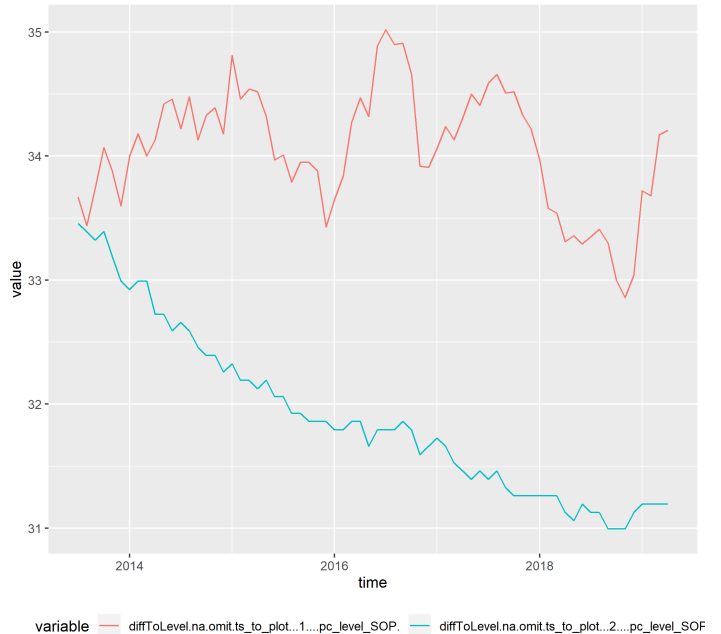
Backtesting on differenced values

FI_IG_mid - Backtesting
R²: 0.12/ 0.6654 * UNRATEMoMDifferenceL0



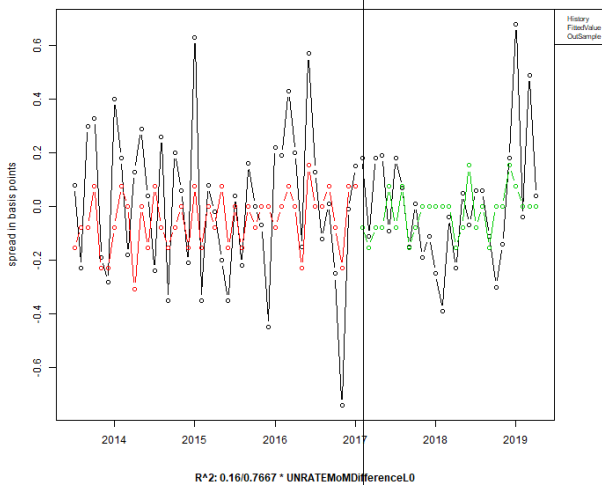
Backtesting on level values

FI_IG_mid - Level Value Backtesting
R²: 0.12/ 0.6654 * UNRATEMoMDifferenceL0



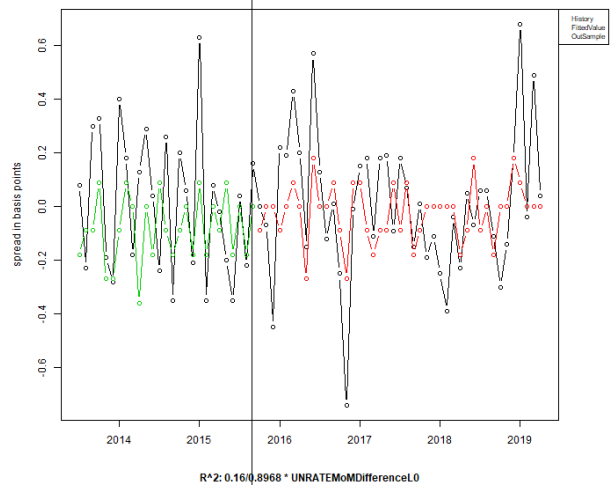
Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



Out-of-Sample testing

FI_IG_mid Jul 2013-Apr 2019



4.10 Two Factor Model for $FI_I G_m id$

4.11 Single Factor Model for $FI_H Y$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	VIX_IndexMoMDifferenceL0	0.37	256.72	-0.29	0.00	0.96			
2	SPX_Index_30DRSIMoMPercentChangeL0	0.36	257.46	7.88	0.00	1.01			
3	BAA10YB_IndexMoMDifferenceL0	0.36	257.47	-0.08	0.00	1.01			
4	VIX_IndexMoMPercentChangeL0	0.34	259.35	-4.10	0.00	1.08			
5	SPX_Index_30DRSIMoMDifferenceL0	0.33	260.03	0.14	0.00	1.10			
6	BAA10YB_IndexMoMPercentChangeL0	0.29	264.97	-10.10	0.00	1.11			
7	SPX_Index_14DRSIMoMPercentChangeL0	0.28	265.60	3.73	0.00	1.15			
8	SPX_Index_14DRSIMoMDifferenceL0	0.27	266.13	0.07	0.00	1.17			
9	IPMANSICSMoMDifferenceL1	0.19	273.89	-0.57	0.00	0.96			
10	IPMANSICSMoMPercentChangeL1	0.18	274.35	-60.05	0.00	0.99			
11	AAIIBEAR_IndexMoMDifferenceL0	0.14	277.66	-0.09	0.00	1.22			
12	MOVE_IndexMoMPercentChangeL0	0.12	279.33	-5.20	0.00	1.28			
13	MOVE_IndexMoMDifferenceL0	0.12	279.90	-0.07	0.00	1.31			
14	AAIIBEAR_IndexMoMPercentChangeL0	0.12	279.92	-2.21	0.00	1.34			
15	IPBUSEQMoMDifferenceL1	0.10	281.00	-0.27	0.00	1.72			
16	USGG2YR_IndexMoMDifferenceL1	0.10	281.02	-5.19	0.01	1.30			
17	MOVE_IndexMoMPercentChangeL1	0.00	288.57	1.66	0.38	1.49			

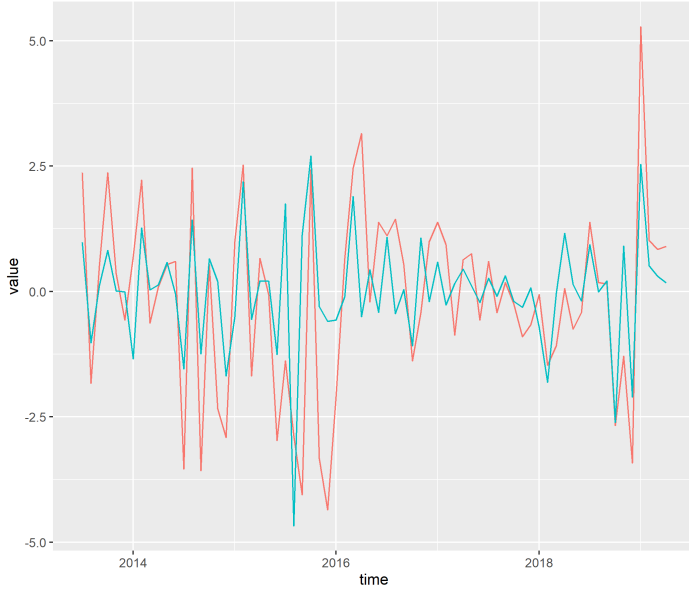
Table 8: Single Factor Analysis- $FI_H Y$

4.11.1 Model - 1

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.37 / -0.2864 * VIX_IndexMoMDifferenceL0$

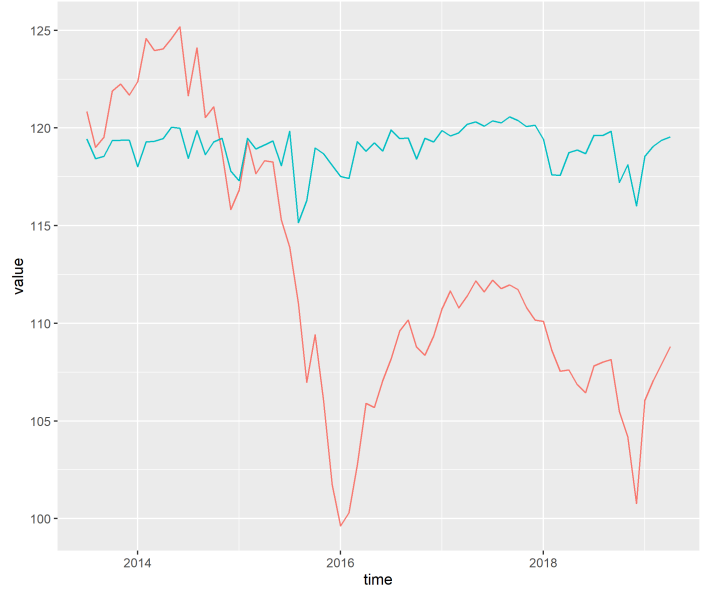


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

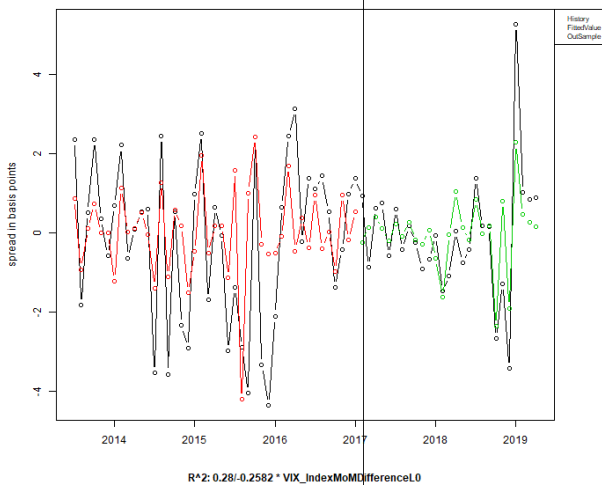
$R^2: 0.37 / -0.2864 * VIX_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

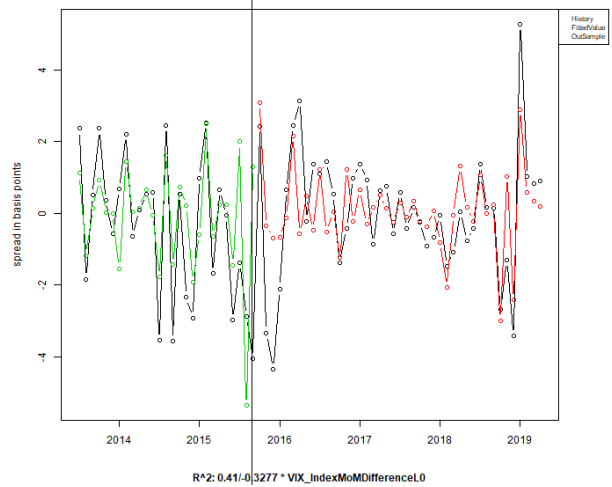
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

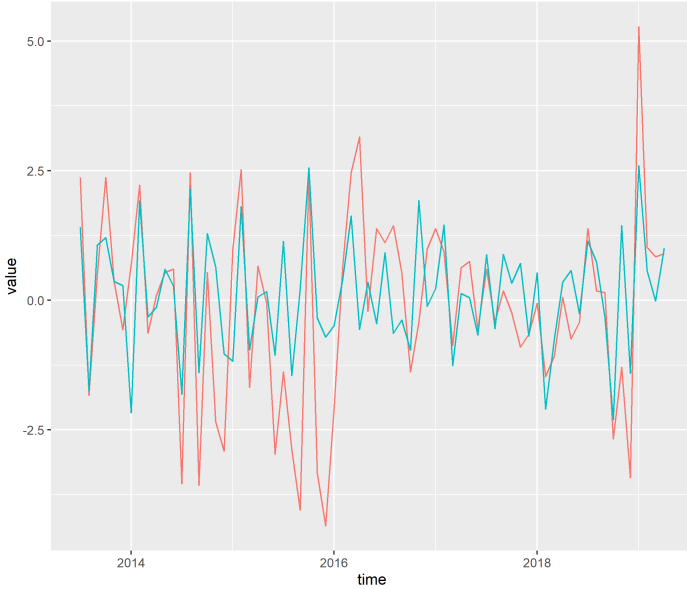


4.11.2 Model - 2

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.36 / 7.883 * SPX_Index_30DRSIMoMPercentChangeL0$

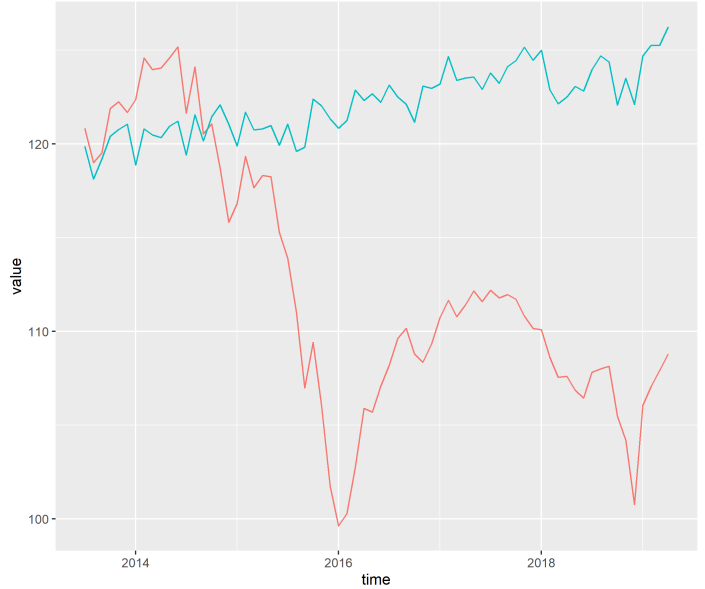


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

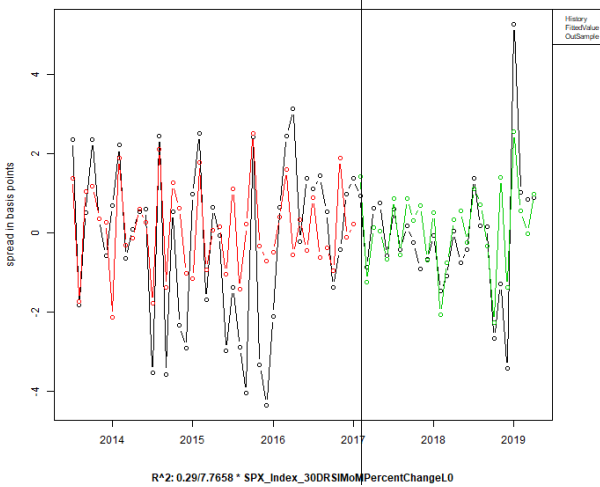
$R^2: 0.36 / 7.883 * SPX_Index_30DRSIMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

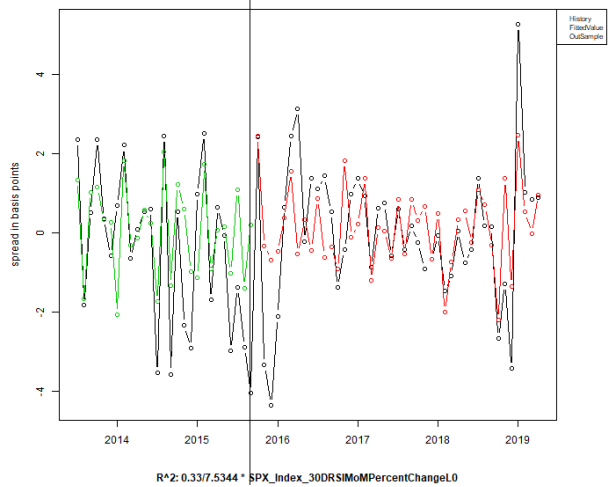
FI_HY Jul 2013-Apr 2019



$R^2: 0.29 / 7.7658 * SPX_Index_30DRSIMoMPercentChangeL0$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



$R^2: 0.33 / 7.5344 * SPX_Index_30DRSIMoMPercentChangeL0$

4.11.3 Model - 3

Backtesting on differenced values

FI_HY - Backtesting

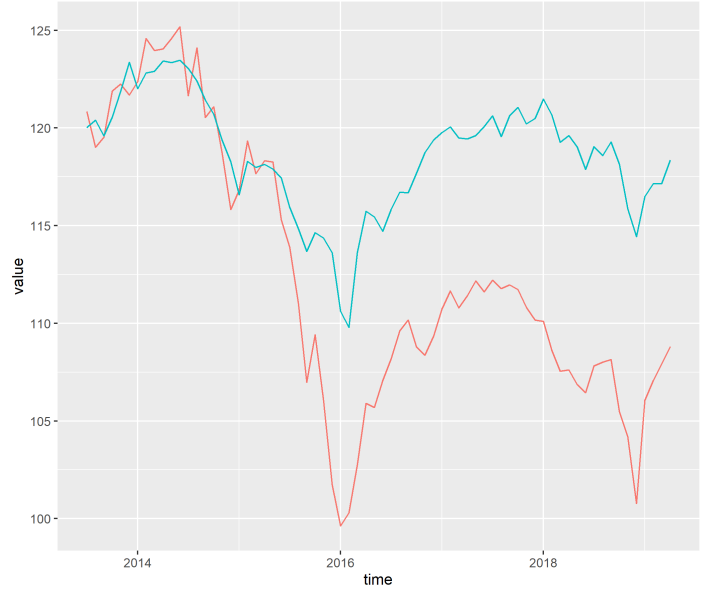
$R^2: 0.36 / -0.0774 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

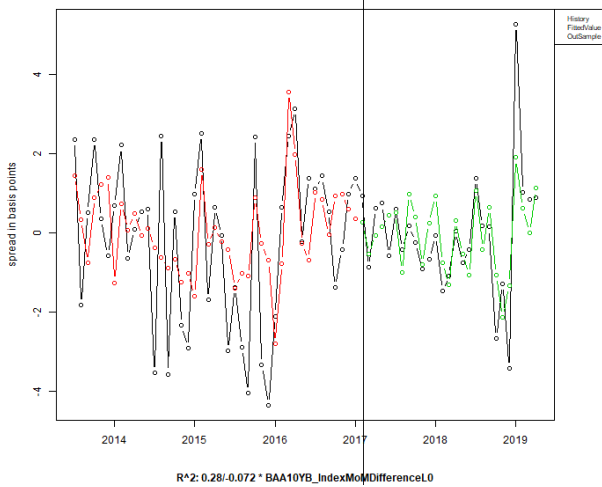
FI_HY - Level Value Backtesting

$R^2: 0.36 / -0.0774 * BAA10YB_IndexMoMDifferenceL0$



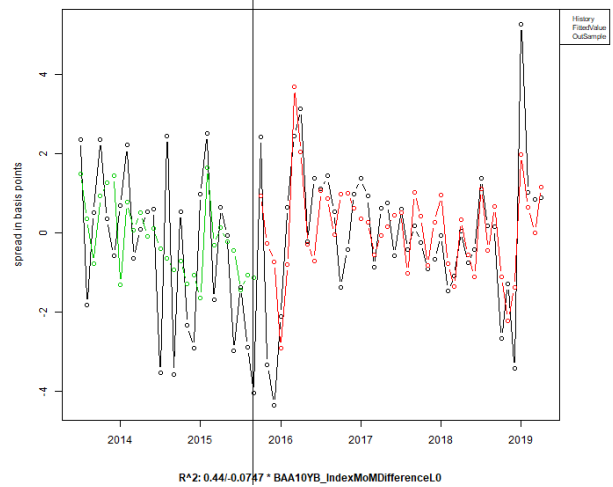
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

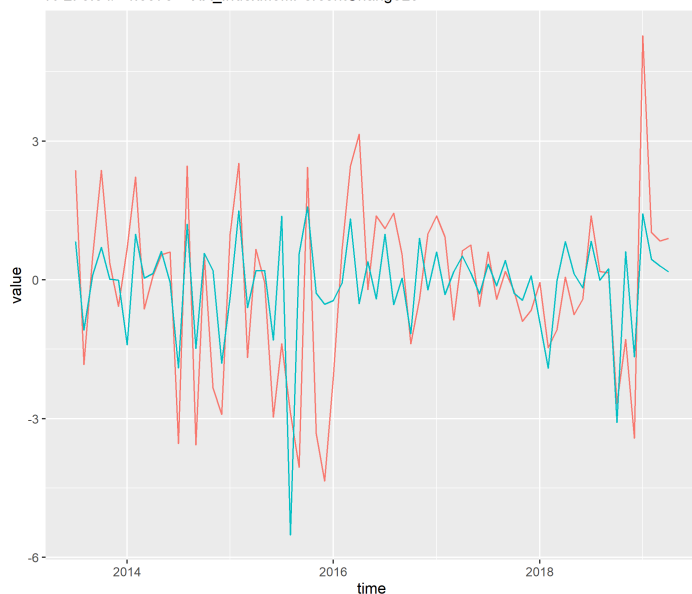
FI_HY Jul 2013-Apr 2019



4.11.4 Model - 4

Backtesting on differenced values

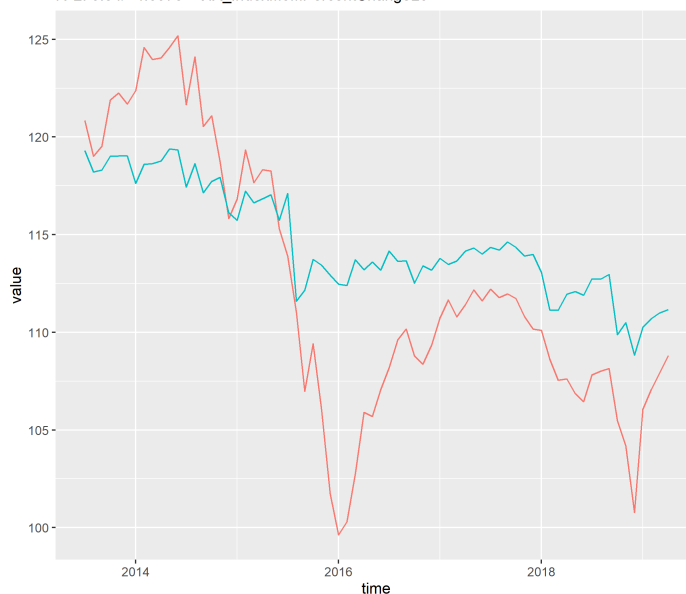
FI_HY - Backtesting

 $R^2: 0.34 / -4.0973 * VIX_IndexMoMPercentChangeL0$ 

variable FI_HY_History FI_HY_FittedValue

Backtesting on level values

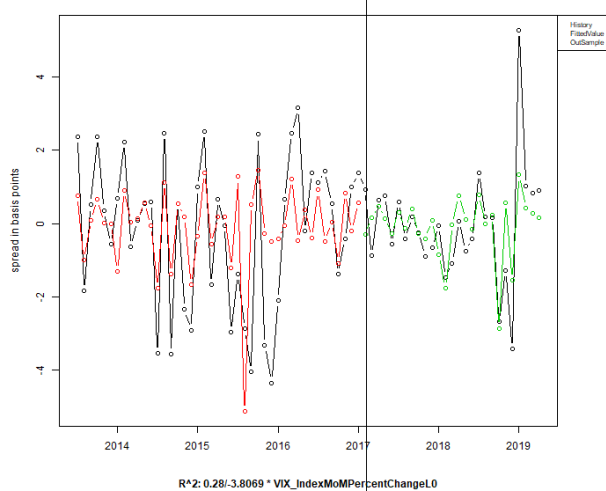
FI_HY - Level Value Backtesting

 $R^2: 0.34 / -4.0973 * VIX_IndexMoMPercentChangeL0$ 

variable diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

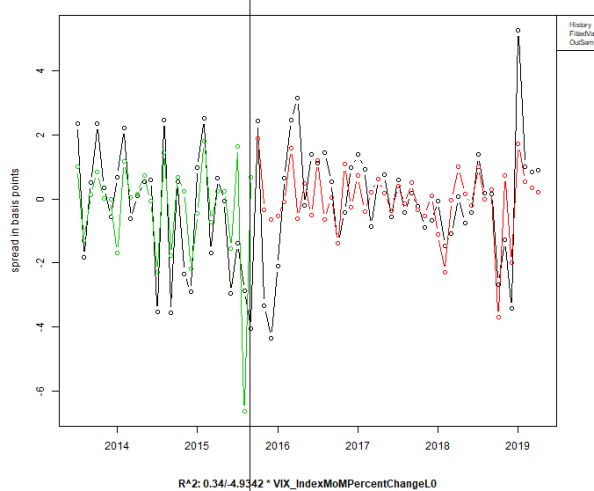
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

 $R^2: 0.28 / -3.8069 * VIX_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

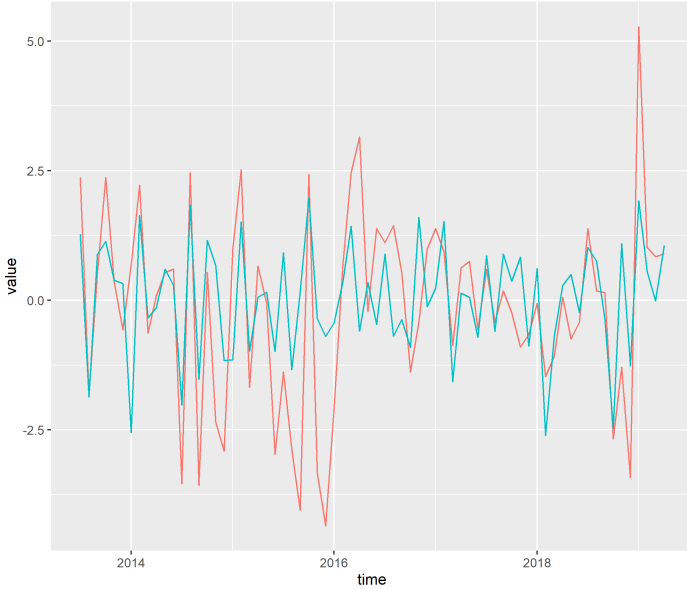
 $R^2: 0.34 / -4.9342 * VIX_IndexMoMPercentChangeL0$

4.11.5 Model - 5

Backtesting on differenced values

FI_HY - Backtesting

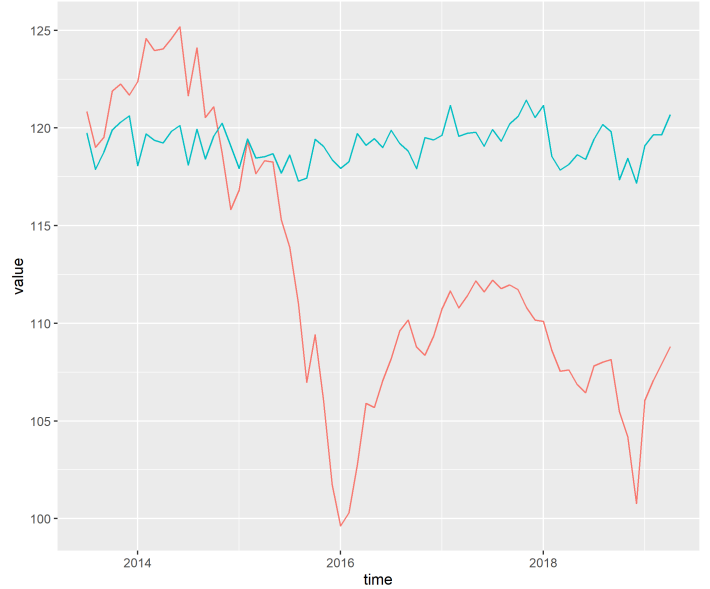
$R^2: 0.33 / 0.1425 * SPX_Index_30DRSIMoMDifferenceL0$



Backtesting on level values

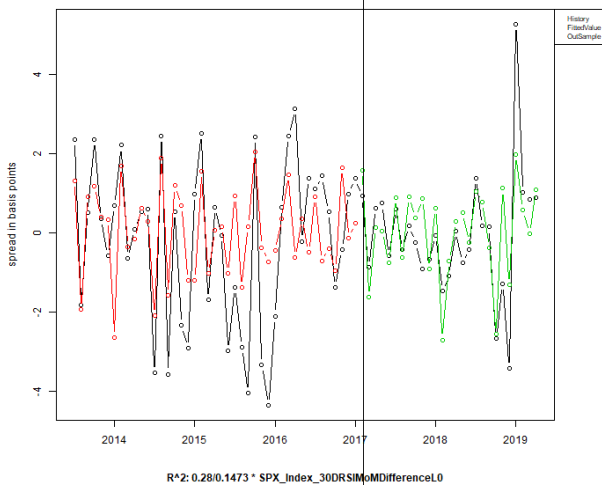
FI_HY - Level Value Backtesting

$R^2: 0.33 / 0.1425 * SPX_Index_30DRSIMoMDifferenceL0$



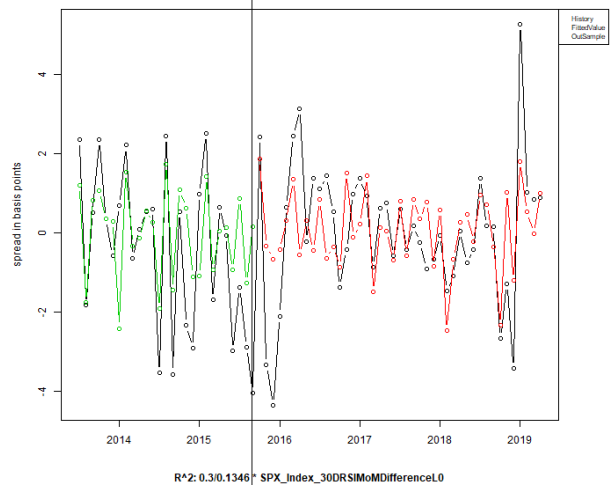
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



4.11.6 Model - 6

Backtesting on differenced values

FI_HY - Backtesting

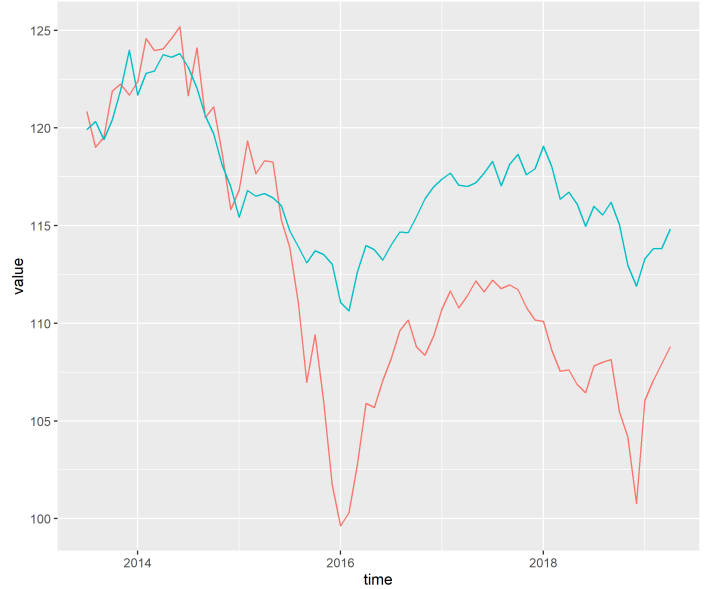
$R^2: 0.29 / -10.1009 * BAA10YB_IndexMoMPercentChangeL0$



Backtesting on level values

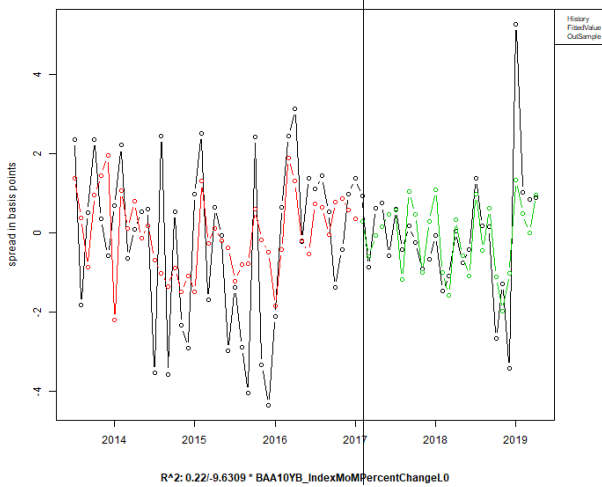
FI_HY - Level Value Backtesting

$R^2: 0.29 / -10.1009 * BAA10YB_IndexMoMPercentChangeL0$



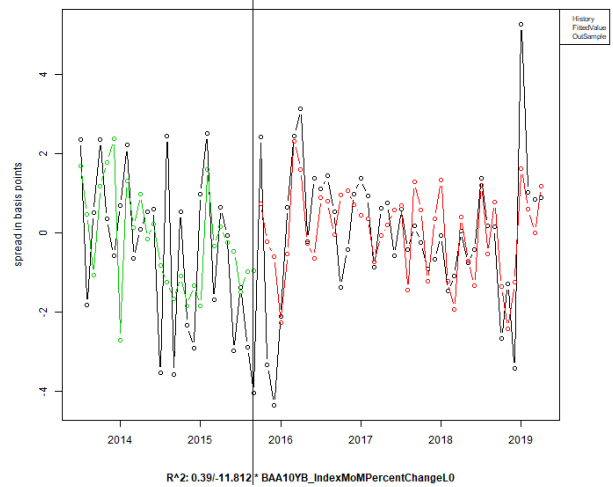
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



4.11.7 Model - 7

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.28 / 3.7277 * SPX_Index_14DRSIMoMPercentChangeL0$

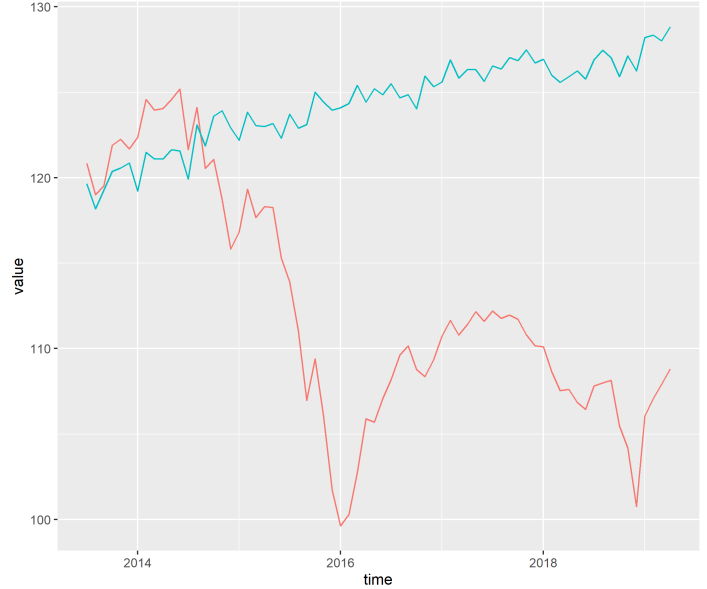


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

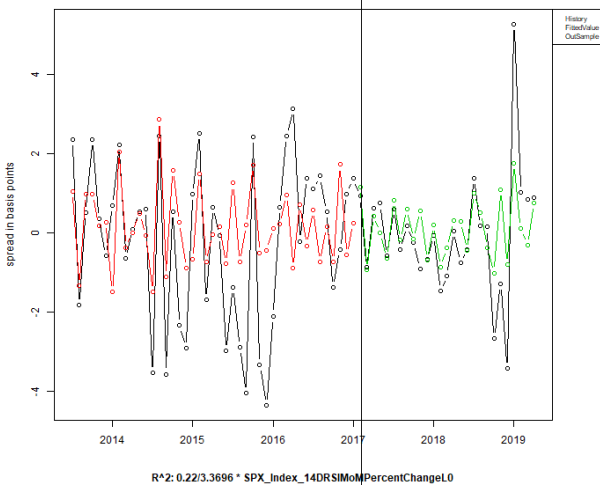
$R^2: 0.28 / 3.7277 * SPX_Index_14DRSIMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

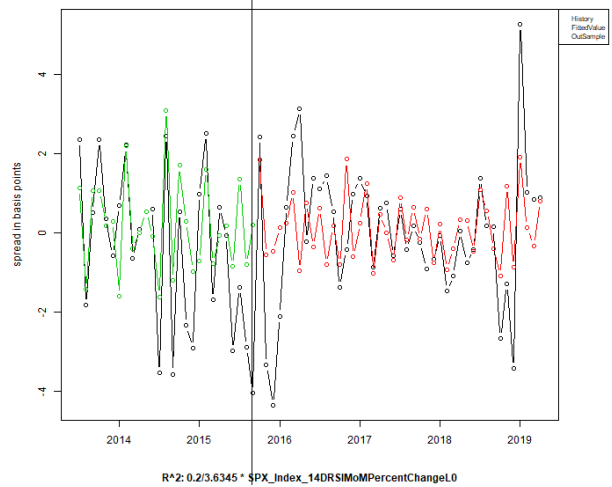
FI_HY Jul 2013-Apr 2019



$R^2: 0.22 / 3.3696 * SPX_Index_14DRSIMoMPercentChangeL0$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



$R^2: 0.2 / 3.6345 * SPX_Index_14DRSIMoMPercentChangeL0$

4.11.8 Model - 8

Backtesting on differenced values

FI_HY - Backtesting

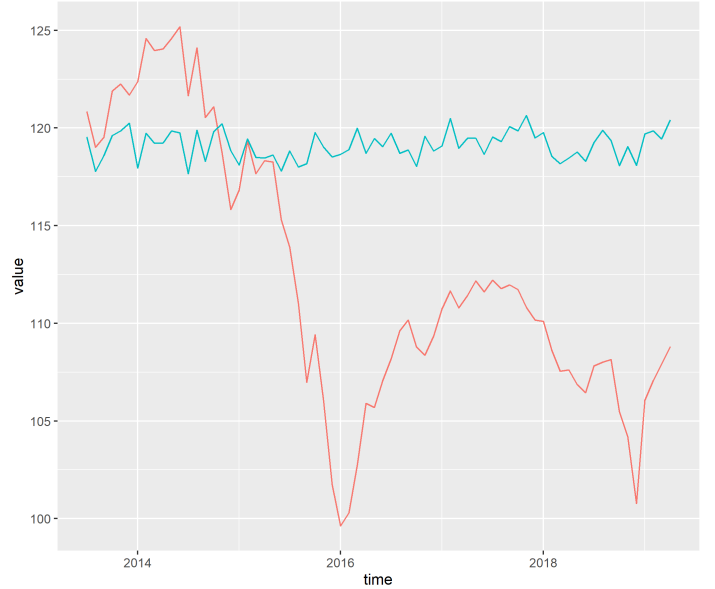
$R^2: 0.27 / 0.073 * SPX_Index_14DRSIMoMDifferenceL0$



Backtesting on level values

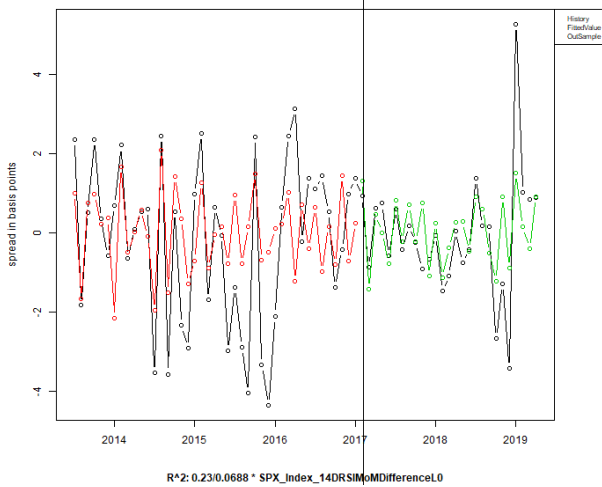
FI_HY - Level Value Backtesting

$R^2: 0.27 / 0.073 * SPX_Index_14DRSIMoMDifferenceL0$



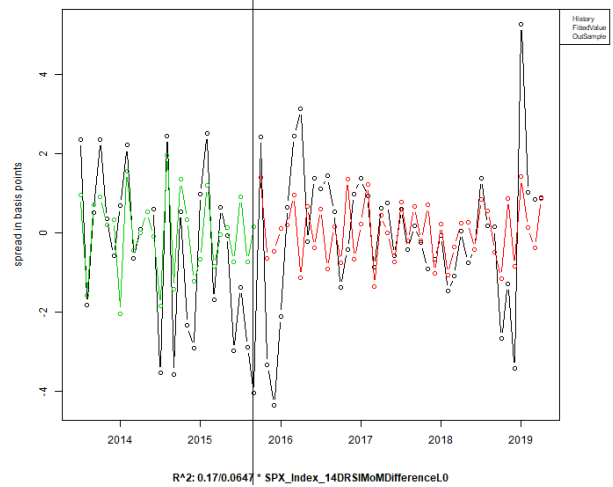
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

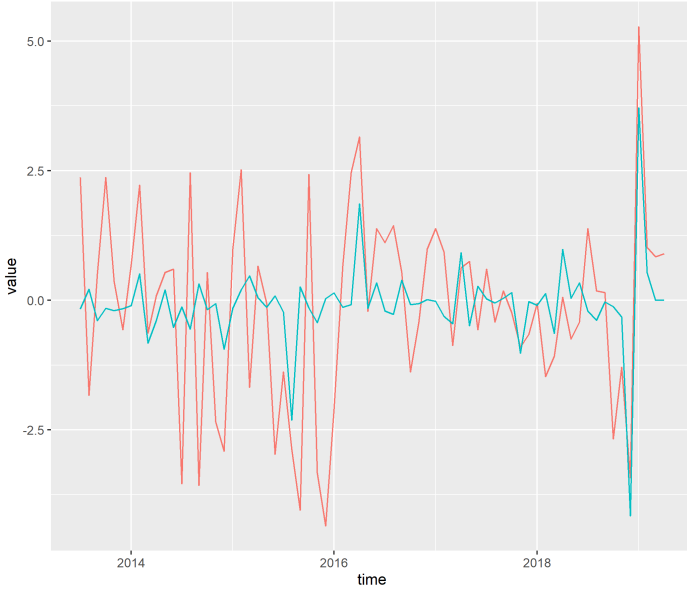


4.11.9 Model - 9

Backtesting on differenced values

FI_HY - Backtesting

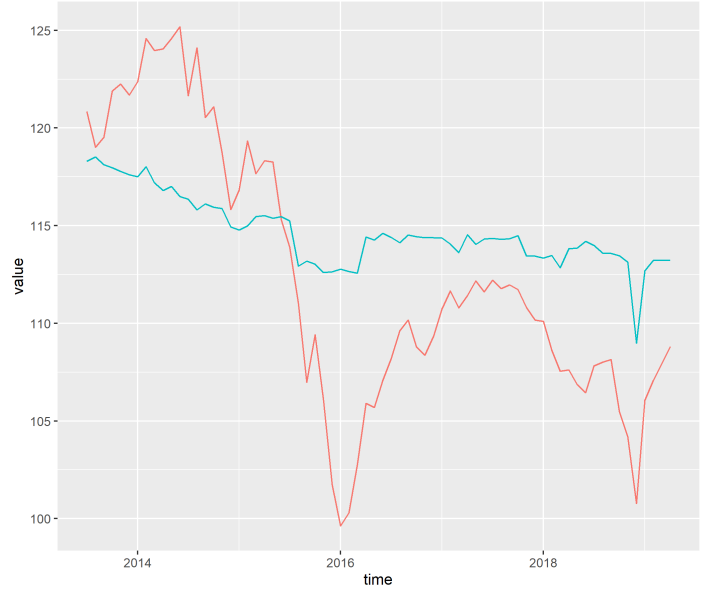
$R^2: 0.19 / -0.5726 * IPMANSICSMoMDifferenceL1$



Backtesting on level values

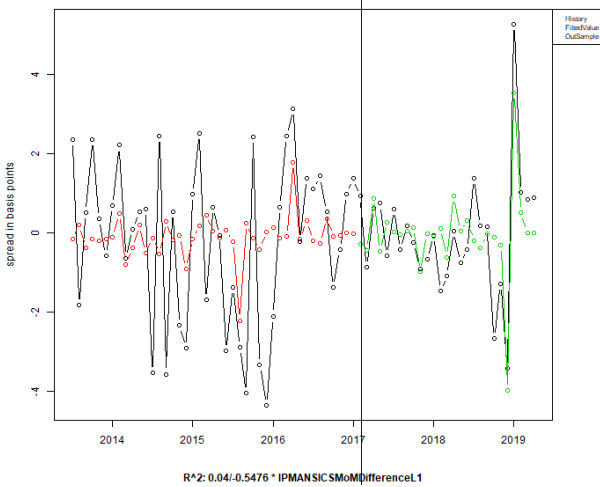
FI_HY - Level Value Backtesting

$R^2: 0.19 / -0.5726 * IPMANSICSMoMDifferenceL1$



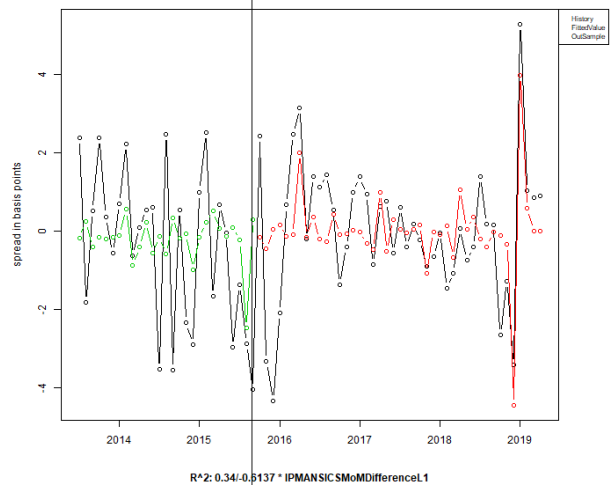
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

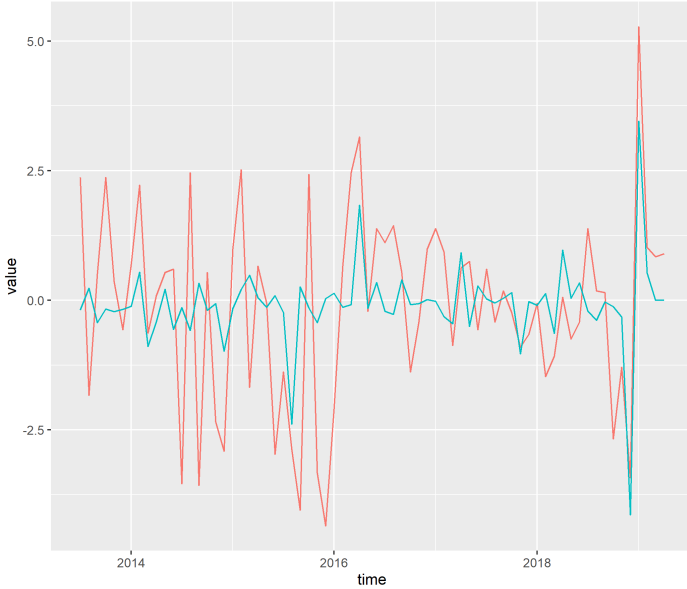


4.11.10 Model - 10

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.18 / -60.053 * IPMANSICSMoMPercentChangeL1$

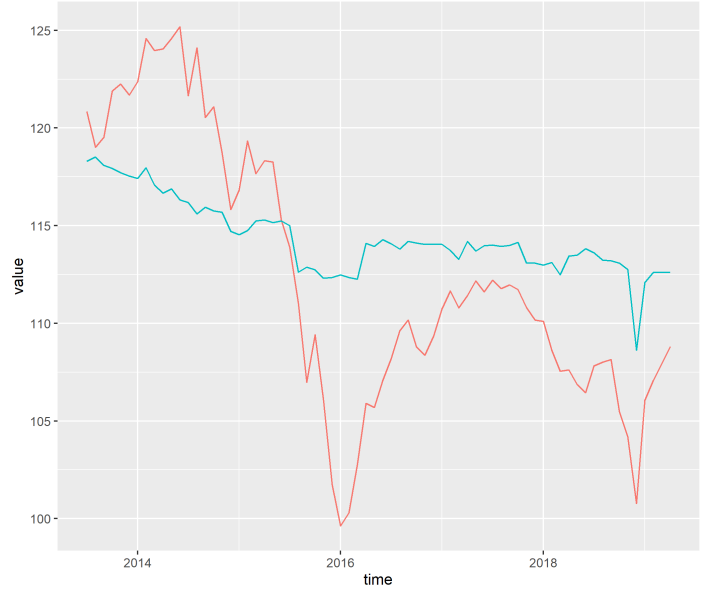


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

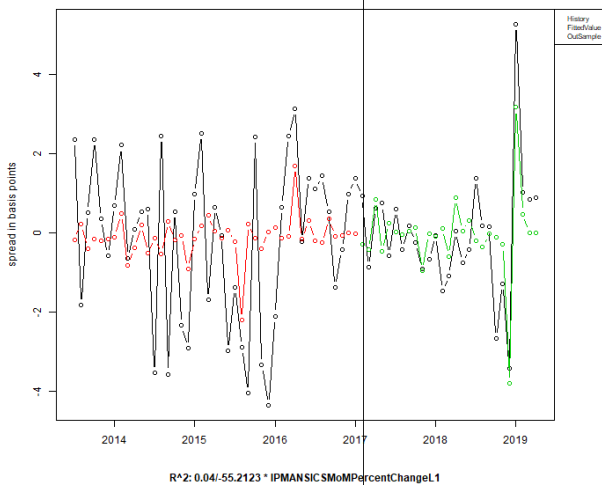
$R^2: 0.18 / -60.053 * IPMANSICSMoMPercentChangeL1$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

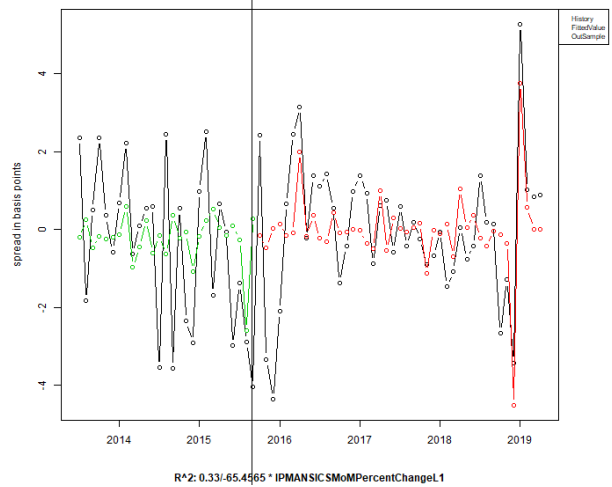
FI_HY Jul 2013-Apr 2019



$R^2: 0.04 / -55.2123 * IPMANSICSMoMPercentChangeL1$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



$R^2: 0.33 / -65.4565 * IPMANSICSMoMPercentChangeL1$

4.11.11 Model - 11

Backtesting on differenced values

FI_HY - Backtesting

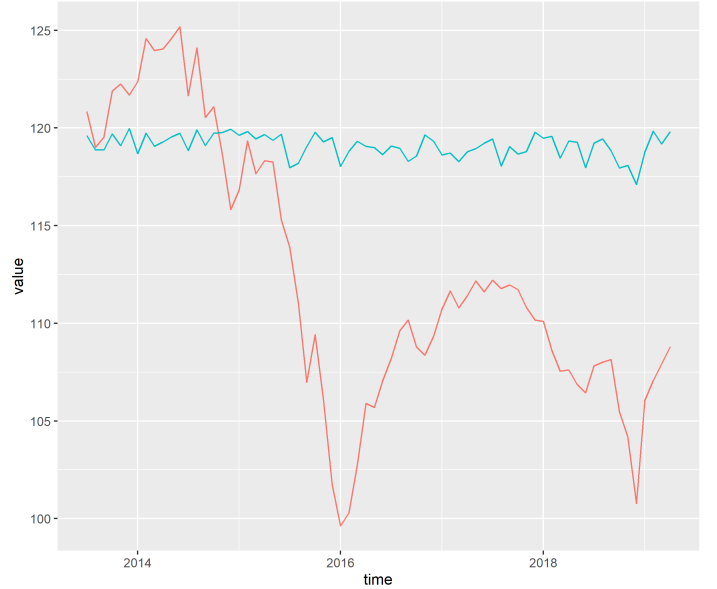
$R^2: 0.14 / -0.0901 * \text{AAIBEAR_IndexMoMDifferenceL0}$



Backtesting on level values

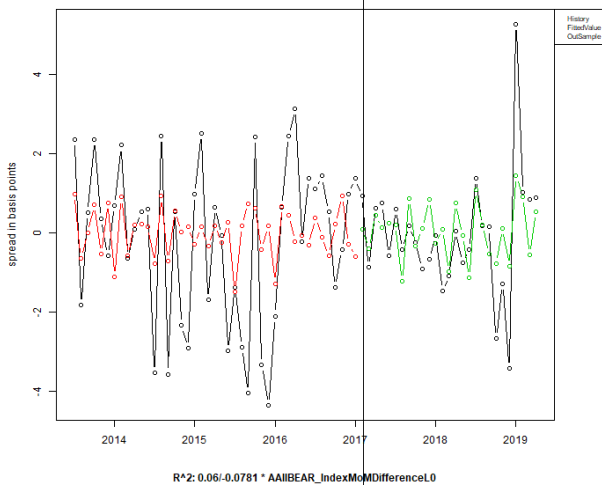
FI_HY - Level Value Backtesting

$R^2: 0.14 / -0.0901 * \text{AAIBEAR_IndexMoMDifferenceL0}$



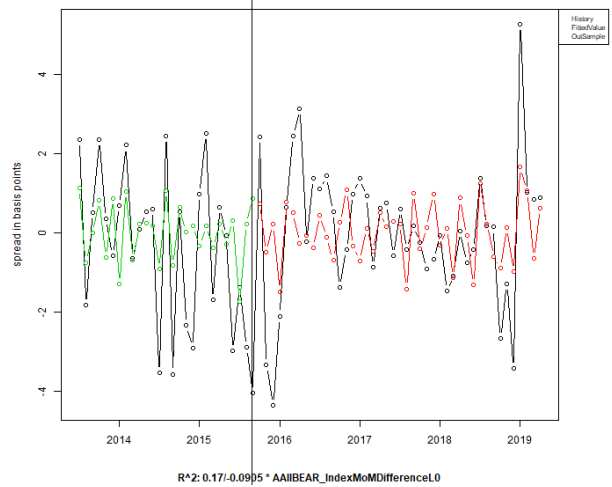
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

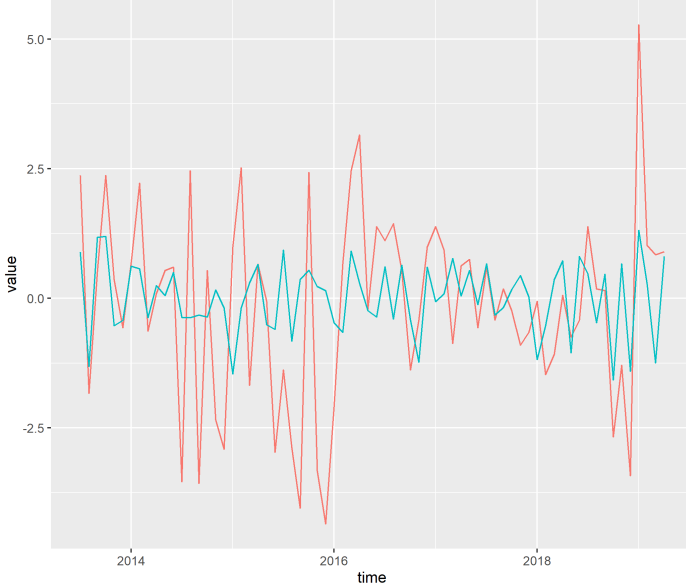


4.11.12 Model - 12

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.12 / -5.2004 * \text{MOVE_IndexMoMPercentChangeL0}$

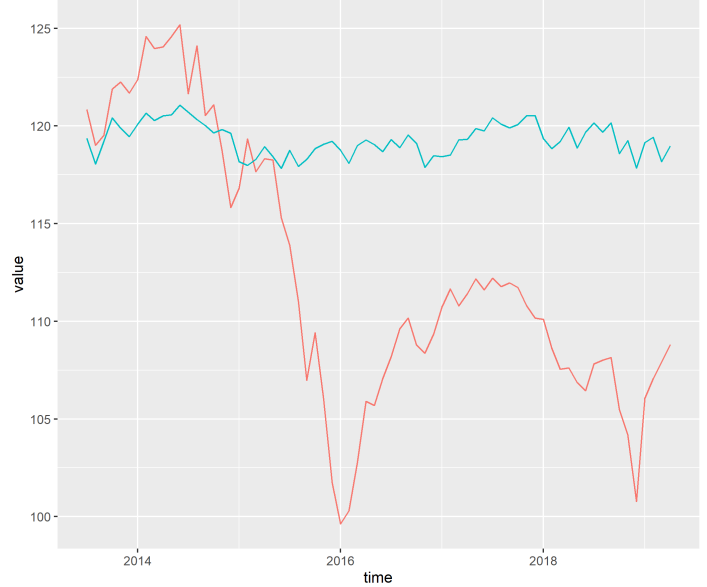


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

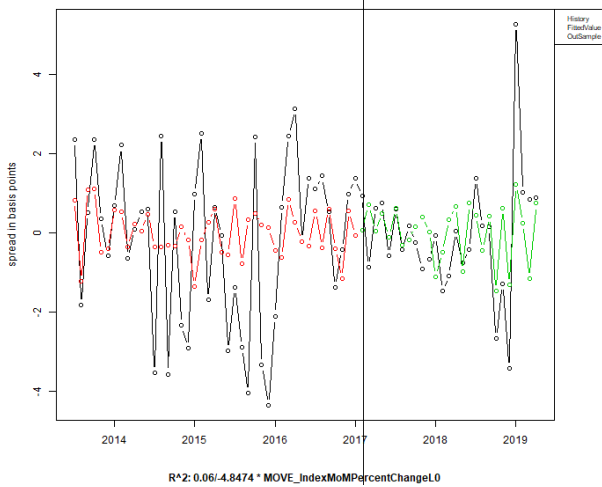
$R^2: 0.12 / -5.2004 * \text{MOVE_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

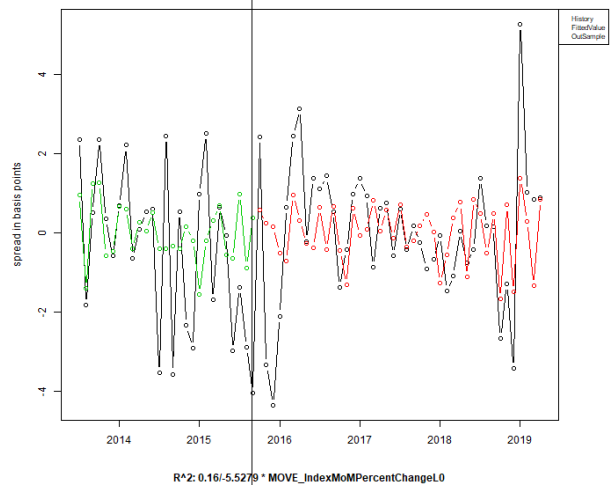
FI_HY Jul 2013-Apr 2019



$R^2: 0.06 / -4.8474 * \text{MOVE_IndexMoMPercentChangeL0}$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



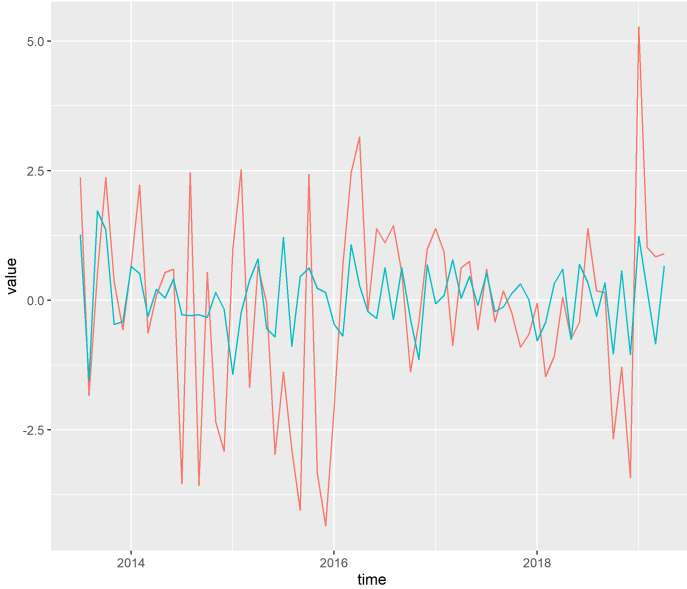
$R^2: 0.16 / -5.5279 * \text{MOVE_IndexMoMPercentChangeL0}$

4.11.13 Model - 13

Backtesting on differenced values

FI_HY - Backtesting

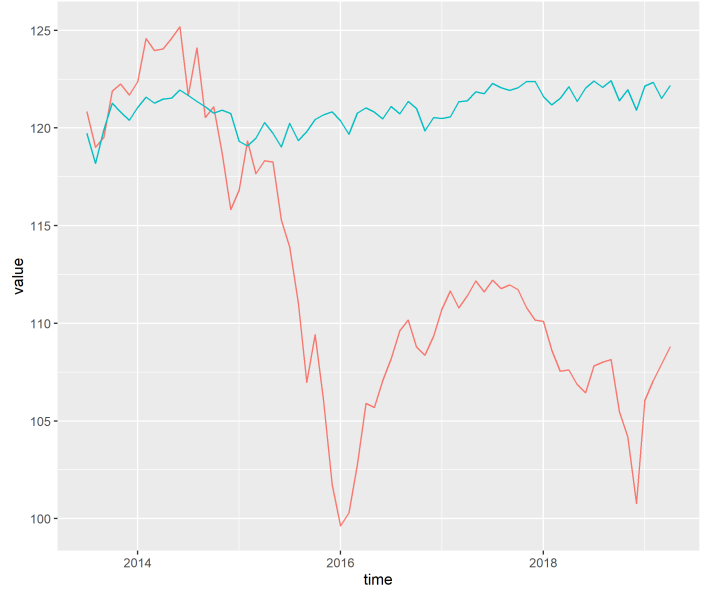
$R^2: 0.12 / -0.0738 * \text{MOVE_IndexMoMDifferenceL0}$



Backtesting on level values

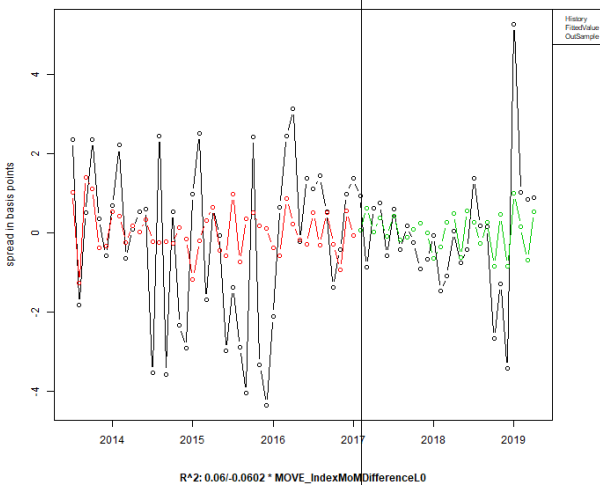
FI_HY - Level Value Backtesting

$R^2: 0.12 / -0.0738 * \text{MOVE_IndexMoMDifferenceL0}$



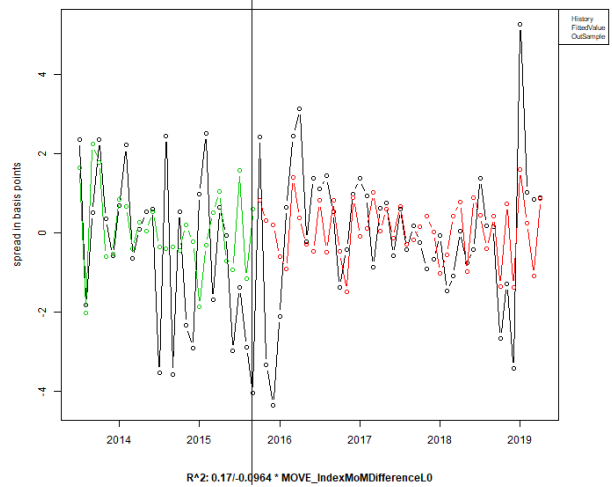
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

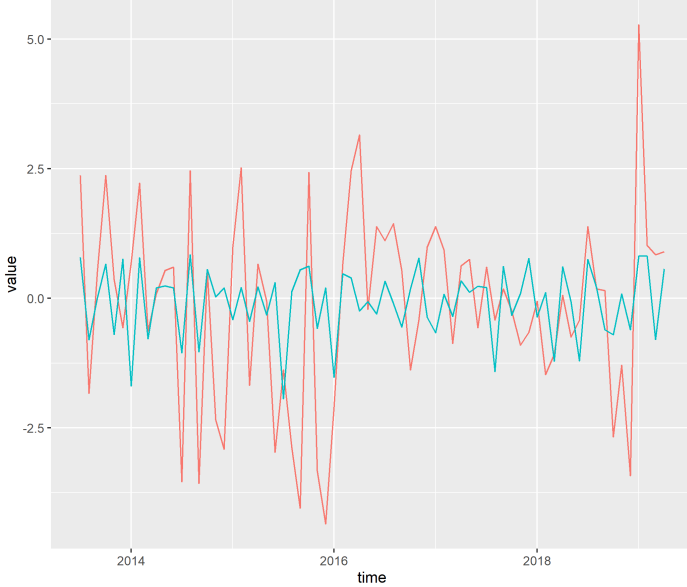


4.11.14 Model - 14

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.12 / -2.2056 * \text{AAIBEAR_IndexMoMPercentChangeL0}$

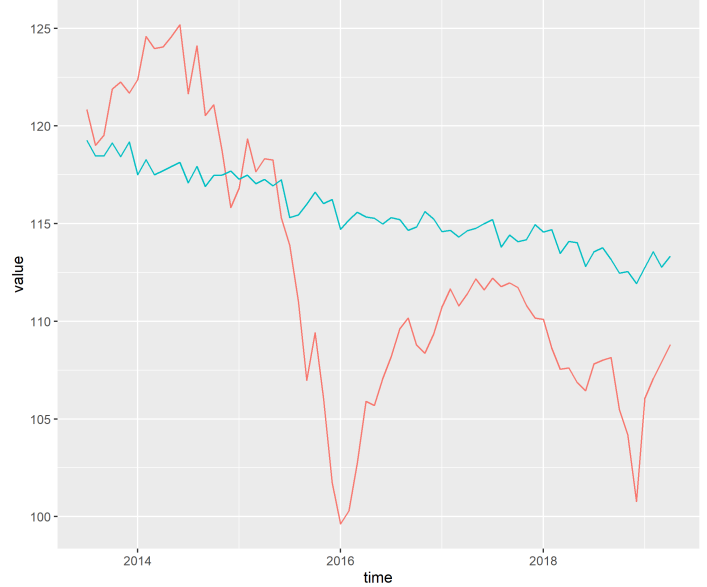


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

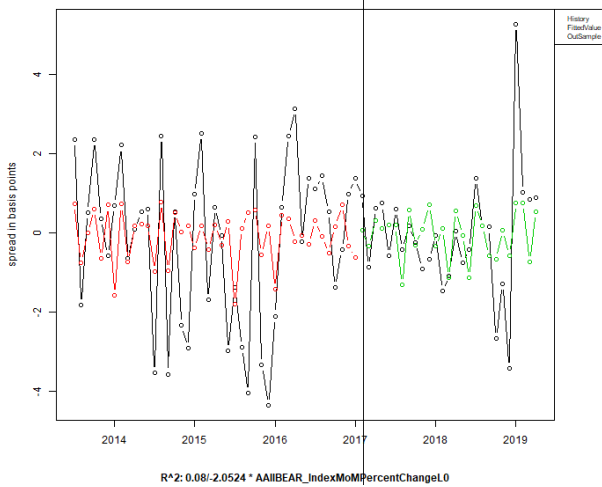
$R^2: 0.12 / -2.2056 * \text{AAIBEAR_IndexMoMPercentChangeL0}$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

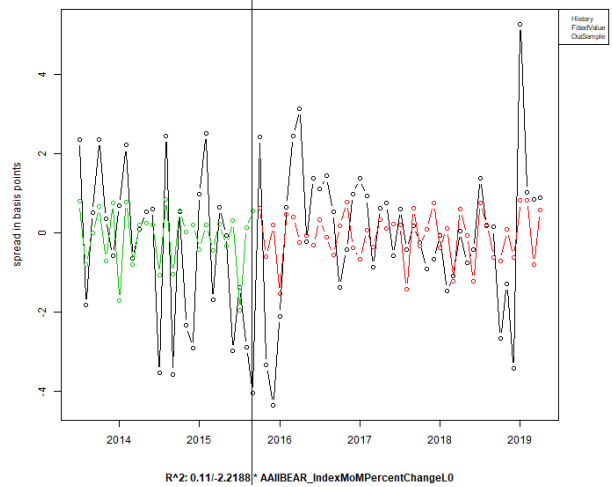
FI_HY Jul 2013-Apr 2019



$R^2: 0.08 / -2.0524 * \text{AAIBEAR_IndexMoMPercentChangeL0}$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



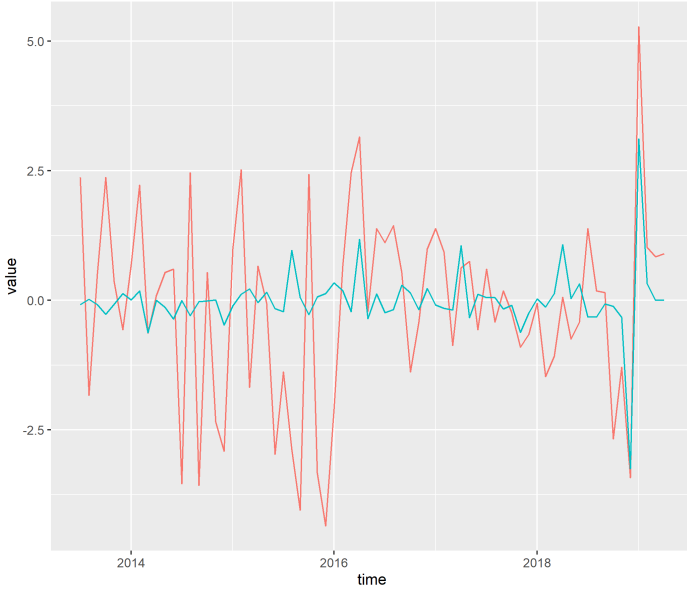
$R^2: 0.11 / -2.2188 * \text{AAIBEAR_IndexMoMPercentChangeL0}$

4.11.15 Model - 15

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0.1 / -0.2715 * IPBUSEQMoMDifferenceL1$

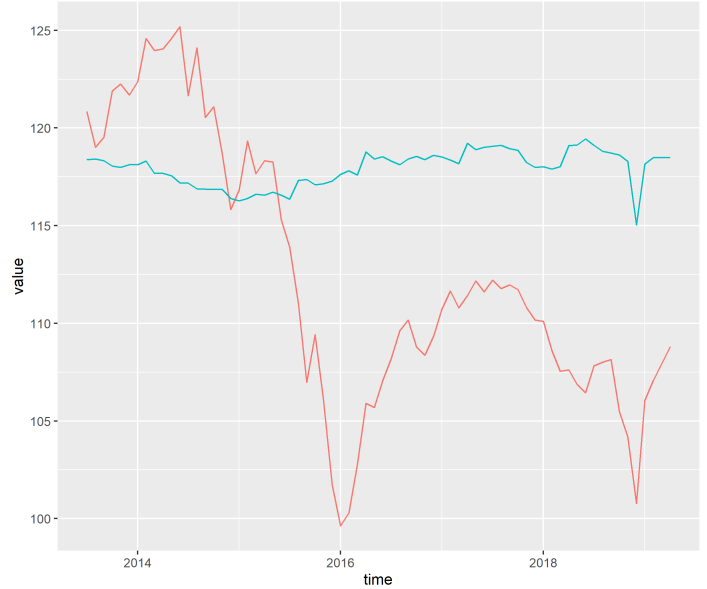


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

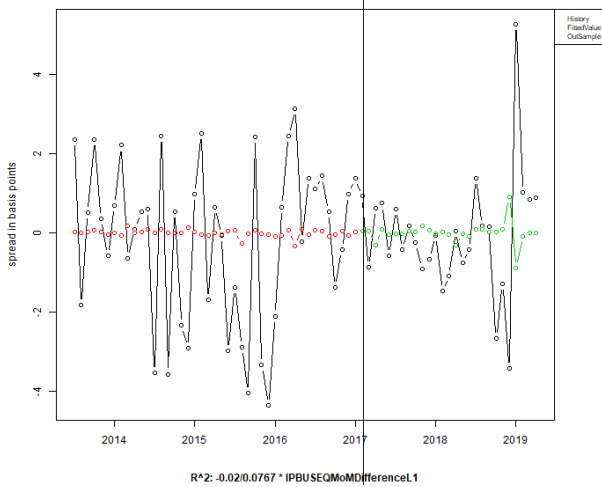
$R^2: 0.1 / -0.2715 * IPBUSEQMoMDifferenceL1$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

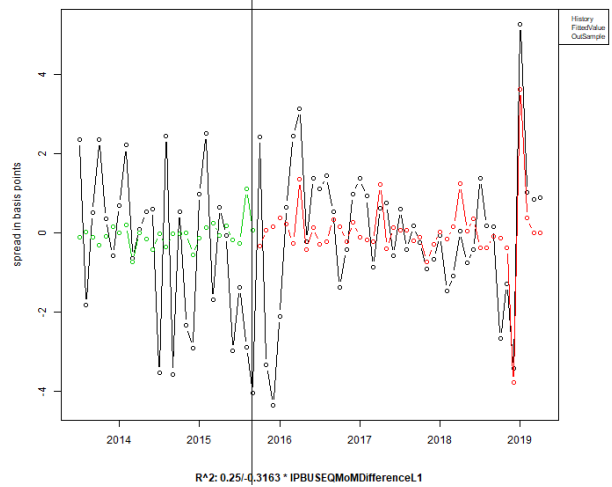
FI_HY Jul 2013-Apr 2019



$R^2: -0.02 / 0.0767 * IPBUSEQMoMDifferenceL1$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



$R^2: 0.25 / 0.3163 * IPBUSEQMoMDifferenceL1$

4.11.16 Model - 16

Backtesting on differenced values

FI_HY - Backtesting

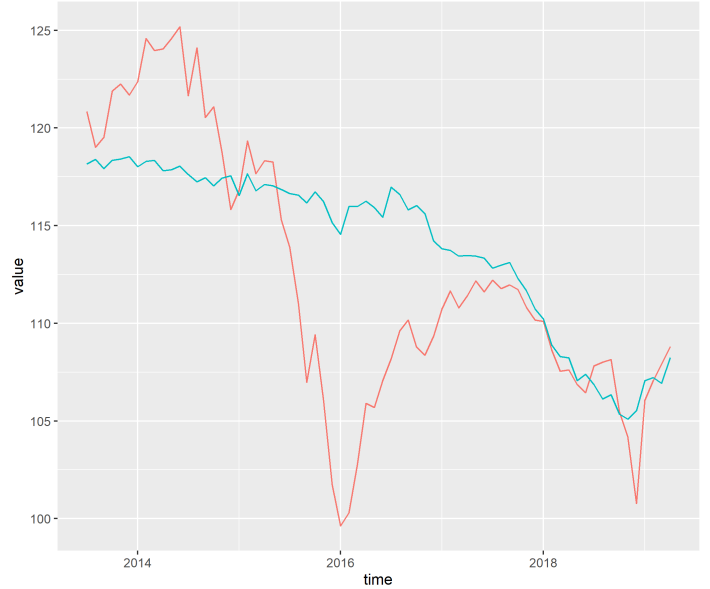
$R^2: 0.1 / -5.1943 * USGG2YR_IndexMoMDifferenceL1$



Backtesting on level values

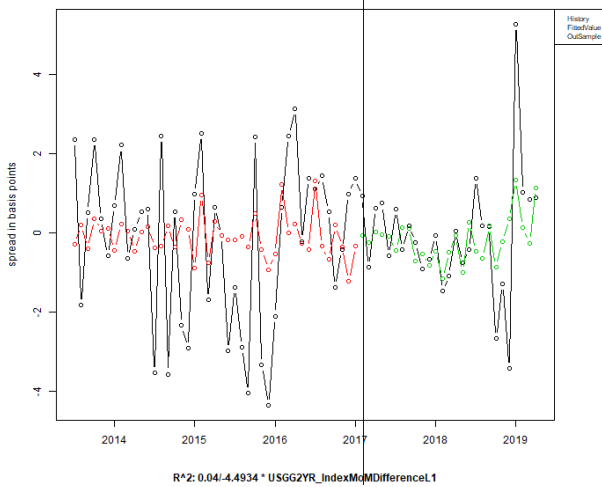
FI_HY - Level Value Backtesting

$R^2: 0.1 / -5.1943 * USGG2YR_IndexMoMDifferenceL1$



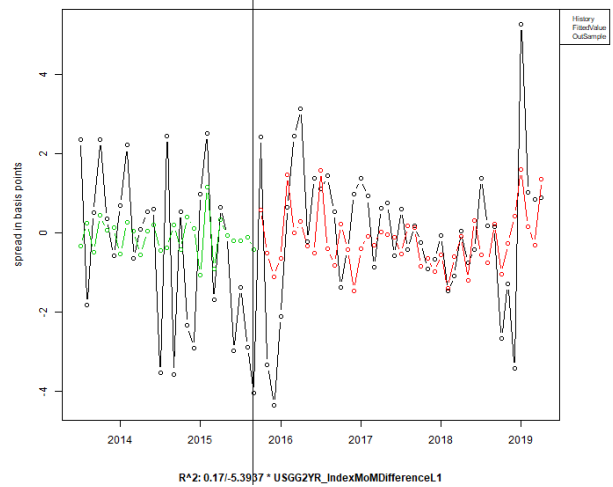
Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



Out-of-Sample testing

FI_HY Jul 2013-Apr 2019

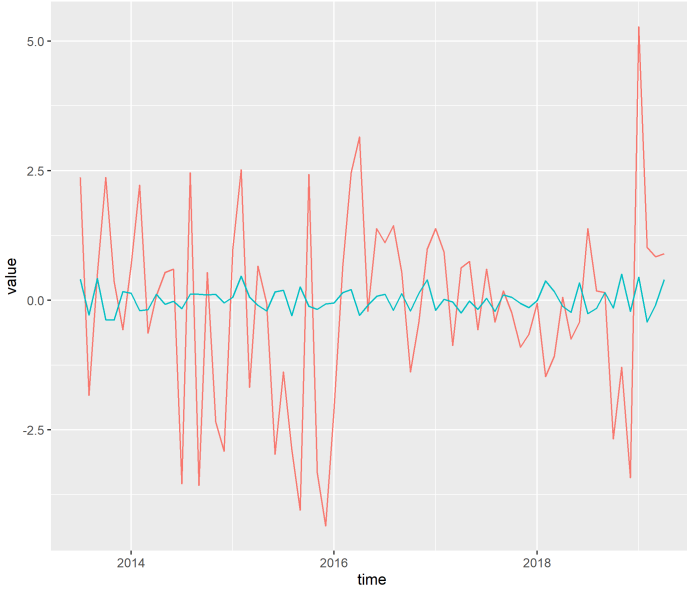


4.11.17 Model - 17

Backtesting on differenced values

FI_HY - Backtesting

$R^2: 0 / 1.6605 * \text{MOVE_IndexMoMPercentChangeL1}$

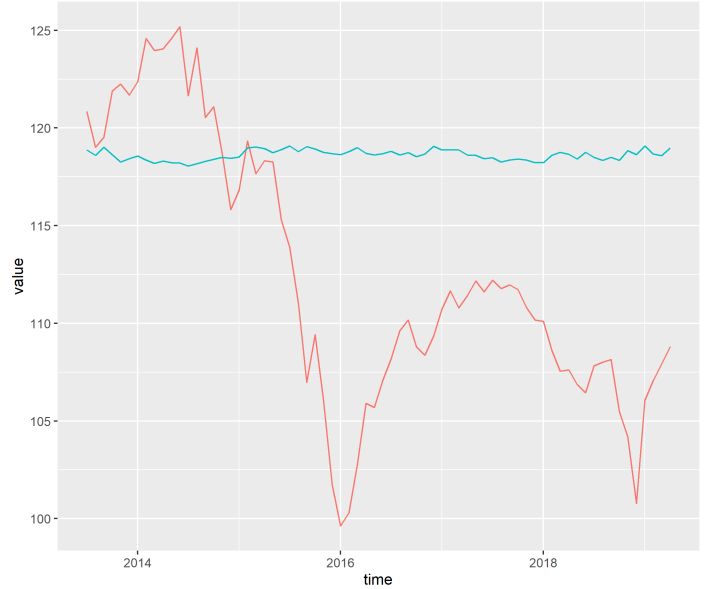


variable — FI_HY_History — FI_HY_FittedValue

Backtesting on level values

FI_HY - Level Value Backtesting

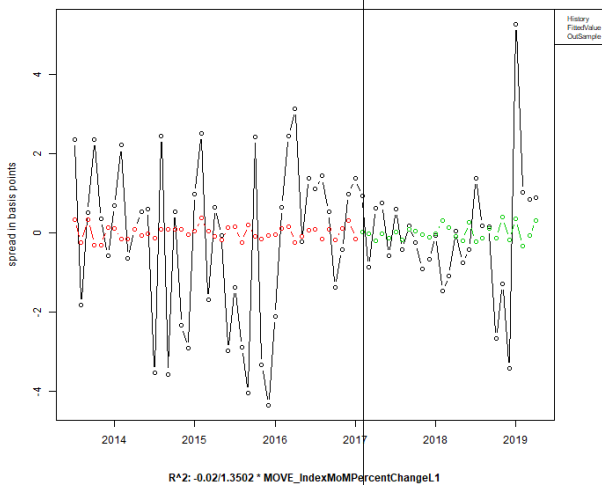
$R^2: 0 / 1.6605 * \text{MOVE_IndexMoMPercentChangeL1}$



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

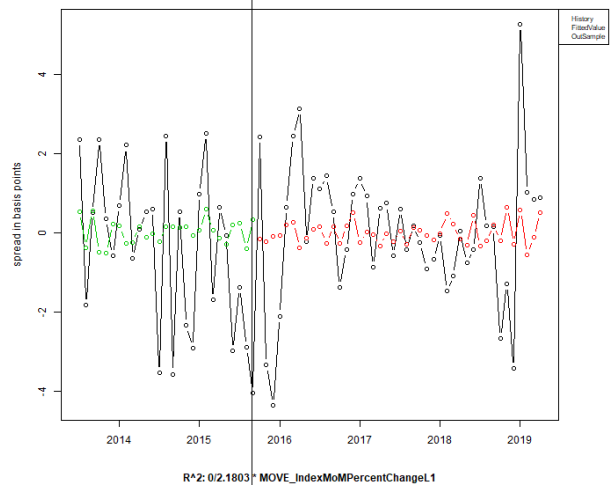
FI_HY Jul 2013-Apr 2019



$R^2: 0.02 / 1.3502 * \text{MOVE_IndexMoMPercentChangeL1}$

Out-of-Sample testing

FI_HY Jul 2013-Apr 2019



$R^2: 0.02 / 1.803 * \text{MOVE_IndexMoMPercentChangeL1}$

4.12 Two Factor Model for $FI_H Y$

4.13 Single Factor Model for $FI_{d^{eved_i}ntl}$

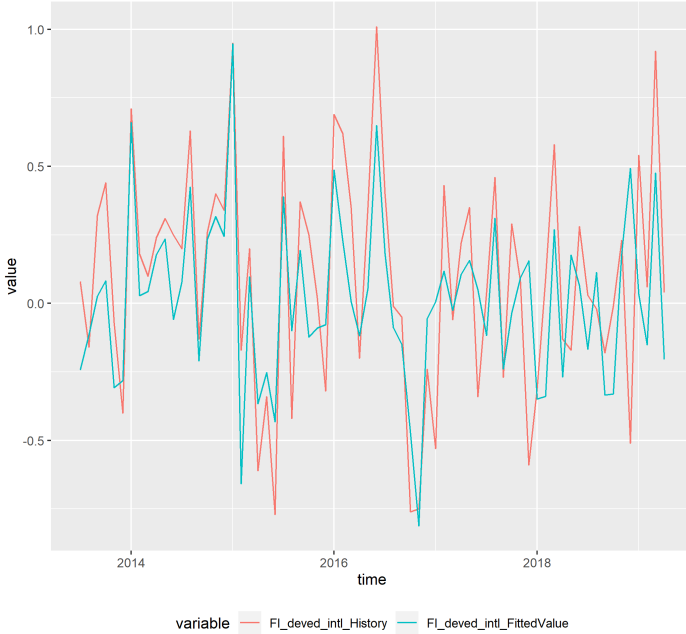
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG30YR_IndexMoMDifferenceL0	0.52	26.61	-1.79	0.00	0.36			
2	USGG30YR_IndexMoMPercentChangeL0	0.51	28.08	-5.00	0.00	0.35			
3	USGG10YR_IndexMoMPercentChangeL0	0.51	28.52	-3.53	0.00	0.31			
4	USGG10YR_IndexMoMDifferenceL0	0.50	29.12	-1.61	0.00	0.33			
5	USGG2YR_IndexMoMDifferenceL0	0.16	65.99	-1.42	0.00	0.40			
6	USGG2YR_IndexMoMPercentChangeL0	0.15	67.10	-1.08	0.00	0.34			
7	ISRATIOMoMPercentChangeL1	0.12	69.31	21.28	0.00	0.36			
8	ISRATIOMoMDifferenceL1	0.12	69.36	15.77	0.00	0.37			
9	BAA10YB_IndexMoMPercentChangeL0	0.12	69.41	1.50	0.00	0.36			
10	WPSID62MoMDifferenceL0	0.11	69.93	-0.02	0.00	0.32			
11	WPSID62MoMPercentChangeL0	0.10	70.83	-4.06	0.00	0.32			
12	M2REALMoMPercentChangeL0	0.10	70.91	28.48	0.00	0.36			

Table 9: Single Factor Analysis- $FI_{d^{eved_i}ntl}$

4.13.1 Model - 1

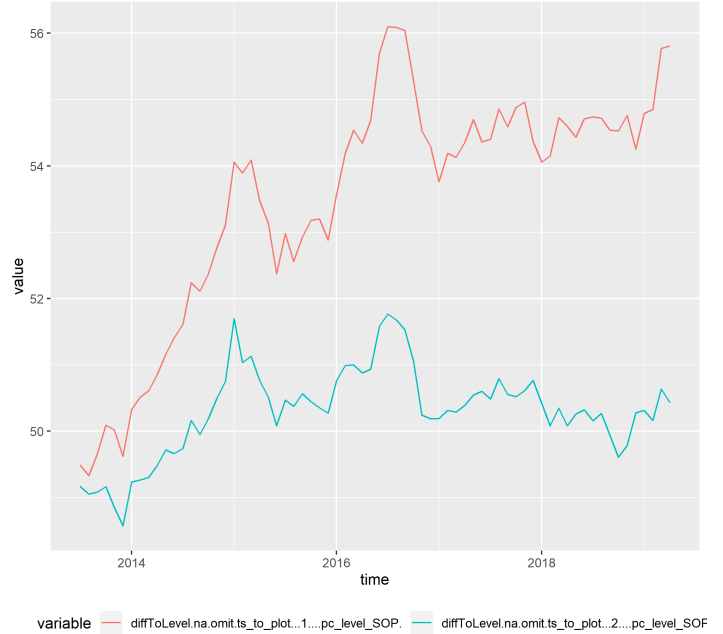
Backtesting on differenced values

FI_deved_intl - Backtesting

 $R^2: 0.52 / -1.7882 * USGG30YR_IndexMoMDifferenceL0$ 

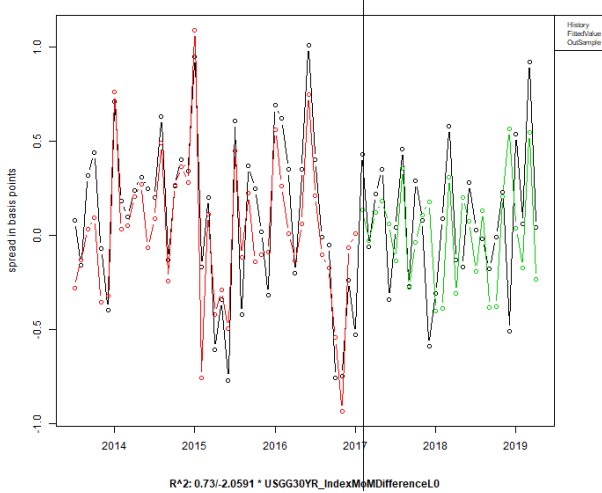
Backtesting on level values

FI_deved_intl - Level Value Backtesting

 $R^2: 0.52 / -1.7882 * USGG30YR_IndexMoMDifferenceL0$ 

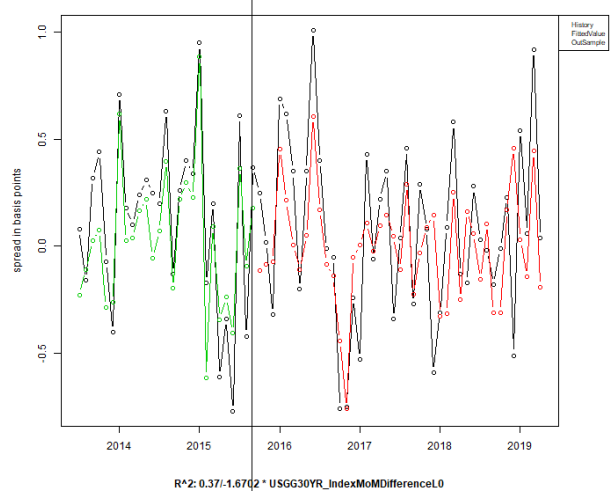
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



4.13.2 Model - 2

Backtesting on differenced values

FI_deved_intl - Backtesting

$R^2: 0.51 / -5.0044 * USGG30YR_IndexMoMPercentChangeL0$

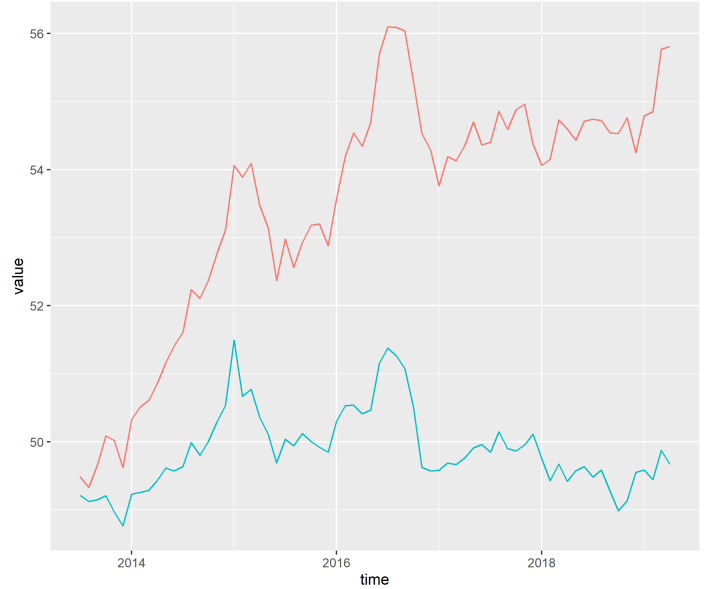


variable — FI_deved_intl_History — FI_deved_intl_FittedValue

Backtesting on level values

FI_deved_intl - Level Value Backtesting

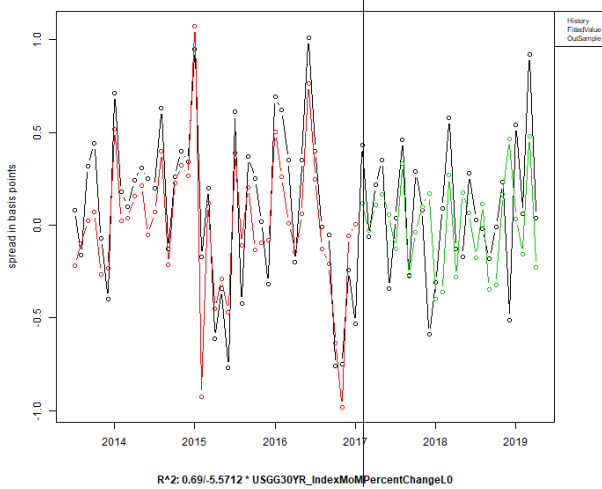
$R^2: 0.51 / -5.0044 * USGG30YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

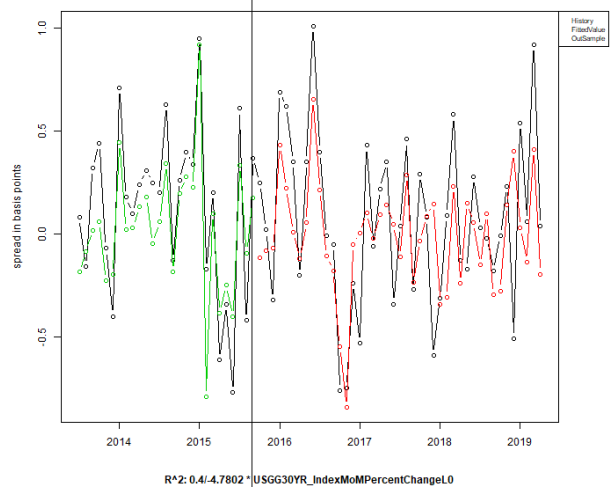
FI_deved_intl Jul 2013-Apr 2019



$R^2: 0.69 / -5.5712 * USGG30YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



$R^2: 0.4 / -4.7802 * USGG30YR_IndexMoMPercentChangeL0$

4.13.3 Model - 3

Backtesting on differenced values

FI_deved_intl - Backtesting

$R^2: 0.51 / -3.5295 * USGG10YR_IndexMoMPercentChangeL0$

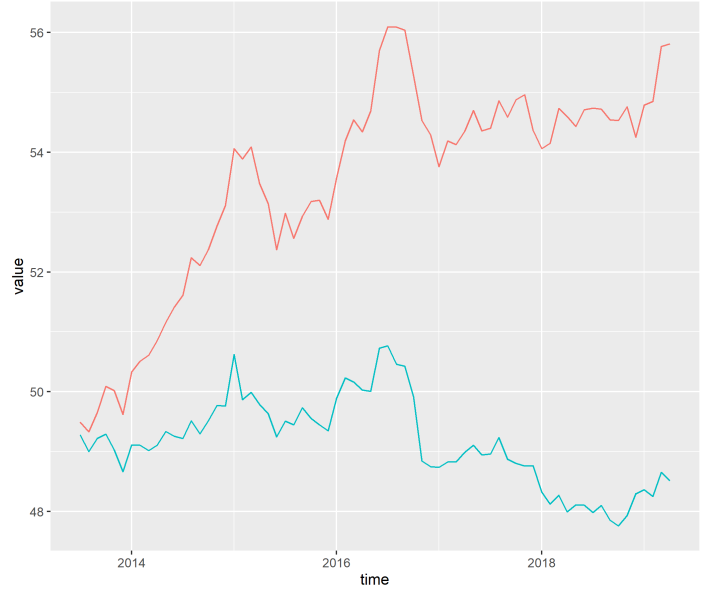


variable — FI_deved_intl_History — FI_deved_intl_FittedValue

Backtesting on level values

FI_deved_intl - Level Value Backtesting

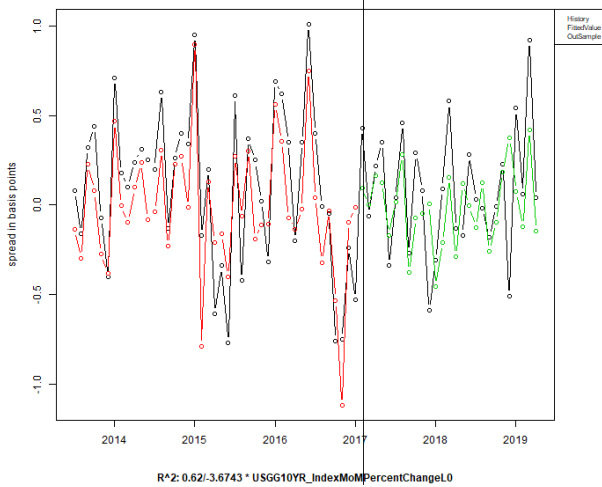
$R^2: 0.51 / -3.5295 * USGG10YR_IndexMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

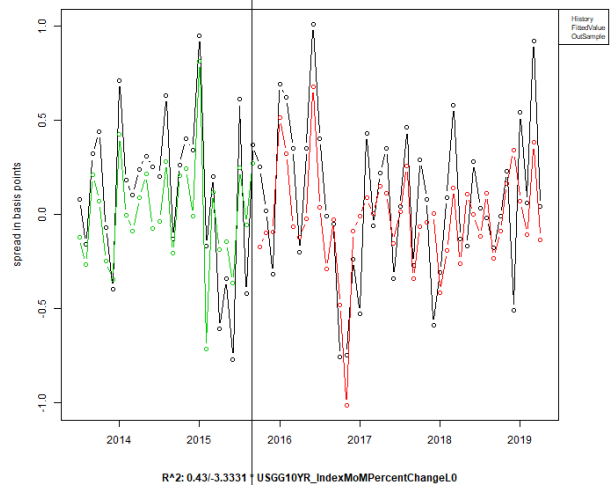
FI_deved_intl Jul 2013-Apr 2019



$R^2: 0.62 / -3.6743 * USGG10YR_IndexMoMPercentChangeL0$

Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019

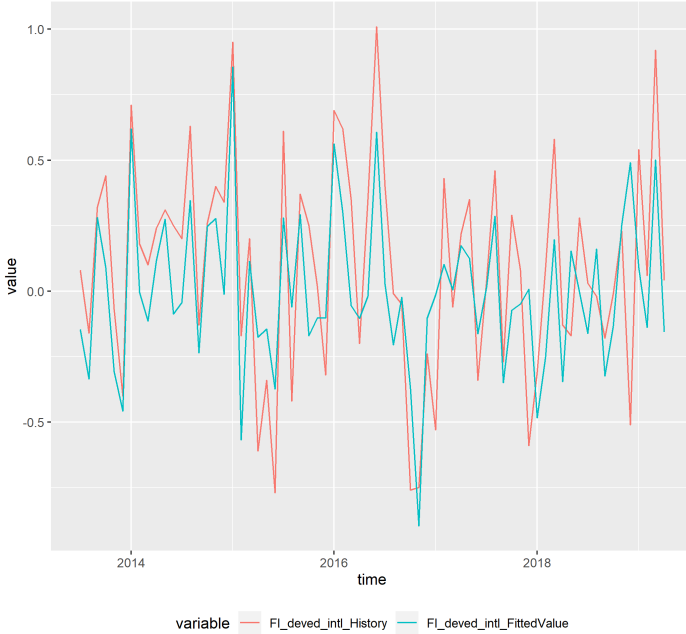


$R^2: 0.43 / -3.3331 * USGG10YR_IndexMoMPercentChangeL0$

4.13.4 Model - 4

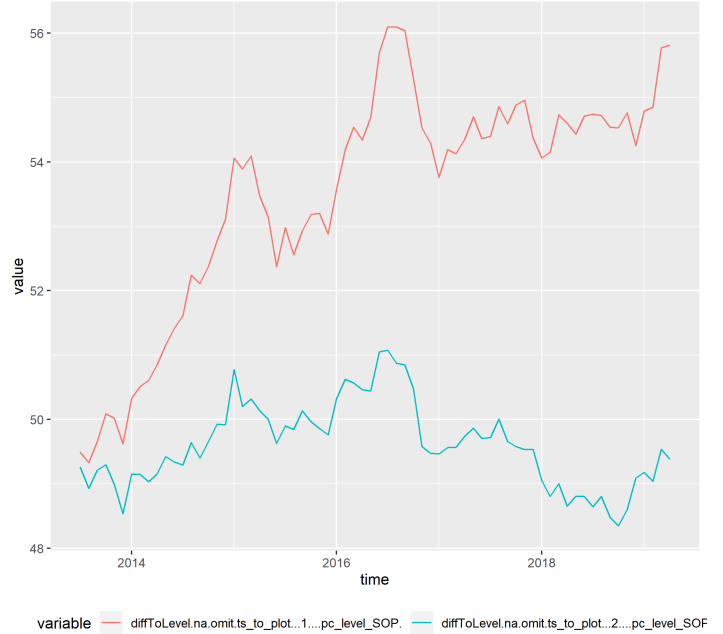
Backtesting on differenced values

FI_deved_intl - Backtesting

 $R^2: 0.5 / -1.6138 * USGG10YR_IndexMoMDifferenceL0$ 

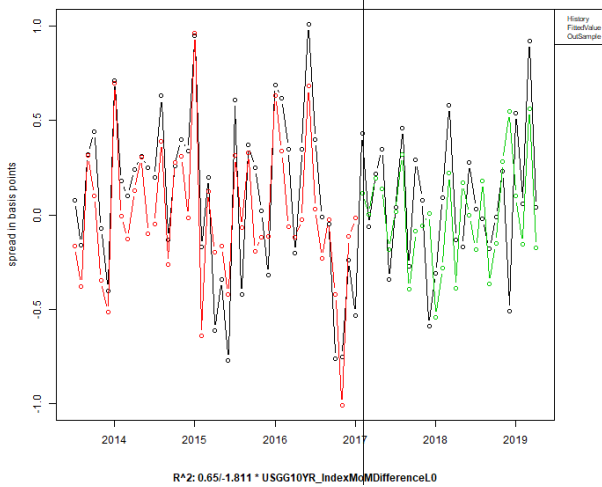
Backtesting on level values

FI_deved_intl - Level Value Backtesting

 $R^2: 0.5 / -1.6138 * USGG10YR_IndexMoMDifferenceL0$ 

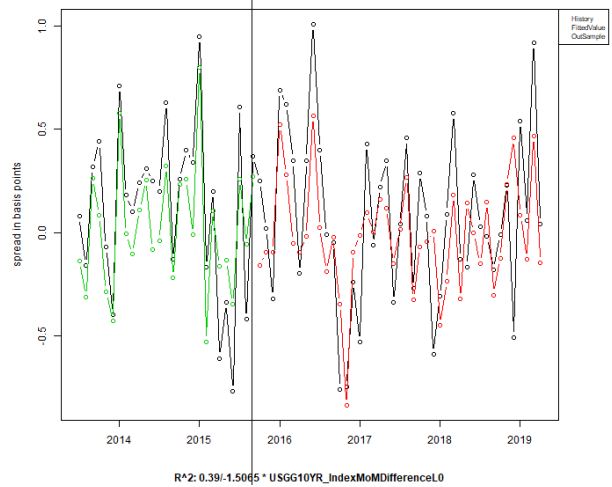
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



4.13.5 Model - 5

Backtesting on differenced values

FI_deved_intl - Backtesting

$R^2: 0.16 / -1.4212 * USGG2YR_IndexMoMDifferenceL0$

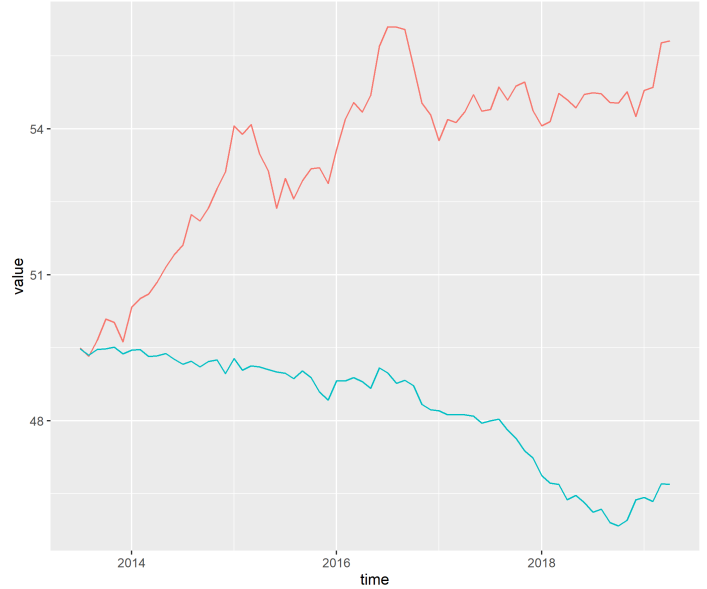


variable — FI_deved_intl_History — FI_deved_intl_FittedValue

Backtesting on level values

FI_deved_intl - Level Value Backtesting

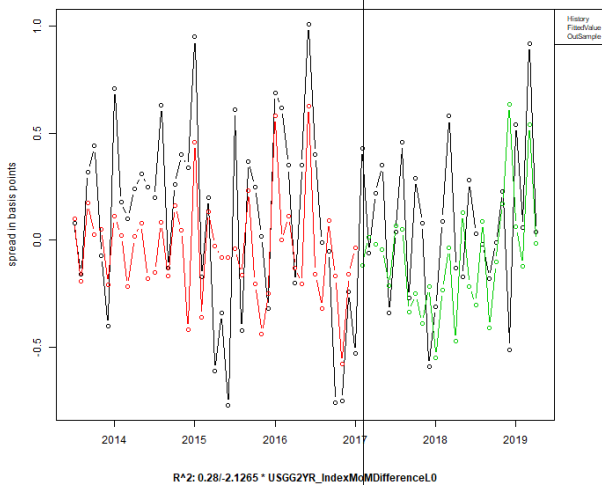
$R^2: 0.16 / -1.4212 * USGG2YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

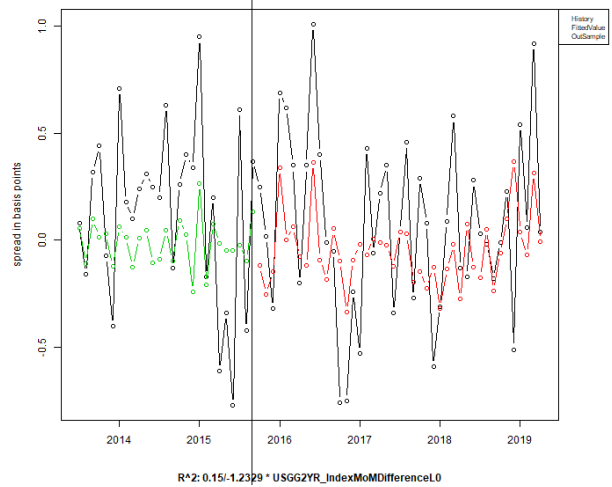
FI_deved_intl Jul 2013-Apr 2019



$R^2: 0.28 / -2.1265 * USGG2YR_IndexMoMDifferenceL0$

Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



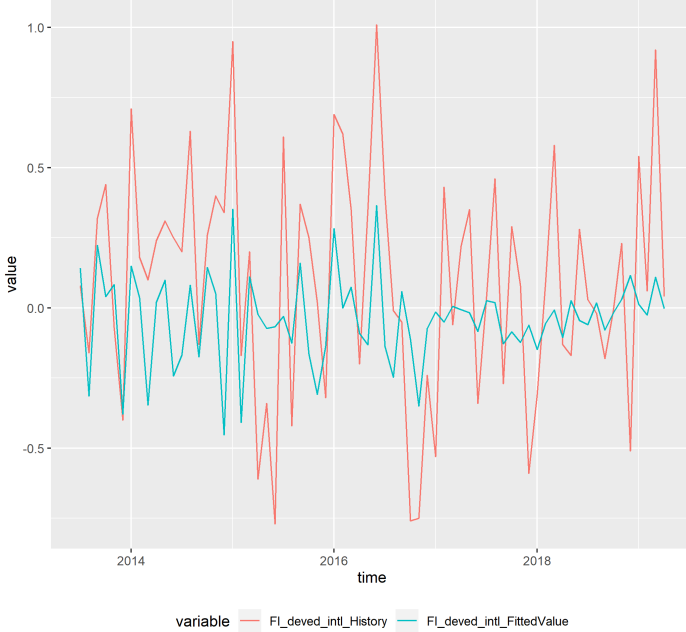
$R^2: 0.15 / -1.2329 * USGG2YR_IndexMoMDifferenceL0$

4.13.6 Model - 6

Backtesting on differenced values

FI_deved_intl - Backtesting

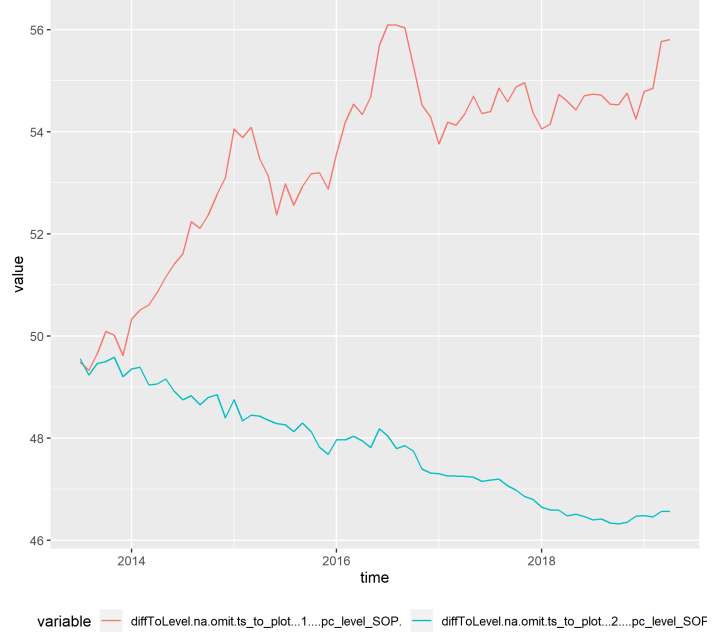
$R^2: 0.15 / -1.0824 * USGG2YR_IndexMoMPercentChangeL0$



Backtesting on level values

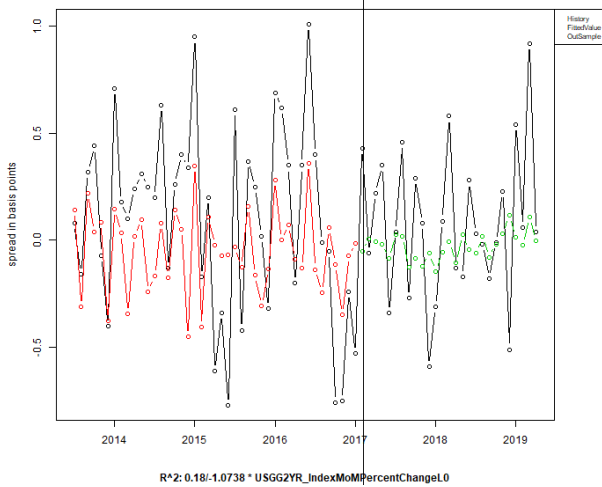
FI_deved_intl - Level Value Backtesting

$R^2: 0.15 / -1.0824 * USGG2YR_IndexMoMPercentChangeL0$



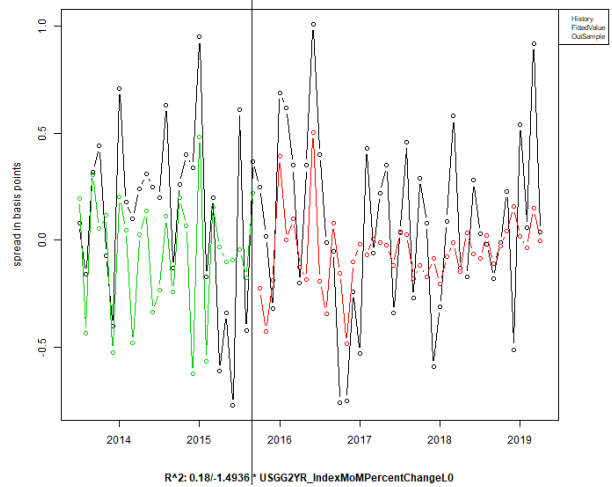
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019

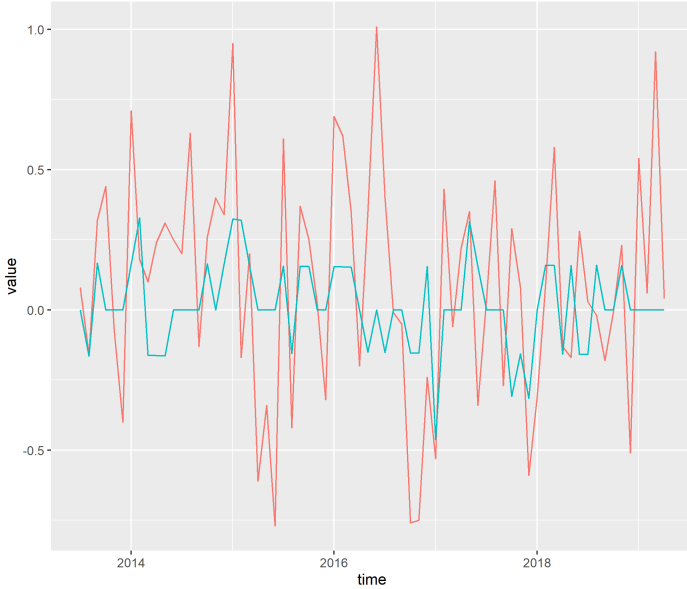


4.13.7 Model - 7

Backtesting on differenced values

FI_deved_intl - Backtesting

R²: 0.12/ 21.2808 * ISRATIOMoMPercentChangeL1

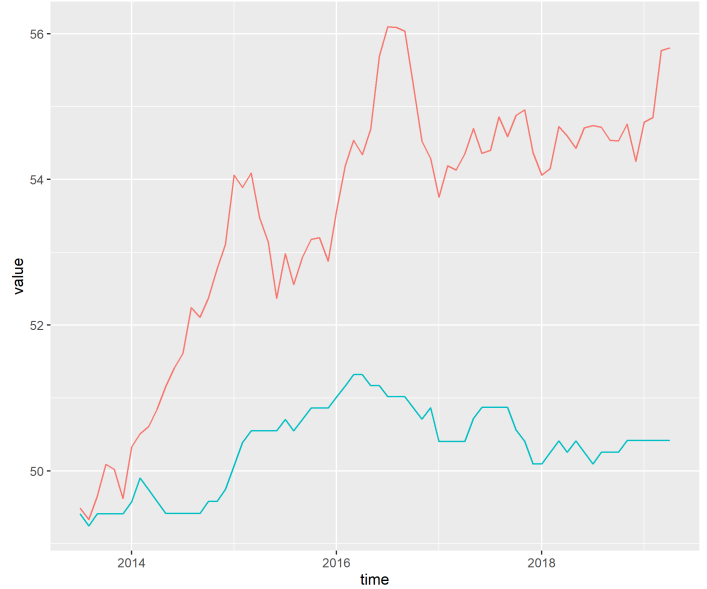


variable — FI_deved_intl_History — FI_deved_intl_FittedValue

Backtesting on level values

FI_deved_intl - Level Value Backtesting

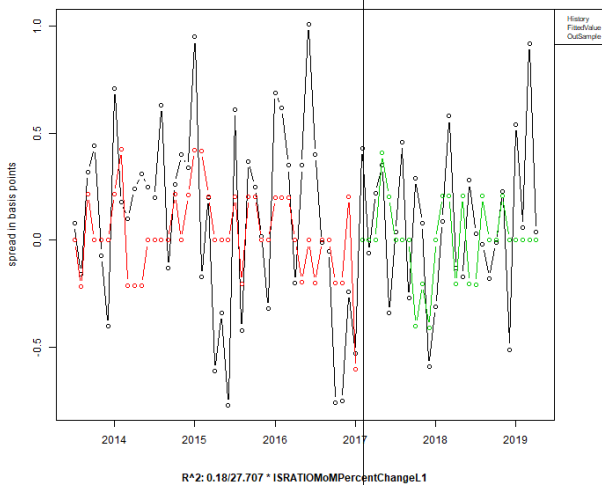
R²: 0.12/ 21.2808 * ISRATIOMoMPercentChangeL1



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

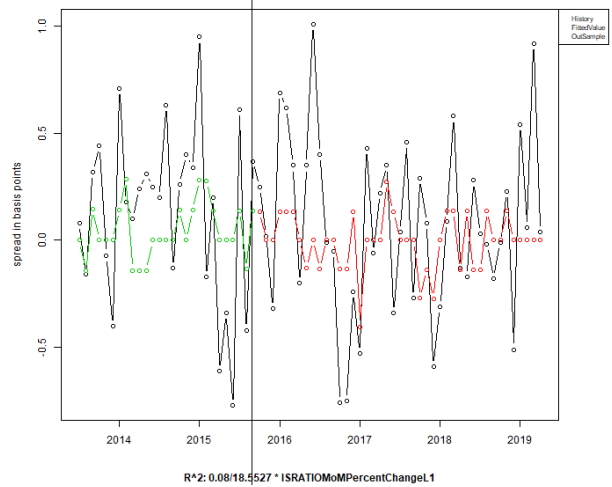
FI_deved_intl Jul 2013-Apr 2019



R²: 0.18/27.707 * ISRATIOMoMPercentChangeL1

Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019

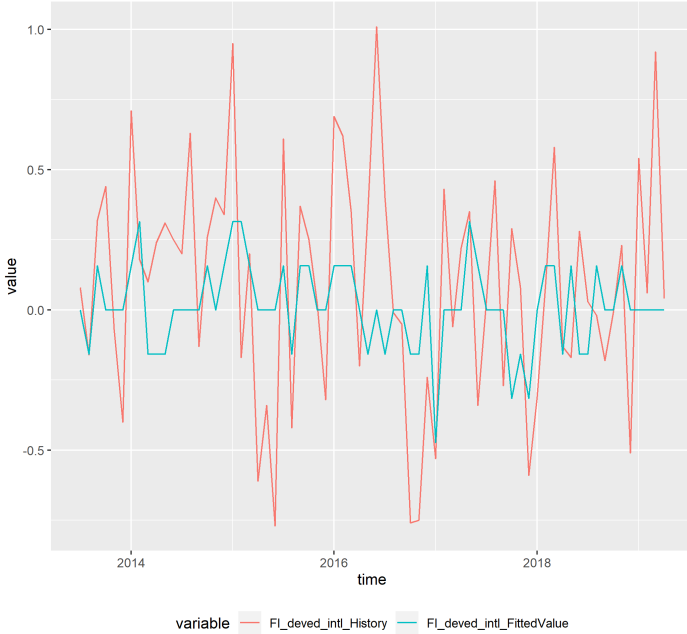


R²: 0.08/18.5527 * ISRATIOMoMPercentChangeL1

4.13.8 Model - 8

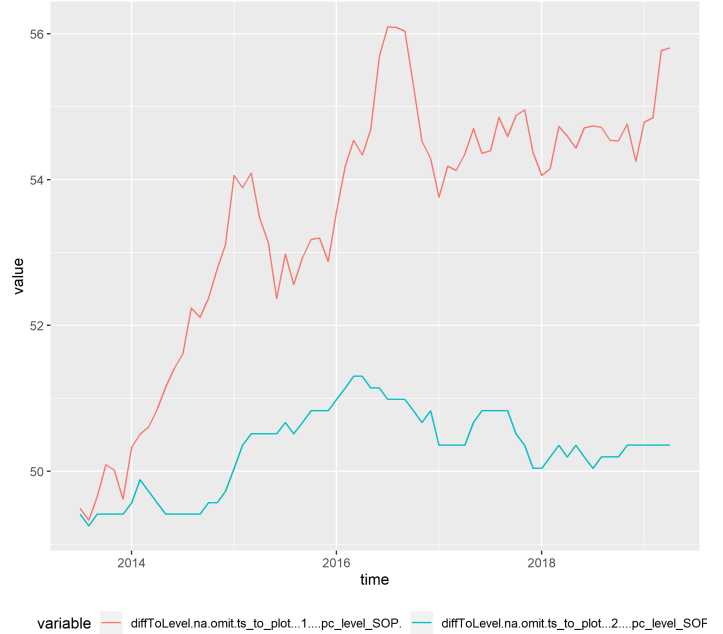
Backtesting on differenced values

FI_deved_intl - Backtesting
R²: 0.12/ 15.7656 * ISRATIOMoMDifferenceL1



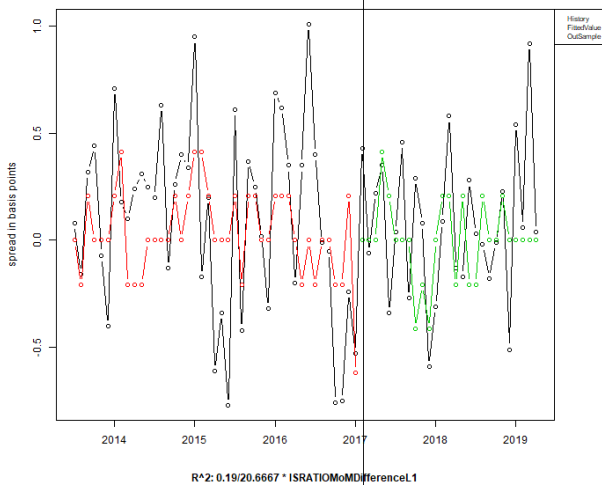
Backtesting on level values

FI_deved_intl - Level Value Backtesting
R²: 0.12/ 15.7656 * ISRATIOMoMDifferenceL1



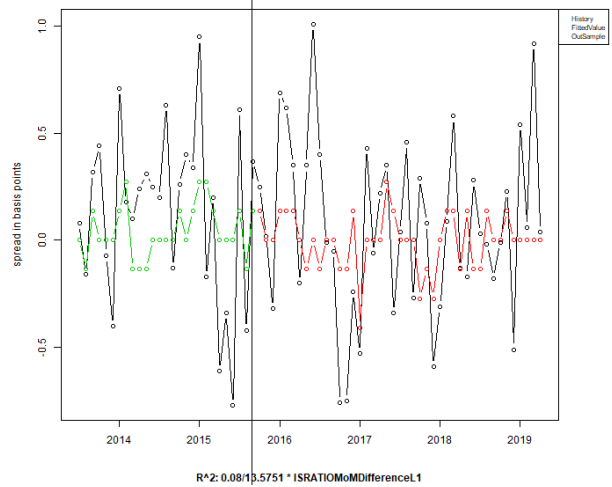
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

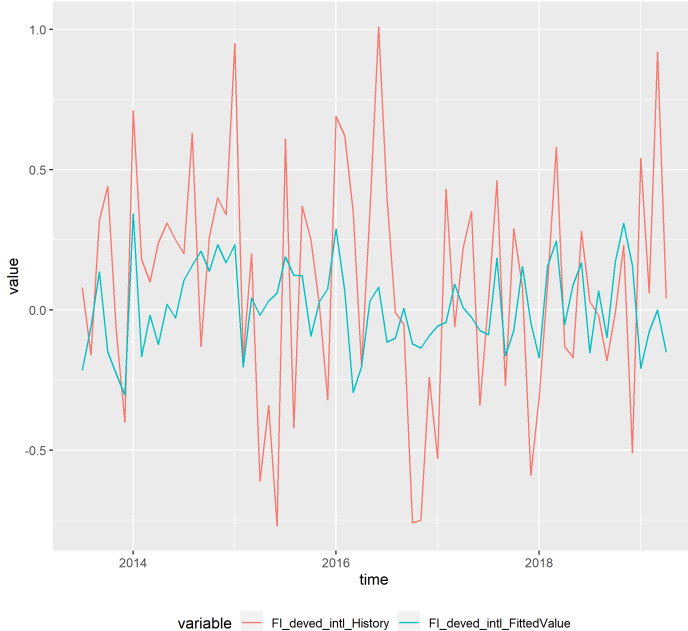
FI_deved_intl Jul 2013-Apr 2019



4.13.9 Model - 9

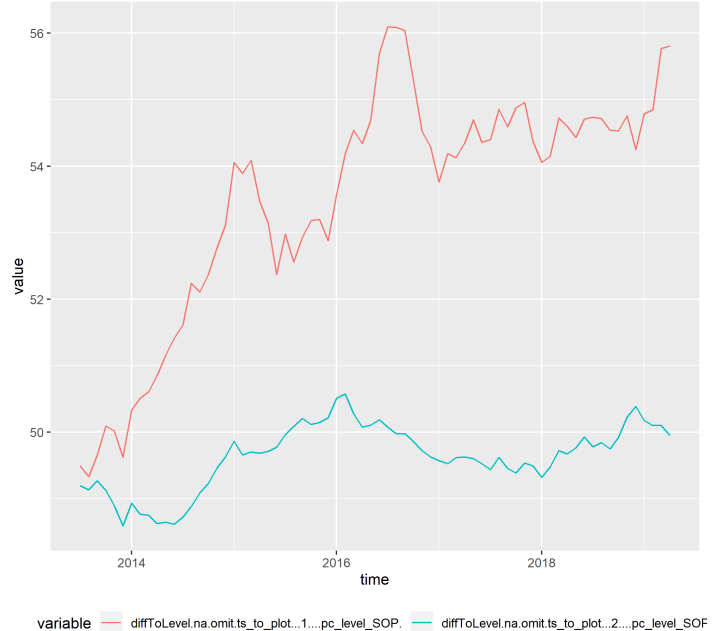
Backtesting on differenced values

FI_deved_intl - Backtesting

 $R^2: 0.12 / 1.4993 * BAA10YB_IndexMoMPercentChangeL0$ 

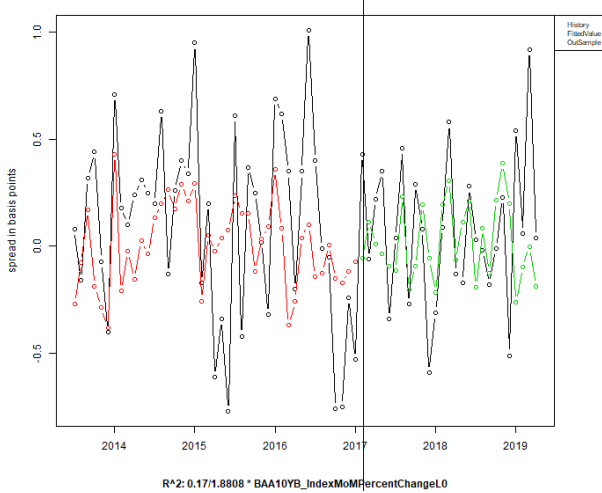
Backtesting on level values

FI_deved_intl - Level Value Backtesting

 $R^2: 0.12 / 1.4993 * BAA10YB_IndexMoMPercentChangeL0$ 

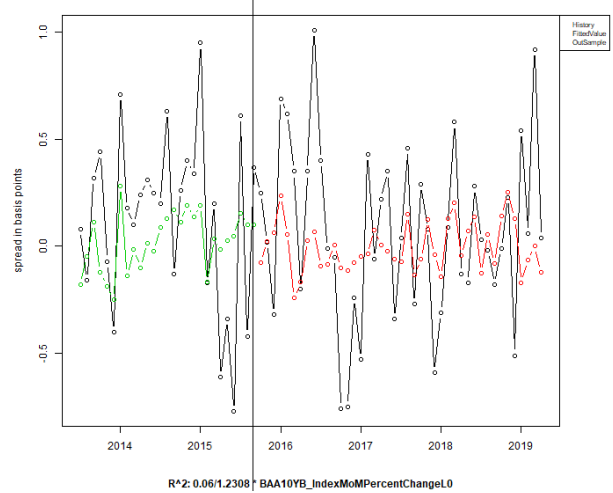
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

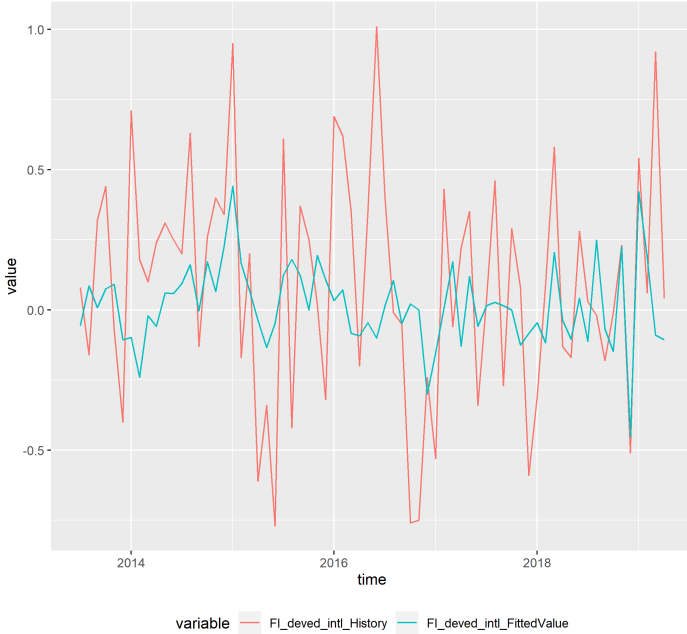
FI_deved_intl Jul 2013-Apr 2019



4.13.10 Model - 10

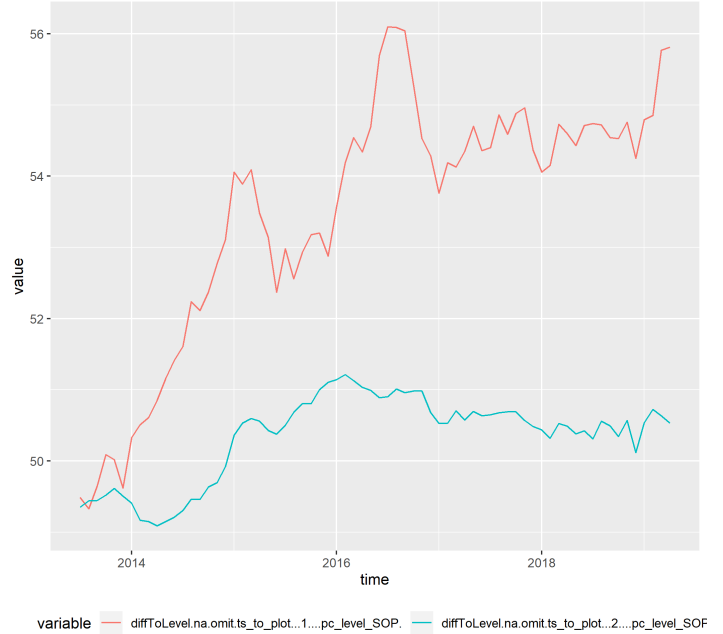
Backtesting on differenced values

FI_deved_intl - Backtesting
R²: 0.11/-0.0209 * WPSID62MoMDifferenceL0



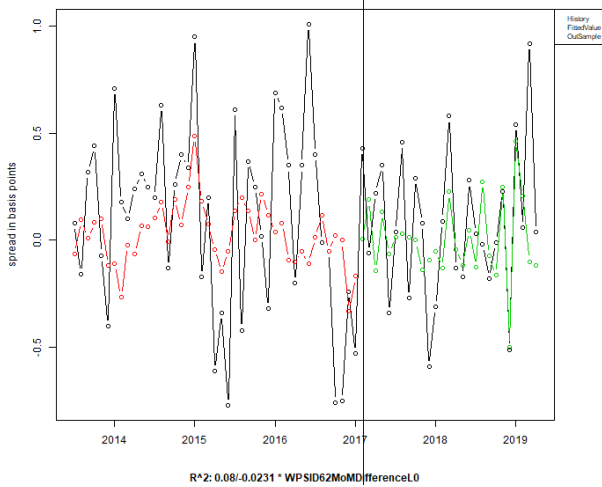
Backtesting on level values

FI_deved_intl - Level Value Backtesting
R²: 0.11/-0.0209 * WPSID62MoMDifferenceL0



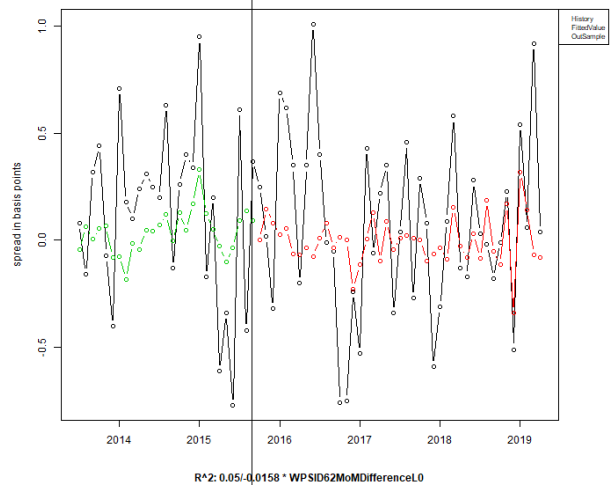
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019

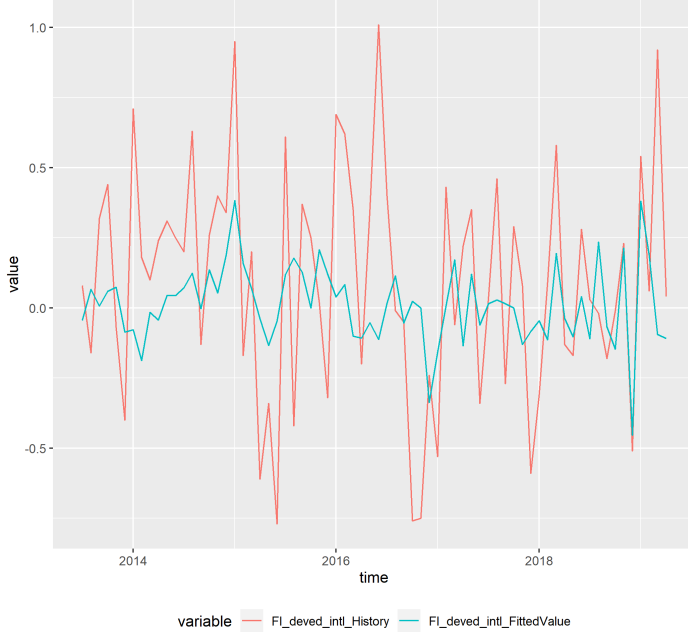


4.13.11 Model - 11

Backtesting on differenced values

FI_deved_intl - Backtesting

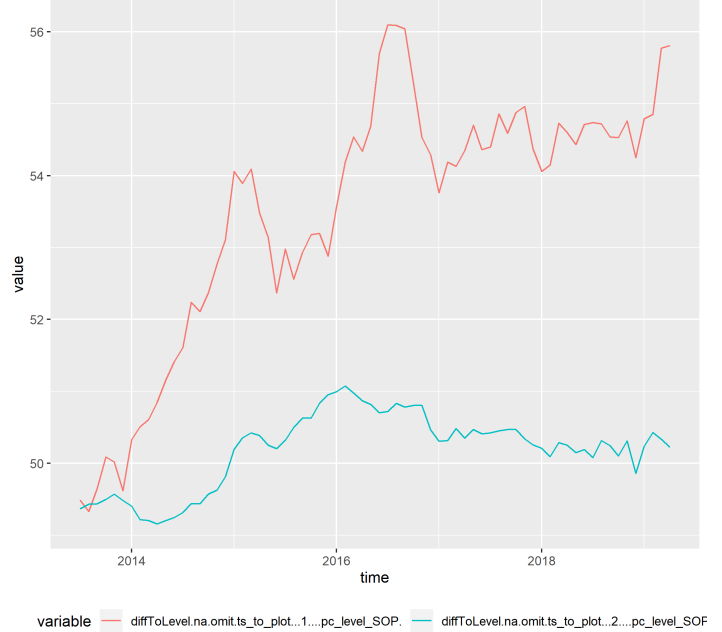
$R^2: 0.1 / -4.0584 * WPSID62MoMPercentChangeL0$



Backtesting on level values

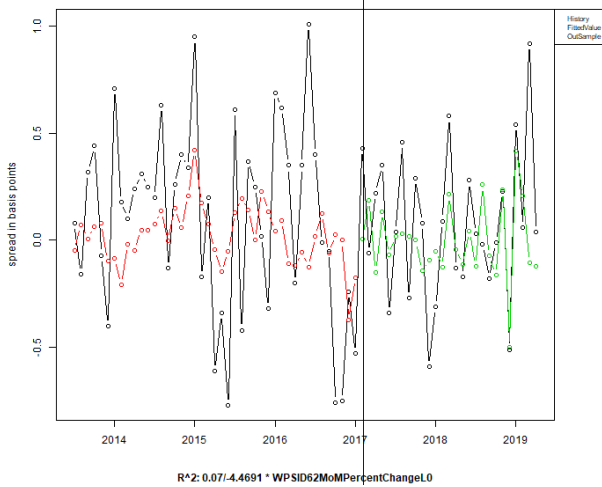
FI_deved_intl - Level Value Backtesting

$R^2: 0.1 / -4.0584 * WPSID62MoMPercentChangeL0$



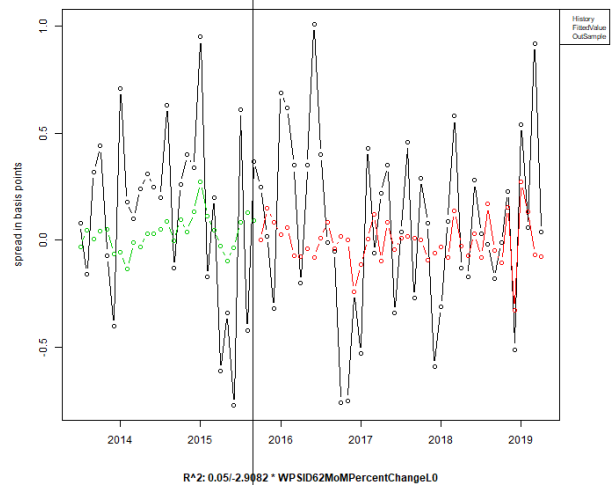
Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019

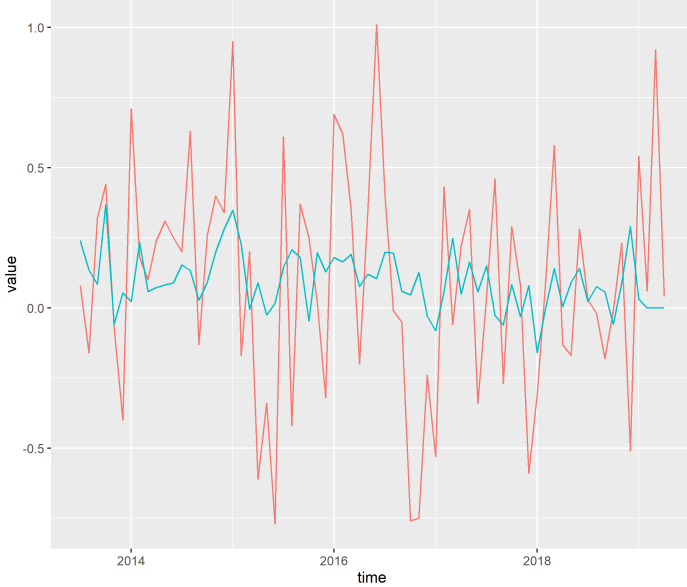


4.13.12 Model - 12

Backtesting on differenced values

FI_deved_intl - Backtesting

$R^2: 0.1/28.4805 * M2REALMoMPercentChangeL0$

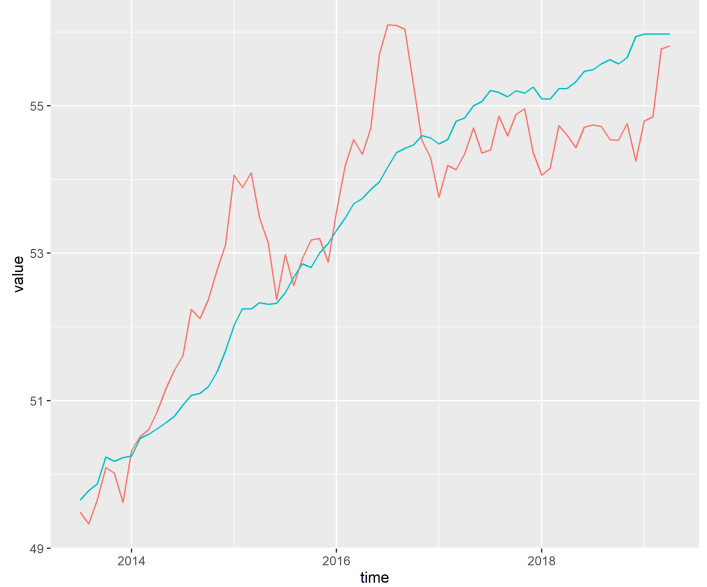


variable — FI_deved_intl_History — FI_deved_intl_FittedValue

Backtesting on level values

FI_deved_intl - Level Value Backtesting

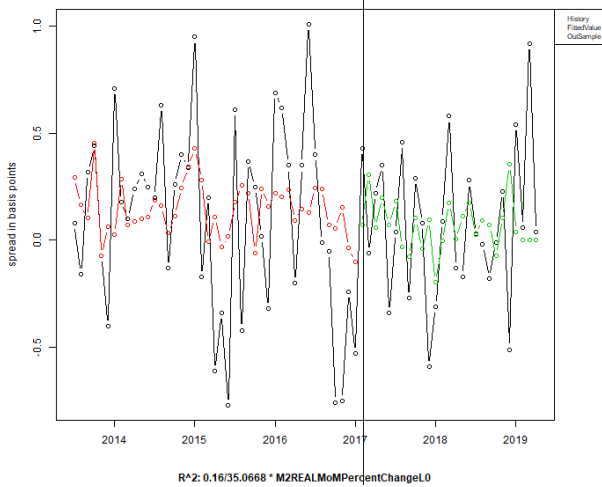
$R^2: 0.1/28.4805 * M2REALMoMPercentChangeL0$



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOF

Out-of-Sample testing

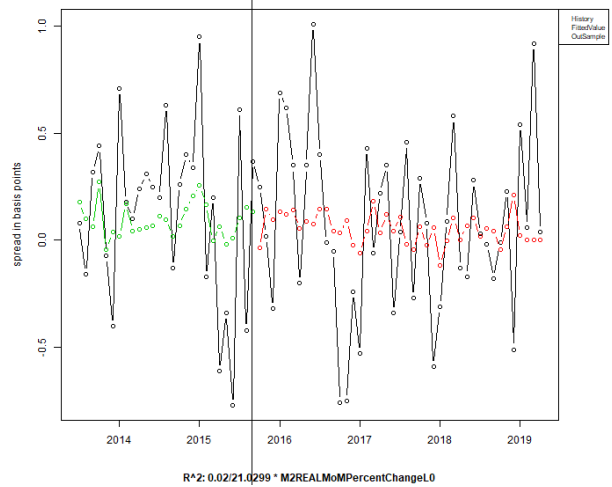
FI_deved_intl Jul 2013-Apr 2019



$R^2: 0.16/35.0668 * M2REALMoMPercentChangeL0$

Out-of-Sample testing

FI_deved_intl Jul 2013-Apr 2019



$R^2: 0.02/21.0299 * M2REALMoMPercentChangeL0$

4.14 Two Factor Model for $\text{FI}_{d\text{eved}_i\text{ntl}}$

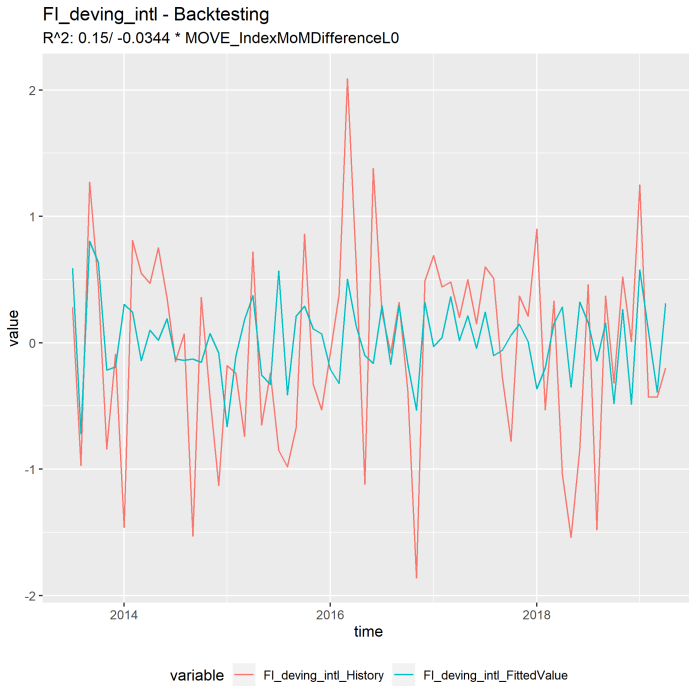
4.15 Single Factor Model for $FI_{deving_i ntl}$

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	MOVE_IndexMoMDifferenceL0	0.15	153.38	-0.03	0.00	0.64			
2	MOVE_IndexMoMPercentChangeL0	0.14	154.38	-2.27	0.00	0.68			

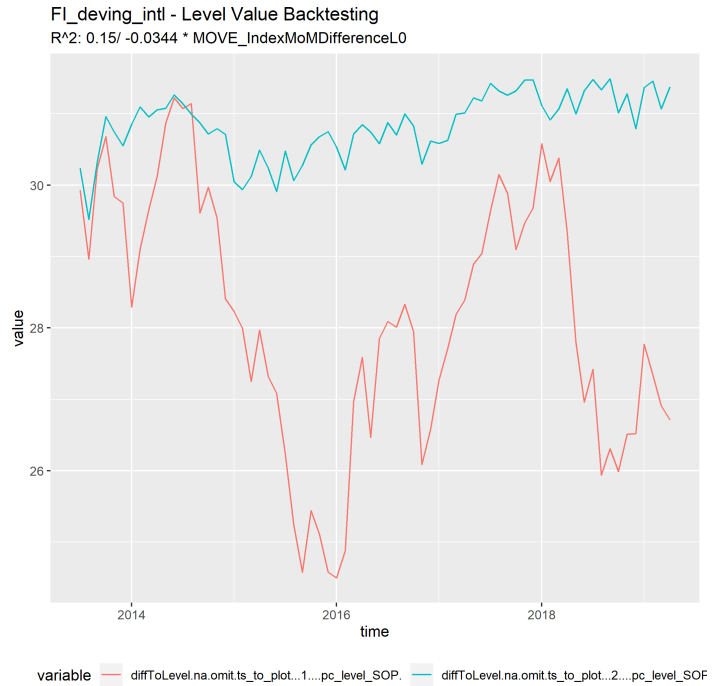
Table 10: Single Factor Analysis- $FI_{deving_i ntl}$

4.15.1 Model - 1

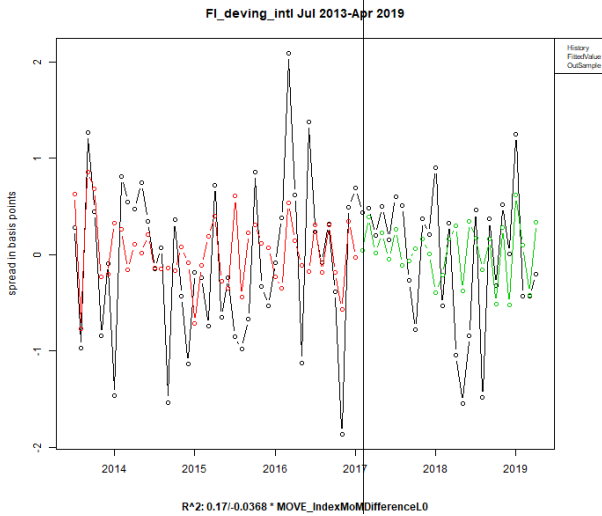
Backtesting on differenced values



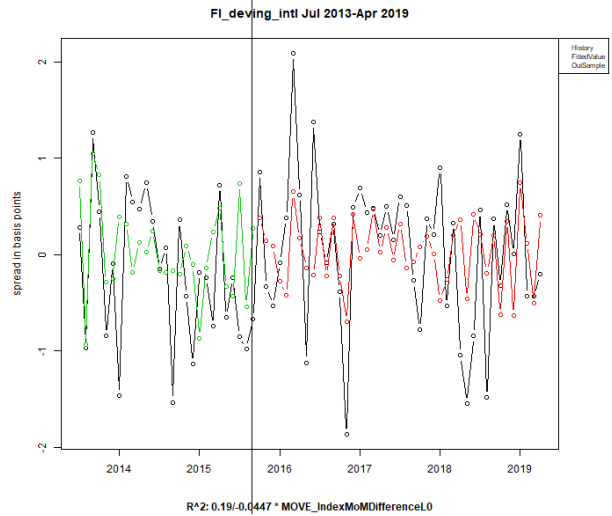
Backtesting on level values



Out-of-Sample testing

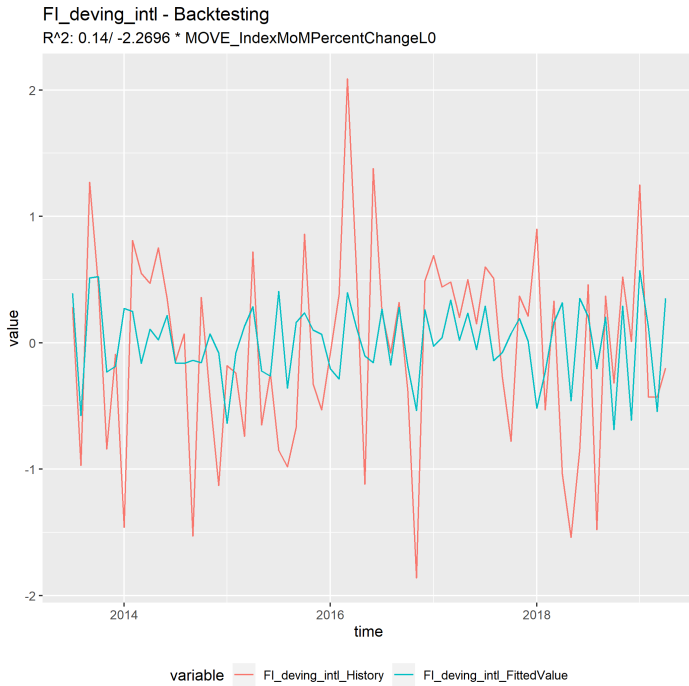


Out-of-Sample testing

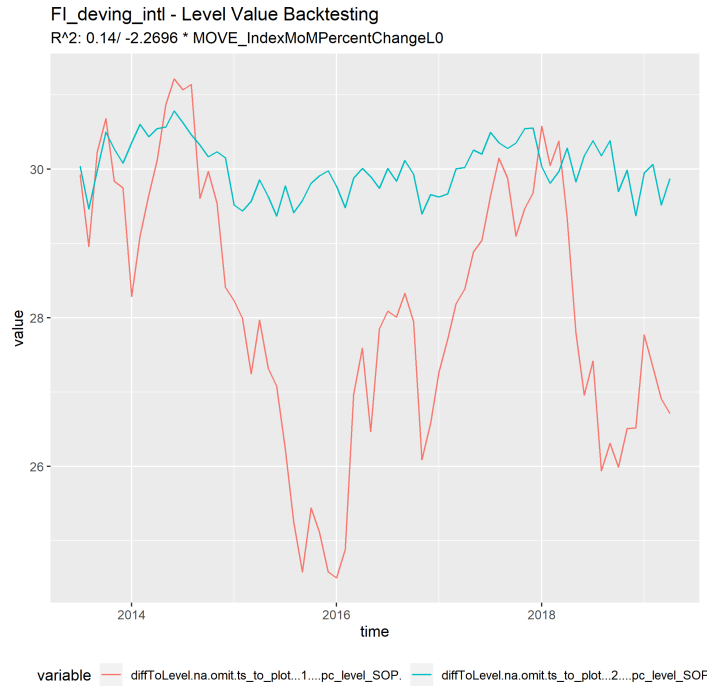


4.15.2 Model - 2

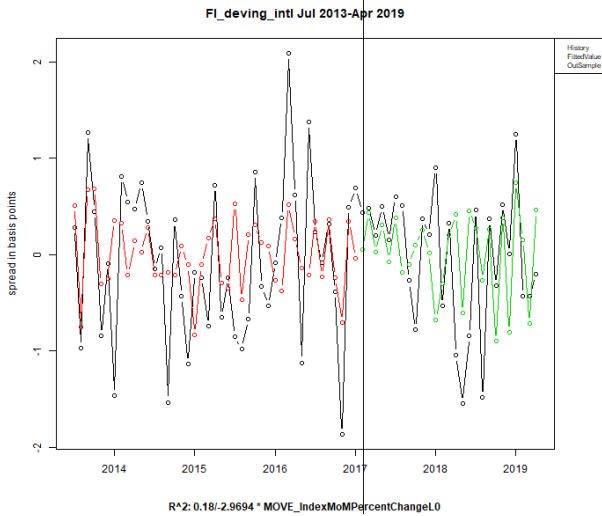
Backtesting on differenced values



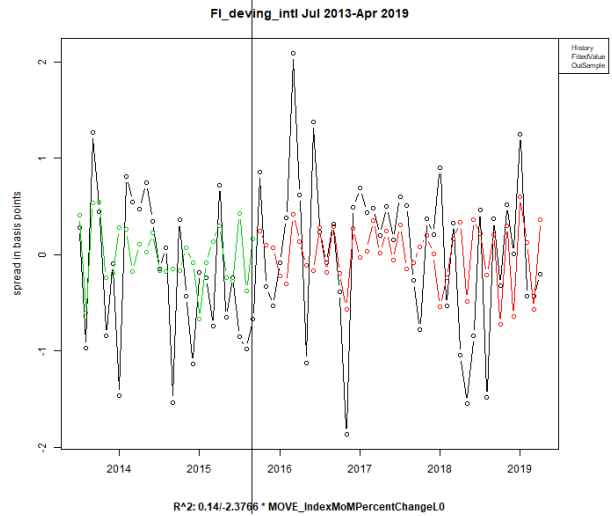
Backtesting on level values



Out-of-Sample testing



Out-of-Sample testing



4.16 Two Factor Model for $FI_{deving_i ntl}$